A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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Contents

1 Topological Preliminaries

This section aims to set the playing ground of the thesis. Since Morse theory needs the topological spaces to be rather restricted and the goal is to relate Morse Homology to topological K-Theory, there is an urge to just not worry about topological nuances. However, for proving things we still need the vocabulary of topological notions. This chapter serves to provide the notions and theorems with references whilst omitting the technical proofs. Steenrod himself writes in his article "A convenient category of topological spaces" [Ste67]:

> "Our need is to be able to make a variety of constructions and to know that the results have good properties without the tedious spelling out at each step of lengthy hypotheses such as countably paracompact normal, completely regular, first axiom of countability, metrizable, and so forth. It may be good research technique and an enjoyable exercise to analyse the precise circumstances for which an argument works; but if a developing theory is to be handy for research workers and attractive to students, then simplicity of the fundamentals must be the goal."

Whilst Steenrod argues for the category of *compactly generated Hausdorff spaces* the standard nowadays is the category of *compactly generated weak Hausdorff spaces*. The latter has direct limits, whilst Steenrod's doesn't. One might wonder why we don't just use the category of CW-complexes, like Hatcher does. There are two reasons for this. The main goal is a Morse theoretic Spectral sequence, which relies on a Morse theoretic filtration. It would be a theoretical drawback to assume that all spaces already omit a CW filtration, as such a filtration naturally already leads to a Atiyah Hirzebruch Spectralsequence. The second reason is, that whilst a lot of pointset topological technicalities can be omitted using CW-complexes, not all of them vanish. This results in the necessity of a discussion of those technicalities, or ignoring the details. Hence we stay away from CW complexes. Let us start constructing this convenient category. All topological constructions that are taught in the compulsory section of a university are assumed to be known. Most proves are omitted but can be found in [Str].

1.1 Compactly Generated Weak Hausdorff Spaces

Remark 1.1.1 (Notation and Convention)**.** The literature on topological spaces in geometric contexts is widely non-uniform. For this work, a topological space is *compact* if it is quasi-compact (every open cover has a finite subcover) and Hausdorff (every pair of points can be separated by disjoint open subsets).

Definition 1.1.2 (Compactly Generated)**.** A topological space X is **compactly generated**, if a subset $A \subseteq X$ is closed, if and only if for all compact K and continuous $f : K \to X$ $f^{-1}(A)$ is closed.

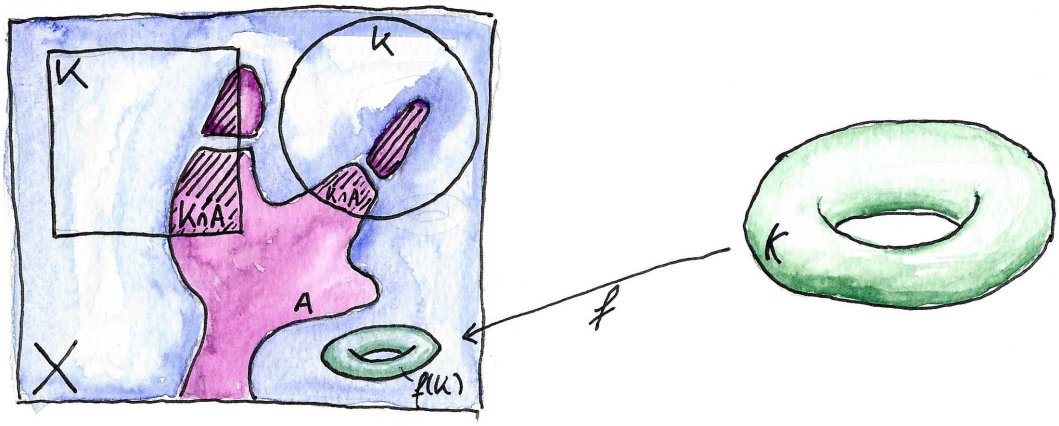

Figure 1: The definition of compactly generated weak Hausdorff spaces.

Definition 1.1.3 (Weak Hausdorff)**.** A topological space X is called weak Hausdorff, if for all compact spaces K and continuous map $f : K \to X$ the image $f(K)$ is closed in X.

Definition 1.1.4. We call the category of compactly generated weak Hausdorff spaces together with continuous functions as morphisms **CGWH**.

Proposition 1.1.5.
1. Every closed subset of a compactly generated weak Hausdorff space is compactly generated weak Hausdorff in the subspace topology.
2. Every locally compact Hausdorff space, and hence every compact space, is compactly generated weak Hausdorff.
3. Every manifold is compactly generated weak Hausdorff.
4. Every metric space is compactly generated weak Hausdorff.
5. Every disjoint union of compactly generated weak Hausdorff spaces is compactly generated weak Hausdorff.
6. If X is compactly generated weak Hausdorff and Y locally compact Hausdorff, then $X \times Y$ is compactly generated weak Hausdorff in the product topology.
7. If X is compactly generated weak Hausdorff and $Y \subseteq X$ a closed subset of X. Then the quotient topology on X/Y is compactly generated and weak Hausdorff.

There is another caveat and reason for the usage of **CGWH**. If we view two CW-complexes as topological spaces, their product might not be a CW-complex. However as objects in **CGWH** their product is a CW-complex. The reason behind this is that the general product doesn't produce the final topology; when it fails to be compactly generated it is strictly coarser than the CW-topology. Passing to the product in **CGWH** (hence making the topologgy finer to become compactly generated) repairs this, and the product of CW-complexes is again a CW-complex.

Remark 1.1.6. From now on all topological spaces will be compactly generated weak Hausdorff. For such spaces the product $X \times Y$ always means the product in **CGWH**.Proposition 1.1.5 (6) tells us: If one of the factors is also locally compact Hausdorff, the product in **CGWH** agrees with the classsical product.

For the construction of topological K-Theory we will heavily rely on the two theorems from pointset topology:

Theorem 1.1.7 (Tietze Extension Theorem)**.** *Let X be a normal space (and hence the theorem is true in* **CGWH***), $Y \subseteq X$ a closed subspace, V a finite dimensional real vector space, and $f : Y \to V$ a continuous map. Then there exists a continuous map $g : X \to V$ such that $g|_Y = f$.*

Theorem 1.1.8 (Existence of Partitions of Unity)**.** *Let X be a compact space, $\{U_i\}$ a finite open covering. Then there exist continuous maps $f_i : X \to \mathbb{R}$ such that:*

1. $f_i(x) \geq 0$ *for all* $x \in X$.
2. $\operatorname{supp}(f_i) \subseteq U_i$.
3. $\sum_i f_i(x) = 1$ *for all* $x \in X$.

Such a collection $\{f_i\}$ is called a ***partition of unity****.*

1.2 Topological Pairs

Definition 1.2.1 (Homotopy Extension Property)**.** Let (X, A) be a topological pair, i.e. $A \subseteq X$. The pair satisfies the homotopy extension property if for any map $f : X \to Y$ and any homotopy $f_t : A \to Y$ with $f_0 = f|_A$ there exists an extension $\tilde{f}_t : X \to Y$ with $\tilde{f}_t|_A = f_t$ and $\tilde{f}_0 = f$. The inclusion $i : A \to X$ of such a pair is called a **cofibration**. We denote the category of pairs of compactly generated weak Hausdorff spaces where the inclusion is a cofibration by $\mathbf{CGWH}^2_{\mathbf{cof}}$.

A more intuitive but (for A closed in X) equivalent definition is:

Definition 1.2.2. If (X, A) is a topological pair, A is closed in X and there is an neighbourhood $U \supseteq A$ such that U deformation retracts onto A. Such a pair is called an **NDR**-pair which stands for neighbourhood deformation retract-pair.

Remark 1.2.3. If we encounter an NDR pair in this thesis that lives in a manifold, we will often omit proving the property. A proof would mostly consist of applying the tubular neighbourhood theorem and the collar neighbourhood theorem.

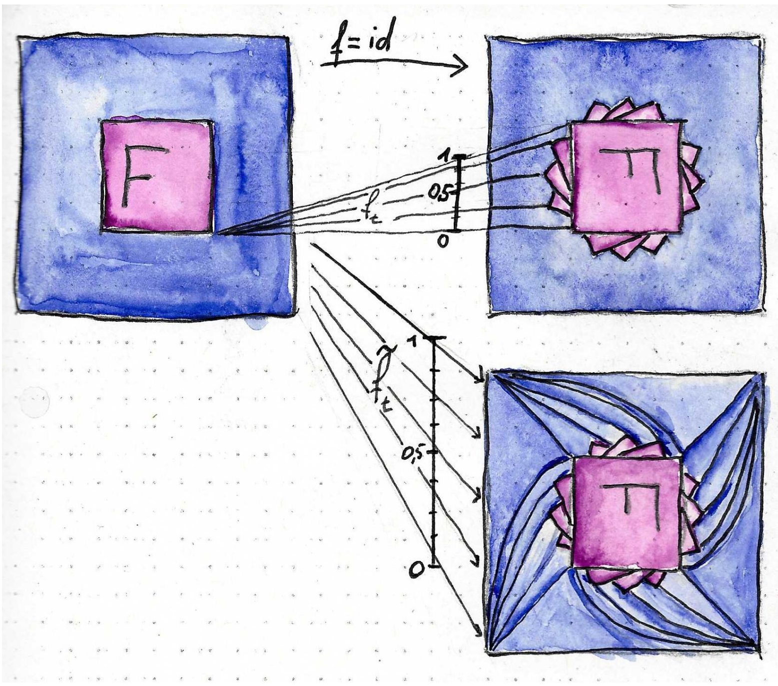

Figure 2: The homotopy extension property for the pair (X, F), where F gets rotated.

> **Lemma 1.2.4.** *If the pair (X, A) satisfies the homotopy extension property and A is contractible, then the quotient map*
>
> $$q : X \to X/A$$
>
> *is a homotopy equivalence.*

Proof. Let $f_t : X \to X$ be a homotopy extending the contraction of A with $f_0 = \mathrm{id}$. Then the composition $q \circ f_t$ maps A to a point and hence factors over X/A for all t giving us the commutative diagram

$$\begin{tikzcd} X \arrow[r, "f_t"] \arrow[d, "q"] & X \arrow[d, "q"] \\ X/A \arrow[r, "\overline{f_t}"'] & X/A \end{tikzcd}$$

Here, $\overline{f_t}$ denotes the induced map on the quotient. Now $f_1(A) = pt$ where pt is a point in A. So f_1 factors over X/A and denote the induced map by $g : X/A \to X$. Now $g \circ q = f_1$ and hence

$$q \circ g(\overline{x}) = q \circ g \circ q(x) = q \circ f_1(x) = \overline{f_1}(\overline{x})\,.$$

In other words we have the commutative diagram

$$\begin{tikzcd} X \arrow[r, "f_1"] \arrow[d, "q"] & X \arrow[d, "q"] \\ X/A \arrow[r, "\overline{f_1}"'] \arrow[ur, "g"] & X/A \end{tikzcd}$$

This concludes the proof, since $g \circ q = f_1 \simeq f_0 = \mathrm{id}$ and $q \circ g = \overline{f_1} \simeq \overline{f_0} = \mathrm{id}$. □

Definition 1.2.5. We call a pointed topological space (X, x_0) well pointed, if the inclusion $x_0 \to X$ is a cofibration. We denote the category of well pointed compactly generated weak Hausdorff spaces by $\mathbf{CGWH}^*_{\mathbf{cof}}$.

1.3 Mapping Spaces

Definition 1.3.1 (Compact-open topology)**.** Let X, Y be topological spaces. Then the set of continuous maps $C(X, Y)$ can be equipped with the topology that has

$$\{V(K,U) | K \subseteq X \text{ compact }, U \subseteq Y \text{ open}\} \text{ where } V(K,U) \coloneqq \{f \in C(X,Y) |\ f(K) \subseteq U\}$$

as a subbasis. For real vector spaces V, W in the standard topology, this coincides with the topology of $\mathrm{Hom}(V, W)$ as a subset of $\mathbb{R}^n$ after identification with the corresponding matrix.

Proposition 1.3.2. If X is compactly generated and Y compactly generated weak Hausdorff, then $C(X, Y)$ is compactly generated weak Hausdorff.

Proposition 1.3.3. Let X, Y, Z be objects in **CGWH**. The natural map

$$C(X, C(Y, Z)) \to C(X \times Y, Z)$$

is a homeomorphism.

2 Vector Bundles and Riemannian Geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another.

2.1 Differentiable Structures on Topological Manifolds

We begin by defining a topological manifold. Afterwards, we equip that space with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

Definition 2.1.1 (Topological Manifold)**.** A second countable Hausdorff space M is called a **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, for all $p \in M$ there exists an open neighbourhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : U \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M, and φ^{-1} a **local coordinate system around** p **on** M.

Definition 2.1.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension m.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$, the **transition function**:

   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$

   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$, we call the pair (M, c) a smooth manifold.

Corollary 2.1.3. Every transition function φ_{ij}, $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i\in I}$ is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that $\varphi_{ij}^{-1} = \varphi_{ji}$ which provides smooth inverses.

Definition 2.1.4 (Smooth Functions)**.** Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any chart (and hence for all) $(\varphi_i)_{i\in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f : U \to \mathbb{R} \text{ continuous} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 2.1.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$−algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \operatorname{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V . \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 2.1.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighbourhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p\in U} \mathcal{E}(U) \right) \Big/ \sim_p$$

for the set of germs, and call it the **stalk in** p. Where $\sum$ denotes the co-product (disjoint union) in **CGWH**.

Corollary 2.1.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$−algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \operatorname{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\operatorname{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. We only need to verify the local ring structure of the stalks. An element $f_p \in \mathcal{E}_p(M)$ is invertible if and only if $f(p) \neq 0$, equivalently if $f_p \notin \ker(\mathrm{eval}_p)$. Thus $\mathcal{E}_p(M)\big/\ker(\mathrm{eval}_p)$ is a field, which proves that $\ker(\mathrm{eval}_p)$ is maximal. □

Definition 2.1.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it an $\mathbb{R}$-vector space structure by pointwise operations. We define the **tangent space of M at p** to be the vector space

$$T_pM := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})\,.$$

Corollary 2.1.9. Let (M, c) be a smooth manifold of dimension m and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ (where $\varphi \in \mathfrak{A} \in c$). Then we define the derivation

$$\begin{aligned} \partial_i(p) : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto \partial_i(p)(f_p) := \frac{\partial(f \circ \varphi^{-1})}{\partial x_i}\Big|_{x_0}\,. \end{aligned}$$

This is well-defined and hence gives a tangent vector. Moreover, the family

$$(\partial_1(p), \dots, \partial_m(p))$$

defines a basis of T_pM. Hence, the dimension of T_pM is m.

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

Analogously to the definition of a smooth manifold, we start by defining families of vector spaces and equip them with increasingly more structure:

Definition 2.1.10 (Vector Bundles)**.** Let $\mathbb{K}$ be a field. A $\mathbb{K}$ **vector bundle** of rank m is a surjection

$$\pi : E \to X$$

such that E and X are topological spaces and π is continuous. Furthermore π satisfies:

1. For all $x \in X$, $\pi^{-1}(x)$ carries the structure of an m-dimensional $\mathbb{K}$-vector space.
2. For all $x \in X$, there exist a neighbourhood $U \subseteq X$ and a homeomorphism

$$\psi : E|_U := \pi^{-1}(U) \to U \times \mathbb{K}^m\,,$$

that is compatible in the following sense: Let π_1 denote the projection $U \times \mathbb{K}^m \to U$ onto the first factor. Then

$$\pi_1 \circ \psi = \pi$$

and

$$\psi_x : E_x := \pi^{-1}(x) \to \{x\} \times \mathbb{K}^m$$

is a vector space isomorphism.

We call E the **total space**, X the **base space**, E_x the **fibre over** x and ψ a **local trivialisation**.

Remark 2.1.11. In the context of differential geometry we will mostly consider $\mathbb{R}$ vector bundles. However, the K-theory that we will develop will depend on $\mathbb{C}$ vector bundles.

Definition 2.1.12. Let $\pi : E \to X$ be a vector bundle. A continuous function $s : X \to E$ such that $\pi \circ s = \mathrm{id}_X$ is called a (global) **section** (if $s : U \subseteq X \to E$ we call it a local section). We denote the space of global sections by $\Gamma(E)$ and by $\Gamma(U)$ the space of local sections over an open subset $U \subseteq X$.

Definition 2.1.13 (Homomorphism of Vector Bundles)**.** A **homomorphism between two vector bundles** $\pi : E \to X$ and $\tilde{\pi} : F \to X$ over the same base space is a continuous function $\varphi : E \to F$ such that:

1. $$\begin{tikzcd} E \arrow[r, "\varphi"] \arrow[d, "\pi"] & F \arrow[dl, "\tilde{\pi}"] \\ X & \end{tikzcd}$$ commutes.
2. For all $x \in X$ the function $\varphi_x : E_x \to F_x$ is linear.

Definition 2.1.14 (Pullback of Vector Bundles)**.** Let $\pi : E \to Y$ be a vector bundle, X be a topological space and $f : X \to Y$ be continuous. Then we define the **pullback** $f^*\pi : f^*(E) \to X$ as follows: The total space $f^*(E)$ is defined as the subspace of $(x, e) \in X \times E$ such that $f(x) = \pi(e)$, and the map $f^*\pi$ is the projection to the second factor restricted to said subspace.

Example 2.1.15. Let $\pi : E \to X$ be a vector bundle. For any $Y \subseteq X$ we have the vector bundle

$$\pi|_Y : \pi^{-1}(Y) \to Y\,,$$

which is defined as the pullback along the inclusion of Y into X.

Remark 2.1.16. We will often omit the notation of the projection and denote a vector bundle by its total space. The restriction $\pi|_Y : \pi^{-1}(Y) \to Y$ is often denoted by $E|_Y$. The restriction may equivalently be defined by restricting the domain and codomain of the projection.

Remark 2.1.17. Notice that this construction of the pullback $f^*\pi : f^*(E) \to X$ is the simplest space to make the square commute:

$$\begin{tikzcd} f^*(E) \arrow[r, "\pi_2"] \arrow[d, "f^*\pi"] & E \arrow[d, "\pi"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

The notion of simplicity is captured by the universal property, that for any other vector bundle $\tilde{\pi} : \tilde{E} \to X$ and function $\tilde{f} : \tilde{E} \to E$ there exists a unique $h : \tilde{E} \to f^*(E)$ (which is given by $h(y) := (\tilde{\pi}(y), \tilde{f}(y))$ such that the following diagram commutes:

$$\begin{tikzcd}
\tilde{E} \arrow[drr, bend left, "\tilde{f}"] \arrow[dr, "h"] \arrow[ddr, bend right, "\tilde{\pi}"'] & & \\
& f^*(E) \arrow[r, "\pi_2"] \arrow[d, "f^*\pi"] & E \arrow[d, "\pi"] \\
& X \arrow[r, "f"] & Y
\end{tikzcd}$$

Remark 2.1.18. Let $\pi : E \to X$ be a vector bundle. Then the space of sections $\Gamma(E)$ is a (in general, infinite-dimensional) $\mathbb{R}$-vector space. Furthermore, and more relevant to the purposes of this thesis, we view $\Gamma(E)$ (and $\Gamma(U)$ for open subsets) as a $C(X)$ (or $C(U)$) module. The latter denotes the ring of continuous functions from E (or U) to $\mathbb{R}$.

Example 2.1.19. For any smooth manifold M of dimension m , we can equip the union

$$TM := \bigsqcup_{p \in M} T_pM$$

with a topology, such that $\pi : TM \to M$ defines a vector bundle which we call **tangent bundle**. For this we first define the trivialisations and then equip TM with a topology to make them homeomorphisms: Let $\varphi : U \to V$ be a chart on M. If we denote the i-th coordinate function of φ by φ_i we can define the bijection

$$\tilde{\varphi} : \pi^{-1}(U) \to V \times \mathbb{R}^m, \quad \xi_p \mapsto \left(\varphi(p), \begin{pmatrix} \xi_p(\varphi_1) \\ \vdots \\ \xi_p(\varphi_m) \end{pmatrix} \right).$$

Now we declare a subset $W \subseteq TM$ to be open, if and only if for all charts φ the set $\tilde{\varphi}(W \cap \pi^{-1}(U)) \subset V \times \mathbb{R}^m$ is open. We can now extend and give meaning to the notation of the local basis vectors $\partial_i(p) \in T_pM$: For any chart $\varphi : U \to V$ we have the local sections

$$\partial_i : U \to TU, \quad p \mapsto \partial_i(p).$$

Furthermore, TM is a $2m$-dimensional manifold. Using a smooth atlas on M yields a smooth atlas on TM making it a smooth manifold. Notice that the family $(\partial_i)_i$ is a basis of the $C(U)$-module $\Gamma(U)$. This follows by expressing any section in local coordinates via $\tilde{\varphi}$.

Notice that TM as a bundle satisfies more than just the bundle definition: Taking two different charts induces two different local trivialisations. Transitioning between them is not just fibrewise linear, but also smooth. This is captured in the definition:

Henceforth we suppress the differentiable structure from the notation.

Definition 2.1.20 (The Derivative)**.** Let M and M' be smooth manifolds. We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth map between the tangent bundles as manifolds as follows:

$$\mathrm{d}f : TM \to TM' \ , \mathrm{d}f_p(\xi_p)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi_p \in T_pM$, $g_p \in \mathcal{E}_p(M')$. The function $\mathrm{d}f$ is called the **derivative** or **pushforward** of f.

Definition 2.1.21 (Smooth Vector Bundles)**.** A **smooth vector bundle of rank** r is a surjection $\pi : E \to M$ where the codomain (or base space) is a smooth manifold together with a family of continuous functions (or local trivialisations)

$$\Phi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{R}^r$$

such that $\bigcup_i U_i = M$ and for all overlapping $U_i \cap U_j \neq \varnothing$, the transition functions

$$\Phi_j \circ \Phi_i^{-1} : \pi^{-1}(U_i \cap U_j) \to (U_i \cap U_j) \times \mathbb{R}^r$$

are of the form

$$\Phi_j \circ \Phi_i^{-1}(p, v) = (p, g_{i,j}(v)), \qquad \text{where} \qquad g_{i,j} : (U_i \cap U_j) \to \mathrm{GL}(r, \mathbb{R})$$

is smooth.

Corollary 2.1.22. A smooth vector bundle is a smooth manifold and also a vector bundle. The latter is ensured by the codomain of $g_{i,j}$ which is also called the **structure group**. We can enforce structure on the bundle by reducing the structure group. As a preview: reducing it to $\mathrm{GL}^+(k, \mathbb{R})$ the linear maps with positive determinant means that we can induce the concept of orientations. Reducing it to $\mathrm{O}(k)$ lets us equip it with a Riemannian metric.

Definition 2.1.23. Let $\pi : E \to M$ be a smooth vector bundle of rank r and let

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r$$

be a local trivialisation. Then a section is called smooth if it is a smooth map of manifolds. A family of smooth local sections $(s_1, \ldots, s_r)$ with $s_i \in \Gamma(U)$ is called a **local frame** if they form a basis of the $C(U)$-module $\Gamma(U)$. Such a family is obtained from Φ by defining $s_i(p) := \Phi^{-1}(p, e_i)$. Similarly, from a local frame we can obtain a local trivialisation by defining:

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r \, , \quad v = \sum_{i=1}^{r} \lambda_i s_i(\pi(v)) \mapsto (\pi(v), \lambda_1, \ldots, \lambda_r) \, .$$

Example 2.1.24. The local trivialisations ∂_i of the tangent bundle of a smooth manifold M form a local frame and make the tangent bundle into a smooth vector bundle.

Definition 2.1.25 (Homomorphism Bundle)**.** Let W, V be vector spaces and $E = V \times X$, $F = W \times X$ the product bundles over X. Now any bundle homomorphism $\varphi : E \to F$ determines a map $\Phi : X \to \hom(V, W)$ by sending $x \mapsto \varphi_x$. With respect to the compact-open topology, this Φ is continuous. Furthermore, a continuous map $\Phi : X \to \hom(V, W)$ determines a bundle homomorphism $\varphi : E \to F$.

Proof. Let $\{e_i\}$ and $\{f_i\}$ be bases of V and W. Then each $\Phi(x)$ is a matrix $\Phi_{i,j}(x)$, such that:

$$\Phi(x)e_i = \sum_j \Phi(x)_{i,j} f_j \,.$$

Now Φ is continuous if and only if $\Phi_{i,j}$ is continuous. Since $\varphi(x, v) = (x, \Phi(x)v)$ this is equivalent to φ being continuous. ☐

Lemma 2.1.26 (Isomorphic Vector Bundles)**.** *Let E and F be vector bundles and $\varphi : E \to F$ be a homomorphism. Then φ is an isomorphism if and only if all φ_x are isomorphisms.*

Proof. To show that it is an isomorphism we need a continuous inverse, equivalently, we need to show that φ is a homeomorphism. First of all we can deduce, that φ is bijective: Assume it wasn't injective, then there would be a fibre in which φ_x is not injective. Assume that it wasn't surjective. Then there would be a $w \in F$ that has no preimage which means that in the fibre containing w there is no isomorphism. Thus, there is a unique inverse and it remains to verify that it is continuous, which is a local condition. Therefore we can assume that E and F are product bundles. Then we have a continuous map $\Phi : X \to \mathrm{Hom}(V, W)$. Furthermore, since all φ_x are isomorphisms, we can restrict the image to $\mathrm{Iso}(V, W)$ which is an open subset of $\mathrm{Hom}(V, W)$. Now we can define the map $x \mapsto \Phi(x)^{-1}$ which is continuous since taking inverses is continuous. By the corollary above, we obtain a continuous map $\psi : F \to E$ by sending $(x, w) \mapsto (x, \Phi(x)^{-1}(w))$, which is the inverse. ☐

Example 2.1.27. Let M be a smooth manifold of dimension m, $\varphi : U \to V$ a chart and $\tilde{\varphi} : TU \to V \times \mathbb{R}^m$ be the corresponding bundle chart. Then it follows from the lemma, that TU is isomorphic to $V \times \mathbb{R}^m$. This is verified by observing that $\partial_i(p)$ is a basis of T_pU for all $p \in U$. In this case $\tilde{\varphi}\,(\partial_i) = e_i$ giving us isomorphisms in each fibre.

Definition 2.1.28 (Sub-bundles)**.** Let $\pi : E \to X$ be a vector bundle. A **sub-bundle** is a subset of E which is a vector bundle in the induced topology and fibrewise vector space structure.

Definition 2.1.29 (Mono- and Epimorphisms)**.** A homomorphism $\varphi : E \to F$ is a **monomorphism** if all φ_x are monomorphisms (i.e. injective). Respectively, it is called an **epimorphism** if all φ_x are epimorphisms (i.e. surjective).

Corollary 2.1.30. If F is a sub-bundle of E, then the inclusion $\varphi : F \to E$ is a monomorphism.

Lemma 2.1.31. *If $\varphi : E \to F$ is a monomorphism, then $\varphi(E)$ is a sub-bundle and $\varphi : E \to \varphi(E)$ is an isomorphism.*

Proof. Once we know that $\varphi(E)$ is a sub-bundle, the map $\varphi : E \to \varphi(E)$ is a bijective homomorphism and fibrewise an isomorphism, hence an isomorphism by Lemma 2.1.26. So we need to check that $\varphi(E)$ is a sub-bundle, which is a local property. Thereby we can assume that E and F are product bundles $E = X \times V$ and $F = X \times W$. Let $x \in X$ and let $W_x \subseteq W$ be a subspace complementary to $\varphi_x(V)$. Then $G := X \times W_x \subseteq F$ and we define

$$\begin{aligned} \theta : E \oplus G &\to F \\ (a,b) &\mapsto \varphi(a) + i(b) \quad \text{where } i \text{ denotes the inclusion } G \hookrightarrow F\,. \end{aligned}$$

In the fibre over x, the map θ_x is an isomorphism and thereby there exists an open neighbourhood U of x such that $\theta|_U$ is an isomorphism. That is because $\mathrm{Iso}(V \oplus W_x, W)$ is open in $\mathrm{Hom}(V \oplus W_x, W)$. Since E is a sub-bundle of $E \oplus G$ we have that $\theta(E)|_U = \varphi(E)|_U$ is a sub-bundle of $F|_U$. □

Corollary 2.1.32. In the proof, we have shown more: locally a sub-bundle is a direct summand. Furthermore, we have shown that the set of points x where φ_x is a monomorphism, is open.

Definition 2.1.33 (Quotient Bundle)**.** If F is a sub-bundle of E we define the **quotient bundle** E/F as the union of the spaces E_x/F_x equipped with the quotient topology.

Corollary 2.1.34. The quotient bundle is a vector bundle, because locally F is a direct summand and thereby E/F is locally trivial.

Lemma 2.1.35. *Quotiend bundles of smooth subbundles are smooth.*

Proof. Let $F \subseteq E$ be a smooth subbundle of rank k, where E has rank n and M is the base manifold. We need to check that the transition functions consist of smooth maps into the structure group. First, we will show that there are smooth local frames of E that restricts to a smooth local frame of F:

Let $p \in M$. Since F is smooth there exists a local frame $(s_1, \dots, s_k)$ of F arround p. In particular, the s_i are smooth local sections into E arround p. Let $(t_1, \dots, t_k)$ be a smooth local frame of F near p. (without restrictions over the same domain as the sections s_i) Now we make use of steinitz exchange lemma to choos $n - k$ many t_i (Without restrictions $t_{k+1}, \dots, t_n$) such that

$$\big(s_1(p), \dots s_k(p), t_{k+1}(x), \dots, t_n(p)\big)$$

is a basis of E_p. Being a basis over a point $q \in M$ near p is equivalent to the invertability of the matrix where the columns are the sections at q, and since invertability (or rather nonzero determinant) is an open condition, there exists a neighbourhood, such that the family

$$\left(s_1, \ldots s_k, t_{k+1}, \ldots, t_n\right)$$

is a local frame, whose first k sections span F. By Definition 2.1.23 it induces a smooth trivialisation Φ_U of E with $\Phi_U\left(F|_U \right) = U \times \mathbb{R}^k \times \{0\}$, and fibrewise passing to the quotient $\mathbb{R}^n/(\mathbb{R}^k \times \{0\}) \cong \mathbb{R}^{n-k}$ yields a trivialisation $\bar{\Phi}_U$ of $(E/F)|_U$.

Now we can cover M by such adapted trivializations. Their transition functions $g_{i,j} : U_i \cap U_j \to \mathrm{GL}(n, \mathbb{R})$ are smooth and pointwise preserve $\mathbb{R}^k \times \{0\}$: for $v \in \mathbb{R}^k \times \{0\}$ we have $\Phi_i^{-1}(p, v) \in F_p$ since Φ_i is adapted, and hence $\Phi_j(\Phi_i^{-1}(p, v)) \in \{p\} \times \mathbb{R}^k \times \{0\}$ since Φ_j is adapted. Hence they have block form

$$g_{i,j} = \begin{pmatrix} A_{i,j} & B_{i,j} \\ 0 & C_{i,j} \end{pmatrix},$$

where all coordniate entries and hence all blocks smoothly depend on the basepoint. Finally, the transition functions of the quotient with respect to the trivialization $\bar{\Phi}_U$ of $(E/F)|_U$ consist of the smooth maps $p \mapsto C(p) = (C_{i,j})_{i,j}(p)$ into the structure group $\mathrm{GL}(n-k, \mathbb{R})$. □

Now we want to construct new bundles from old ones. There are many operations to construct new vector spaces such as the tensor product, direct sum, Hom-Spaces and many more. Instead of individually proving that those operations applied fibrewise yield new bundles, we subsume these operations within the following definition:

Definition 2.1.36 (Continuous Functor)**.** Let T be a covariant endofunctor in the category of finite-dimensional vector spaces. We call T a **continuous functor**, if for all V, W the map $T : \hom(V, W) \to \hom(T(V), T(W))$ is continuous. All vector spaces are given the topology induced by their isomorphism to real Euclidean space. If E is a vector bundle we define the set:

$$T(E) := \bigcup_{x \in X} T(E_x).$$

Now we want to equip this with a topology such that $T(E)$ is a bundle over the same base space as E.

Definition 2.1.37 (Constructing new bundles)**.** We start by assuming that $E = X \times V$. In that case we define the topology on $T(E)$ as the product topology $X \times T(V)$. Now suppose that $F = X \times W$ is another product bundle and $f : E \to F$ is a homomorphism. Let $\Phi : X \to \hom(V, W)$ be the corresponding map. Then $T\Phi : X \to \hom(T(V), T(W))$ is continuous by assumption. Thereby, $T(\varphi) : X \times T(V) \to X \times T(W)$ is continuous. If φ is an isomorphism, then $T(\varphi)$ is also an isomorphism. That follows because it is continuous and an isomorphism fibrewise.

Now assume that E is trivial. Then we can choose an isomorphism $\alpha : E \to X \times V$ and induce a topology on $T(E)$ by pulling it back via $T(\alpha) : T(E) \to X \times T(V)$. This topology does not depend on α by the above. For general vector bundles we do this construction locally.

If $f : X \to Y$ is continuous, there is a natural isomorphism

$$f^*T(E) \cong Tf^*(E)$$

This construction is carried out in detail by Atiyah in [Ati89, Chapter 1.2] and can be generalised for continuous functors with several variables both co- and contravariant. Hence this procedure gives us for vector bundles E and F over the same base space the new bundles:

(i) $E \oplus F$, their direct sum

(ii) $E \otimes F$, their tensor product

(iii) $\mathrm{Hom}(E, F)$

(iv) E^*, the dual bundle of E

(v) $\Lambda^i(E)$ the i-th exterior power

and by the natural isomorphism we have natural isomorphisms:

(i) $E \oplus F \cong F \oplus E$

(ii) $E \otimes F \cong F \otimes E$

(iii) $E \otimes (F \oplus F') \cong E \otimes F \oplus E \otimes F'$

(iv) $\mathrm{Hom}(E, F) \cong E^* \otimes F$

Corollary 2.1.38. Let $\pi_E : E \to X$ and $\pi_F : F \to X$ be two vector bundles over X. Then, the space $\Gamma(\mathrm{Hom}(E, F))$ can be identified with the space of bundle morphisms $\hom(E, F)$ since for any section $s : X \to \mathrm{Hom}(E, F)$ we can define the bundle homomorphism $\tilde{s} : E \to F$ by sending $v \mapsto s_{\pi_E(v)}(v)$. This is a homomorphism since $\pi_F(\tilde{s}(v)) = \pi_E(v)$.

Example 2.1.39 (Cotangent bundle)**.** Let M be a smooth manifold of dimension m. Then we can define the **cotangent bundle** T^*M to be the dual of the tangent bundle. A local frame of said bundle is given by the fibrewise dual of the frame (∂_i). Sections into the tangent bundle are called **vector fields** and sections into the cotangent bundle are called **one-forms**.

Definition 2.1.40. Let V be a linear manifold (i.e. a finite-dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$\begin{aligned} R_v : V &\to T_vV \\ x &\mapsto \left(f_v \mapsto \frac{\partial}{\partial t}\Big|_{t=0} f(v + tx) \right) \end{aligned}$$

Let $L : V \to V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$\mathrm{d}L_v \circ R_v = R_{L(v)} \circ L\,.$$

This is proved by a short calculation:

$$\Big(\mathrm{d}L_v\big(R_v(x)\big)\Big)(f_{L(v)}) = R_v(x)\,(f \circ L) = \frac{\partial}{\partial t}\bigg|_{t=0} f \circ L(v + tx) = \underbrace{\frac{\partial}{\partial t}\bigg|_{t=0} f\big(L(v) + tL(x)\big)}_{=R_{L(v)} \circ L(x)}$$

Now let $v, v' \in V$. Then we have the translation

$$tr_{v,v'} : V \to V; \quad x \mapsto (x + (v' - v))$$

This allows us to relate R_v and $R_{v'}$ as follows:

$$\big(\mathrm{d}tr_{v,v'}|_v\,(R_v(x))\big)\,(f_{v'}) = \frac{\partial}{\partial t}\bigg|_{t=0} (f \circ tr_{v,v'}(v + tx)) = \frac{\partial}{\partial t}\bigg|_{t=0} \big(f(v' + tx)\big) = R_{v'}(x)(f_{v'})\,.$$

Hence:

$$R_{v'} = \mathrm{d}tr_{v,v'} \circ R_v\,.$$

The **linear identification** is the map Q given on the whole tangent bundle:

$$Q : TV \to V; \quad \xi_p \mapsto R_p^{-1}(\xi_p)\,,$$

and satisfies for any linear $L : V \to V$

$$Q \circ \mathrm{d}L = L \circ Q\,.$$

Corollary 2.1.41. Let $f : M \to \mathbb{R}$ be smooth. Then we can interpret $\mathrm{d}f$ as a one-form. To be precise, let $Q : T\mathbb{R} \to \mathbb{R}$ denote the linear identification with respect to the identity on $\mathbb{R}$. For $v \in \Gamma(TM)$ we have

$$(Q \circ \mathrm{d}f)(v) = v(f)$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We retain this notation so that vector fields can take functions as an input.

Proof. Let $\varphi : U \to V$ be a chart of M. We prove this by showing it for the local sections ∂_i and the general case follows by $C(U)$-linearity, since those sections form a local frame. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f_p(\partial_i(p))(g_p) = \partial_i((g \circ f)_p) = \frac{\partial}{\partial x_i}\bigg|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial_{x\, f_p(p)}}(g_p) \cdot \partial_i(p)(f_p)$$

Hence, applying Q to the calculation yields $(Q \circ \mathrm{d}f)(v) = v(f)$. $\square$

Definition 2.1.42 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(T^*M \otimes T^*M)$ into the tensor product of the dual tangent bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non-degenerate**, meaning if for a $v_p \in T_pM$ we have $g_p(v_p, w_p) = 0$ for all $w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ for all $v_p, w_p \in T_pM$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0$ for all v_p with equality only for $v_p = 0$.

We call a tuple (M, g) a **Riemannian manifold**.

Remark 2.1.43. Equipping a smooth manifold M of dimension m with a Riemannian metric is equivalent to reducing the structure group of the tangent bundle from $\mathrm{GL}(m)$ to $\mathrm{O}(m)$. To be precise: A Riemannian metric g lets us canonically choose an atlas with structure group $\mathrm{O}(m)$ and an atlas with such a structure group lets us canonically choose a Riemannian metric.

Proof. We start by assuming that we have an atlas with structure group $\mathrm{O}(m)$. In Definition 2.1.23 we saw that a covering of local trivialisations is equivalent to a covering of smooth frames. Now in a single trivialisation

$$\Phi : \pi^{-1}(U) =: TU \to U \times \mathbb{R}^m$$

we call the projection to the $\mathbb{R}^m$-part (the second factor) π_2 and define

$$\begin{aligned} g : U &\to T^*U \otimes T^*U \\ p &\mapsto \begin{array}{ll} g_p^\Phi : T_pU \times T_pU & \to \mathbb{R} \\ (\nu_p, \xi_p) & \mapsto g_p^\Phi(\nu_p, \xi_p) := \pi_2\left(\Phi(\nu_p)\right) \cdot \pi_2\left(\Phi(\xi_p)\right) , \end{array} \end{aligned}$$

where $\cdot$ denotes the standard scalar product. This gives us locally a Riemannian metric. If we do this for all trivialisations we have to make sure, that the result is well defined. So assume that two trivialisations Φ and Ψ overlap in V. The structure group is $\mathrm{O}(m)$, $\Phi = R\Psi$ for some function

$$R : V \to \mathrm{O}(m) .$$

Let p be in said overlapping and (ν_p, ξ_p) be from $T_pM \times T_pM$. Then since R is orthogonal we can calculate the independence from the trivialisation:

$$g_p^\Phi(\nu_p, \xi_p) = \pi_2\left(R\Psi(\nu_p)\right) \cdot \pi_2\left(R\Psi(\xi_p)\right) = \pi_2\left(\Psi(\nu_p)\right) \cdot \pi_2\left(\Psi(\xi_p)\right) = g_p^\Psi(\nu_p, \xi_p) .$$

To get an orthonormal atlas from a smooth one given a Riemannian metric, we start by building a covering of local frames from the trivialisations. We then apply Gram–Schmidt to the local frames to get orthonormal frames. Now a covering of orthonormal frames gives a covering of trivialisations, where the structure group consists of basis changes between orthonormal bases and hence is $\mathrm{O}(m)$. □

Definition 2.1.44 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function and $Q : T\mathbb{R} \to \mathbb{R}$ the linear identification. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vector field V:

$$Q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Notice how the non-degeneracy of g ensures existence and uniqueness.

Definition 2.1.45. Let (M, g) be a Riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be:

$$g_{ij}(p) = g_p(\partial_i(p), \partial_j(p))\,.$$

Then if two vector fields v, w over U are given with $v = \sum v_i \partial_i$ and $w = \sum_i w_i \partial_i$ we can calculate the metric locally:

$$g(v, w) = \sum_{i,j} g_{ij} v_i w_j \ : U \to \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure as can be seen by the construction via Cramer's rule) to get smooth functions

$$\begin{aligned} g^{ij} &: U \to \mathbb{R} \\ &p \mapsto g^{ij}(p) = \left((g_{kl}(p))^{-1}_{kl} \right)_{ij} \end{aligned}$$

Remark 2.1.46. Due to the rare appearance of big summations with vectors and co-vectors in this work, we omit the Einstein summation convention and assign all coordinate entries lower indices.

Lemma 2.1.47 (Local Description of the Gradient)**.** *Let (M, g) be a smooth Riemannian manifold of dimension m and $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\mathrm{grad}(f) = \sum_{i,j} g^{ij} \left(\partial_i(f)\right) \cdot \partial_j\,.$$

Proof. Assume that $v, w \in \Gamma(TU)$ such that $v = \sum_i v_i \partial_i$ and $w = \sum_i w_i \partial_i$ (Here v_i and w_i are smooth functions $U \to \mathbb{R}$). Then $g(v, w) = \sum_{i,j} g_{ij} v_i w_j$. Suppose that $\mathrm{grad}(f) = \sum_j G^j \partial_j$ and Q is the linear identification $T\mathbb{R} \to \mathbb{R}$ in each tangent space. Then by definition of the gradient and corollary 2.1.41 we have:

$$\partial_j(f) = Q \circ \mathrm{d}f(\partial_j) = g(\mathrm{grad}(f), \partial_j) = g(\sum_i G^i \partial_i, \partial_j) = \sum_i g_{ij} G^i$$

Reading this as a pointwise matrix product:

$$(G^1, \cdots G^m)(g_{ij}) = (\partial_1(f), \cdots, \partial_m(f))$$
$$\text{which gives } (G^1, \cdots G^m) = (\partial_1(f), \cdots, \partial_m(f))\,(g^{ij})\,.$$

But then we can conclude that $G^j = \sum_i g^{ij}\partial_i(f)$.

□

Definition 2.1.48 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M (sometimes also called a **flow on** M) is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cap \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:
$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$
3. For all p we require $\varphi(0,p) = p$.

We will write $\varphi_t(p) := \varphi(t,p)$ to keep the notation manageable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 2.1.49. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff}(M), \quad t \mapsto \varphi_t$$

is a homomorphism. To verify well-definedness notice how for all $t \in \mathbb{R}$, φ_{-t} is an inverse of φ_t. The map ρ is called a **one-parameter group of diffeomorphisms**.

Definition 2.1.50 (Vector Field of a Dynamical System)**.** Let $\varphi : \Omega \to M$ be a dynamical system on a smooth manifold M. For a chart $\psi : U \to V$ of M we have the chart $\mathrm{id} \times \psi$ of Ω. For this chart we denote the associated local frame by $(\partial_t, \partial_1, \ldots, \partial_m)$. Using this we define the mapping

$$\begin{aligned} M &\to TM \\ p &\mapsto X(p) := \mathrm{d}\varphi(\partial_t(0,p)) \end{aligned}$$

and call it the **associated vector field**. This is in fact a smooth vector field.

Definition 2.1.51 (Dynamical System of a Vector Field)**.** Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M. Then we call a dynamical system φ on M the **associated dynamical system** to X, if for all $t \in \mathbb{R}$ and $x \in M$

$$\mathrm{d}\varphi\left(\partial_t(t,x)\right) = X(\varphi_t(x))\,.$$

Remark 2.1.52. Regarding the left-hand side of the definition $\mathrm{d}\varphi\left(\partial_t(t,x)\right)$ we could also differentiate the map

$$t \mapsto \varphi_t(x)\,,$$

for fixed x at t and evaluate this at the constant vector field $t \mapsto R_t(1) \in T_t\mathbb{R}$ from definition 2.1.40.

These two constructions are mutually inverse, giving the following theorem:

Theorem 2.1.53. *Let M be a smooth manifold. Then there is a one-to-one correspondence between global dynamical systems and vector fields given by the constructions above.*

3 Complex K Theory

3.1 Mathematical Goals

Theorem 3.1.1 (The Theorems and Mathematical Structures). • *Definition of The K-Functor*

- *Cohomology structures*
  - *Additivity (pointed sum)*
  - *Long exact sequences*
  - *Excision*

3.2 Classification of Vector Bundles

The first goal is to classify all vector bundles over sufficiently nice spaces. The latter refers to spaces admitting partitions of unity for which Tietze's extension theorem holds. For simplicity we just consider compact spaces. Along the way we will discover that the set of isomorphism classes of vector bundles is a homotopy invariant.

Before we go back to the analytical reals, we will further develop the theory of vector bundles:

Definition 3.2.1. For a topological space X we define Vect_n to be the set of isomorphism classes of complex vector bundles of rank n and $\mathrm{Vect}(X)$ to be the set of isomorphism classes of complex vector bundles of any rank. By declaring $\mathrm{Vect}(f) = f^*$ to be the pullback we get a contravariant functor Vect from **CGWH** to **Set**. We can equip the set $\mathrm{Vect}(X)$ with the direct sum and tensor product operation. It is then easy to see that these operations descend to isomorphism classes and hence we get an abelian semiring $(\mathrm{Vect}(X), \oplus, \otimes)$ (abelian semigroup with associative and distributive multiplication). In fact we even get a contravariant functor from **CGWH** to the category of abelian semirings.

Lemma 3.2.2. *Let X be a compact space, $Y \subseteq X$ a closed subspace and E a vector bundle over X. Then any section $s : Y \to E|_Y$ can be extended to a global section of X.*

Proof. For a detailed proof see Chapter 1.4 of [Ati89]. The general argument however is easy to follow: By Tietze's extension theorem we can extend the section to around every $x \in X$. By compactness we can pick finitely many of those extensions. Now using a partition of unity with respect to their domains we can build the sum of all the extensions to get a well defined global one. □

Corollary 3.2.3. Let Y be a closed subspace of a compact space X, and let E, F be bundles over X. If $f : E|_Y \to F|_Y$ is an isomorphism, then there is an open neighbourhood $U \supset Y$ and an extension $f : E|_U \to F|_U$ which is an isomorphism.

Proof. By corollary 2.1.38 we can view f as a section into the bundle $\mathrm{Hom}(E|_Y, F|_Y)$, which can be extended to a global one in $\mathrm{Hom}(E,F)$. Let U be the open set for which the section maps to isomorphisms. Since U contains Y this yields the required extension. □

Theorem 3.2.4 (Homotopy Invariance of the Pullback)**.** *Let X, Y be two compact spaces, $I = [0,1]$, and*

$$f : Y \times I \to X$$

be a homotopy. For a fixed $t \in I$, we define the function $f_t : Y \to X$, $f_t(y) = f(t,y)$. Let $\pi : E \to X$ be a vector bundle. Then

$$f_0^*(E) \cong f_1^*(E)\,.$$

Proof. Consider the following two vector bundles over $Y \times I$:

$$f^*E \quad \text{and} \quad \pi_1^* f_t^* E\,,$$

where $\pi_1 : Y \times I \to Y$ denotes the projection and $t \in I$. Since the restriction is by definition the pullback along the inclusion $i : Y \times \{t\} \to Y \times I$. The equality of the two maps:

$$f \circ i = f_t \circ \pi_1 \circ i : Y \times \{t\} \to X$$

together with the functoriality gives us an isomorphism

$$\Phi : f^*E|_{Y\times\{t\}} \to \pi_1^* f_t^* E|_{Y\times\{t\}}\,.$$

By Corollary 3.2.3 we can extend the isomorphism for some open neighbourhood of $Y \times \{t\}$ and by the product topology we can deduce that there exists a δ_t such that U contains the thickened line $Y \times (t-\delta_t, t+\delta_t)$. The isomorphism class of $f^*E|_{Y\times\{t\}}$ as a bundle over Y is locally constant in t: Let $0 < \varepsilon < \delta$ and $r \in I$. Then define the inclusions

$$\begin{aligned} i_t : Y &\to Y \times \{t\} \subseteq Y \times I \\ y &\mapsto (y,t) \end{aligned} \qquad \text{and} \qquad \begin{aligned} i_{t+\varepsilon} : Y &\to Y \times (t-\delta_t, t+\delta_t) \\ y &\mapsto (y, t+\varepsilon) \end{aligned}$$

Then we have the identities $f_t = f \circ i_t$ and $\pi_1 \circ i_{t+\varepsilon} = \mathrm{id}_Y$ and we can calculate:

$$f_{t+\varepsilon}^* E = i_{t+\varepsilon}^* f^* E = i_{t+\varepsilon}^* \pi_1^* f_t^* E = (f_t \circ \pi_1 \circ i_{t+\varepsilon})^* E = f_t^* E\,.$$

Finally, since I is connected, any locally constant map from I is globally constant and therefore $f_0^* E \cong f_1^* E$. □

Corollary 3.2.5.

1. If $f : X \to Y$ is a homotopy equivalence, $f^* : (Y) \to (X)$ is a semiring isomorphism.
2. If X is contractible, every bundle over X is trivial and $(X) \cong \mathbb{N}_0$

Definition 3.2.6 (Trivialisation of a Vector Bundle)**.** Let Y be a closed subspace of X, E a vector bundle over X and $\alpha : E|_Y \to Y \times V$ be an isomorphism. Then we call α **trivialisation of E over Y**.

Lemma 3.2.7. *A trivialisation α of a bundle E over $Y \subseteq X$ defines a bundle E/α over X/Y. The isomorphism class of E/α depends only on the homotopy class of α.*

Proof. We define the bundle E/α as follows: Let $\pi_2 : Y \times V \to V$ denote the projection and define an equivalence relation on $E|_Y$ by

$$e \sim e' \;\Leftrightarrow \pi_2\alpha(e) = \pi_2\alpha(e').$$

We extend this relation by the identity on $E|_{X \setminus Y}$ and we define E/α to denote the quotient space of E defined by that relation. To see why this is a bundle we need to check the local triviality only around the base point Y/Y of X/Y. This follows from the fact that by Corollary 3.2.3 we can extend α to an isomorphism $\tilde{\alpha} : E|_U \to U \times V$. Then $\tilde{\alpha}$ induces an isomorphism

$$E/\alpha\big|_{U/Y} \cong (U/Y) \times V$$

proving the local triviality. Now for the homotopy invariance of said bundle suppose that α_0 and α_1 are homotopic trivialisations of E over Y. This means that we have a trivialisation β of $E \times I$ over $Y \times I \subset X \times I$ that induces the α_i at the endpoints. Let $f : (X/Y) \times I \to (X \times I)/(Y \times I)$ be the natural map. Then $f^*((E \times I)/\beta)$ is a bundle on $(X/Y) \times I$ whose restriction to $(X/Y) \times \{i\}$ is E/α_i $(i = 0, 1)$. Hence by the homotopy invariance of the pullback (Theorem 3.2.4) we have

$$E/\alpha_0 \cong E/\alpha_1 .$$

$\square$

Corollary 3.2.8. The trivialisation acts as a right inverse of the pullback along the collapse: If we call $f : X \to X/Y$ the collapse, and fE can be trivialised over Y via α, then

$$f^*(E/\alpha) \cong E .$$

Proof. Intuitively this is clear, as the pullback along the collapse builds a bundle, that is trivial over Y and agrees with E outside of Y. To prove the statement we will construct a bundle map, that is an isomorphism in each fibre: Let $q : E \to E/\alpha$ denote the quotient map from the construction and let $\pi_E : E \to X$, $\pi_{E/\alpha} : E/\alpha \to X/Y$ be the bundle projections. By construction of E/α we have $\pi_{E/\alpha} \circ q = f \circ \pi_E$, so

$$\Phi : E \to f^*\big(E/\alpha\big), \qquad \Phi(e) = \big(\pi_E(e), q(e)\big)$$

is a well-defined bundle morphism. We show Φ is a fibrewise linear isomorphism, and hence a bundle isomorphism by Lemma 2.1.26.

Case $x \notin Y$: Here the equivalence relation defining E/α restricts to the identity on E_x, so

$$q_x : E_x \xrightarrow{\cong} (E/\alpha)_{f(x)} = (E/\alpha)_x$$

is a linear isomorphism.

Case $x \in Y$: By definition of the equivalence relation, two points $e, e' \in E_x$ are identified with $\pi_2\alpha(e) = \pi_2\alpha(e') \in V$, and all fibres E_x $(x \in Y)$ are glued to the single fibre $(E/\alpha)_{y_0} \cong V$ over the basepoint $y_0 = Y/Y$. Hence

$$q_x = \pi_2 \circ \alpha_x : E_x \to V = (E/\alpha)_{y_0} ,$$

where $\alpha_x : E_x \to V$ is the (linear) restriction of α to the fibre over x. Since α_x is a linear isomorphism, so is q_x.

Thus Φ is a bundle morphism which is a linear isomorphism on every fibre. □

Lemma 3.2.9. *Let $Y \subseteq X$ be a closed contractible subspace of a compact space X. Then*

$$f : X \to X/Y \quad \textit{induces a semiring isomorphism} \quad f^* : \mathrm{Vect}(X/Y) \to \mathrm{Vect}(X) .$$

Proof. Since Y is contractible, $E|_Y$ is trivial and thereby a trivialisation $\alpha : E|_Y \to Y \times V$ exists. Two such trivialisations only differ by the automorphism of $Y \times V$ which in turn corresponds to a map $Y \to \mathrm{GL}(V) \cong \mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$. Now since Y is contractible those two maps factor (up to homotopy) over any one point $y \in Y$ and are thereby homotopically equivalent to the constant map. Finally since $\mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$ is connected all constant maps are homotopically equivalent. Thereby α is unique up to homotopy. By Lemma 3.2.7 the isomorphism class of E/α only depends on the homotopy class of α we have a well defined map

$$\mathrm{Vect}(X) \to \mathrm{Vect}(X/Y) ,$$

which in turn is an inverse of f^* concluding the proof. □

Definition 3.2.10 (Gluing and Clutching of Vector Bundles)**.** Let

$$X = X_1 \cup X_2 , \quad A = X_1 \cap X_2 ,$$

and all be compact spaces. Assume that E_i is a vector bundle over X_i and $\varphi : E_1|_A \to E_2|_A$ is an isomorphism. Then we define a vector bundle $E_1 \cap_\varphi E_2$ on X as follows: We topologise the space $E_1 \cup_\varphi E_2$ as the quotient of $E_1 + E_2/\sim$, where $\sim$ is the identification of points with their image under φ. Identifying X with $X_1 + X_2$ in the quotient space we get the natural projection $p : E_1 \cap_\varphi E_2 \to X$ and the fibres have a natural vector space structure. It remains to show that this is locally trivial. This is obvious outside of A so let $a \in A$ and V_1 be a closed neighbourhood of a in X_1 such that $E_1|_{V_1}$ is trivial giving us the isomorphisms

$$\theta_1 : E_1|_{V_1} \to V_1 \times \mathbb{C}^n$$
$$\theta_1^A : E_1|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

Finally, define

$$\theta_2^A : E_2|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

by composition $\theta_1^A \circ \varphi^{-1}$. Let V_2 be a neighbourhood of a in X_2 such that

$$\theta_2 : E_2|_{V_2} \to V_2 \times \mathbb{C}^n$$

is an extension of θ_2^A. This extension exists again by extending the section into the homomorphism bundle corresponding to the isomorphism θ_2^A. Finally θ_1 and θ_2 define a well defined isomorphism

$$\theta_1 \cap_\varphi \theta_2 : E_1 \cap_\varphi E_2 \to (V_1 \cap V_2) \times \mathbb{C}^n \, .$$

This proves the local triviality.

We call such a triple (E_1, E_2, φ) a **clutching datum** and φ the **transition function**.

Corollary 3.2.11 (Elementary Properties of the Gluing and Clutching Construction)**.** Elementary properties of the gluing and clutching construction are:

1. If E is a bundle over X, then the identity defines an isomorphism and
$$E_1 \cup_{\mathrm{id}_A} E_2 \cong E.$$
2. If $\beta_i : E_i \to E_i'$ are isomorphisms on X_i and $\varphi' \beta_1 = \beta_2 \varphi$, then
$$E_1 \cup_\varphi E_2 \cong E_1' \cup_{\varphi'} E_2' \, .$$
3. If (E_i, φ) and (E_i', φ') are two “clutching data” on X_i, then
$$\begin{aligned}(E_1 \cup_\varphi E_2) \oplus (E_1' \cup_{\varphi'} E_2') &\cong E_1 \oplus E_1' \bigcup\nolimits_{\varphi \oplus \varphi'} E_2 \oplus E_2' \, , \\ (E_1 \cup_\varphi E_2) \otimes (E_1' \cup_{\varphi'} E_2') &\cong E_1 \otimes E_1' \bigcup\nolimits_{\varphi \otimes \varphi'} E_2 \otimes E_2' \, , \\ (E_1 \cup_\varphi E_2)^* &\cong E_1^* \bigcup\nolimits_{(\varphi^*)^{-1}} E_2^* \, .\end{aligned}$$
$(-)^*$ denotes the dual here.

Lemma 3.2.12. *The isomorphism class of $E_1 \cup_\varphi E_2$ depends only on the homotopy class of the isomorphism $\varphi : E_1|_A \to E_2|_A$.*

Proof. A homotopy of isomorphisms is a map

$$\Phi : E_1|_A \times I \to E_2|_A \, ,$$

such that for all t the map

$$\Phi_t : E_1|_A \to E_2|_A \quad \Phi_t(\xi) := \Phi(\xi, t)$$

is a bundle isomorphism. We can encode this Φ into a bundle isomorphism as follows: Let $\pi : X \times I \to X$ be the projection. Then elements $\xi \in \pi^* E_1$ are by definition 2.1.14 of the form $\xi_{x,t} = ((x,t), \xi_x) \in (\pi^* E_1)_{x,t}$. With this notation define the bundle isomorphism:

$$\Psi : \pi^*(E_1)|_{A\times I} \to \pi^*(E_2)|_{A\times I} \ , \quad \Psi(\xi_{x,t}) := ((x,t), \Phi_t(\xi_x))$$

This is a bundle isomorphism, since Φ_t is one. Now denote the inclusion

$$i_t : X \to X \times I \ , \quad i_t(x) := (x,t) \ ,$$

and notice that the pullback along i_t is the same as restricting to $X \times \{t\}$.

With those definitions it is clear by the construction of the gluing, that

$$E_1 \cup_{\Phi_t} E_2 \cong i_t^* \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right) . \tag{1}$$

The isomorphism can be seen by part (2) of Corollary 3.2.11 as follows: First notice the isomorphism $\pi^*(E_i)|_{A\times\{t\}} \cong E_i$ for $i = 1,2$ and any t. By construction of the gluing this leads to the isomorphism

$$E_1 \cup_{\Phi_t} E_2 \cong \pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t, \mathrm{id})} \pi^*(E_2)|_{A\times\{t\}} \ .$$

Now again by construction we notice, that gluing restricted spaces is the same as gluing the whole space and then restricting. Furthermore the map (Φ_t, id) is the restriction of Ψ to $X \times \{t\}$. Therefore:

$$\pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t, \mathrm{id})} \pi^*(E_2)|_{A\times\{t\}} \cong \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right)|_{X\times\{t\}}$$

Finally, pulling back via i_t is the same as restricting $X \times i$ to $X \times \{t\}$ and identifying X with $X \times \{t\}$. This concludes the chain of arguments:

$$\left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right)|_{X\times\{t\}} \cong i_t^* \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right) .$$

Since the pullback is homotopy invariant we can conclude by the isomorphism (1):

$$E_1 \cup_{\Phi_0} E_2 \cong E_1 \cup_{\Phi_1} E_2 \, .$$

□

Definition 3.2.13. Let X, Y be topological spaces. By $[X, Y]$ we denote the set of homotopy classes of maps from X to Y.

Lemma 3.2.14. *Let $X = X_1 \cup X_2$ with X_1 and X_2 closed contractible spaces. Then there is a canonical bijection*

$$\mathrm{Vect}_n(X) \to [A, \mathrm{GL}_n(\mathbb{C})] \, .$$

Proof. Let $[E]$ be any vector bundle over X of rank n. Then E can be represented by a vector bundle obtained by gluing that is trivial on both X_i: Since X_i is contractible we can choose two trivialisations

$$\alpha_1 : E|_{X_1} \to X_1 \times \mathbb{C}^n, \quad \text{and} \quad \alpha_2 : E|_{X_2} \to X_2 \times \mathbb{C}^n \,.$$

With part (2) of Corollary 3.2.11 we can conclude:

$$E \cong E|_{X_1} \cup_{\mathrm{id}_A} E|_{X_2} \cong X \times \mathbb{C}^n \cup_{\alpha_2 \circ \alpha_1^{-1}|_A} X \times \mathbb{C}^n$$

Since a map between the two trivial bundles $A \times \mathbb{C}^n \to A \times \mathbb{C}^n$ can be identified with a map $A \to \mathrm{GL}_n(\mathbb{C})$ we get an assignment from bundles over X together with the trivialisations to maps $A \to \mathrm{GL}_n(\mathbb{C})$. The next step is to show that two isomorphic vector bundles give homotopic maps: Assume that we are given two vector bundles E, F over X that are isomorphic via $T : E \to F$ and without loss of generality we assume that they are given as the gluing of two trivial bundles:

$$E : X_1 \times \mathbb{C}^n \cup_\varphi X_2 \times \mathbb{C}^n, \quad \text{and} \quad F : X_1 \times \mathbb{C}^n \cup_\psi X_2 \times \mathbb{C}^n \,.$$

We now want to define a homotopy between ψ and φ. For this we start by splitting the isomorphism T into its two parts over X_1 and X_2:

$$T|_{X_i} : X_i \times \mathbb{C}^n \to X_i \times \mathbb{C}^n \,.$$

These maps are given by maps

$$T_i : X_i \to \mathrm{GL}_n(\mathbb{C}) \,,$$

where the gluing compatibility of the $T|_{X_i}$ yield for any $a \in A$:

$$\psi(a) T_1(a) = T_2(a) \varphi(a) \,, \qquad \text{equivalently} \qquad \psi(a) = T_2(a)\varphi(a) T_1^{-1}(a) \,.$$

By the contractibility of X_i we have for any fixed $x_i \in X_i$ a homotopy $h_t^i : X_i \to X_i$ such that $h_0^i = c_{x_i}$ is the collapsing map whilst $h_1^i = \mathrm{id}_{X_i}$. Furthermore, since $\mathrm{GL}_n(\mathbb{C})$ is connected we have a path $\gamma^i : I \to \mathrm{GL}_n(\mathbb{C})$ that connects $T_i(x_i) = \gamma^i(1)$ to $\mathbb{1}_n = \gamma^i(0)$. With this we can define the homotopies

$$\begin{aligned} T_{i,t} : X_i &\to \mathrm{GL}_n(\mathbb{C}) \\ x &\mapsto \begin{cases} T_i \circ h_{2t}^i(x) & \text{if } t \le \frac{1}{2} \,, \\ \gamma^i(2(t - \frac{1}{2})) & \text{if } t > \frac{1}{2} \,, \end{cases} \end{aligned}$$

that satisfy $T_{i,1} = T_i$ and $T_{i,0}(x) = \mathbb{1}_n$ for all x. Finally we define the homotopy $H_t : A \to \mathrm{GL}_n(\mathbb{C})$ by

$$H_t(a) := T_{2,t}(a) \varphi(a) T_{1,t}^{-1}(a)$$

to get a homotopy from $\varphi = H_0$ to $\psi = H_1$. Hence, we get a well defined map from isomorphism classes of bundles over X of rank n to the homotopy classes of maps from A to $\mathrm{GL}_n(\mathbb{C})$ that is the inverse of the gluing construction and hence is bijective. □

Example 3.2.15. We now want to analyse the tautological line bundle l over $\mathbb{CP}^1$. For this we cut $\mathbb{CP}^1$ into the two pieces which are contractible as they come with homeomorphisms into the unit disc in $\mathbb{C}$:

$$H^+ := \{[x:y] \in \mathbb{CP}^1 \,|\, |x| \geq |y|\} \to D \subset \mathbb{C}\,, \quad [x:y] \mapsto \frac{y}{x}$$
$$H^- := \{[x:y] \in \mathbb{CP}^1 \,|\, |x| \leq |y|\} \to D \subset \mathbb{C}\,, \quad [x:y] \mapsto \frac{x}{y}$$

Those maps give us unique representations of elements in H^+ and H^-. We now want to evaluate the bijection $\mathrm{Vect}_1(\mathbb{CP}^1) \to [H^+ \cap H^-, \mathrm{GL}_1(\mathbb{C})]$ at $[l]$. The image of the intersection $H^+ \cap H^-$ under the two maps is the boundary of the disc and hence a circle. Furthermore, $\mathrm{GL}_1(\mathbb{C}) \cong C^*$. Hence, we know that the image is isomorphic to $[S^1, \mathbb{C}^*] \cong \pi_1(S^1) \cong \mathbb{Z}$. This tells us, that there must be nontrivial vector bundles on $\mathbb{CP}^1$. By the proof of 3.2.14 we know that $[l]$ gets mapped to the function gluing together trivialisations over H^+ and H^- along their intersection. To define the trivialisations we need to define local frames over H^+ and H^-. Those frames are given by a single section defined as:

$$s_+ : H^+ \to l\,, \quad [x:y] \mapsto (1, \frac{y}{x})\,,$$
$$s_- : H^- \to l\,, \quad [x:y] \mapsto (\frac{x}{y}, 1)\,.$$

With those we define the trivialisations:

$$t_+ : \, l|_{H^+} \to H^+ \times \mathbb{C}\,, \quad ([x:y], \lambda \cdot s_+[x:y]) \mapsto ([x:y], \lambda)$$
$$t_- : \, l|_{H^-} \to H^- \times \mathbb{C}\,, \quad ([x:y], \mu \cdot s_-[x:y]) \mapsto ([x:y], \mu)$$

in each fibre over the intersection we have

$$\lambda \cdot s_+([x:y]) = \lambda(1, \frac{y}{x}) = \lambda \cdot \frac{y}{x}(\frac{x}{y}, 1) = \lambda \cdot \frac{y}{x} s_-([x:y])$$

Hence in each fibre over $[x:y] \in H^+ \cap H^-$ the gluing function can be calculated:

$$\lambda \mapsto t_-|_{[x:y]} \circ t_+|_{[x:y]}^{-1}(\lambda) = t_-|_{[x:y]}(\lambda \cdot s_+([x:y])) = t_-|_{[x:y]}(\lambda \cdot \frac{y}{x} \cdot s_-([x:y])) = \lambda \cdot \frac{y}{x}$$

Hence, as a map from $H^+ \cap H^- \to GL_1(\mathbb{C}) \equiv \mathbb{C}^*$ this map reads:

$$H^+ \cap H^- \to \mathbb{C}^* \quad [x:y] \mapsto \frac{y}{x}$$

viewing this as an element of $\pi(S^1)$ by identifying the intersection with a circle via the homeomorphism from $H^+ \to D \subseteq \mathbb{C}$ we notice that the gluing map corresponds to the identity in $\pi_1(S^1)$. If the bundle l would be trivial, the gluing function would be homotopically equivalent to the constant one-function $H^+ \cap H^- \to \mathbb{C}^*$ that maps all $[x:y] \mapsto 1$. This cannot be, because interpreted as elements in $\pi(S)$ they are different.

Definition 3.2.16 (Metrics on Bundles)**.** A **metric** on a complex bundle E is a section $h : X \to \mathrm{Herm}(E)$ such that $h(x)$ is positive definite for all x. $\mathrm{Herm}(E)$ denotes the bundle of hermitian forms (conjugate-symmetric sesquilinear) in each fibre. A bundle together with a metric is a **Hermitian bundle**.

Definition 3.2.17 (Projection Operator)**.** A **projection operator** $P : E \to E$ is a bundle homomorphism in E such that $P^2 = P$.

Lemma 3.2.18. *Any projection operator $P : E \to E$ determines a direct sum decomposition $E = PE \oplus (1 - P)E$.*

Proof. Observe that fibrewise $1 - P$ maps each ξ to the kernel of P, and hence fibrewise $P(E)$ and $(1-P)(E)$ are disjoint. Furthermore $\mathrm{im}(P)+\mathrm{im}(1-P) = E$, because $v = P(v)+(v - P(v))$. Therefore in each fibre we have the direct sum $E_x = P(E_x) \oplus (1 - P)(E_x)$. Now the question is, if that gives a global splitting, meaning that $P(E)$ and $(1 - P)(E)$ are continuous as subbundles of E. For this to be true, we need local triviality, which is clear as long as the rank of P (and $(1 - P)$) is locally constant. In each fibre we know that $\mathrm{rank}(P_x) + \mathrm{rank}(1 - P) = \dim(E_x)$. Furthermore, $\mathrm{rank}(P_x)$ and $\mathrm{rank}(1 - P)$ are locally constant. To see this first notice that they are lower semi-continuous, meaning that for y from a small enough neighbourhood of x $\mathrm{rank}(P_x) \leq \mathrm{rank}(P_y)$. To see this, pick a $\mathrm{rank}(P_x)_x \times \mathrm{rank}(P_x)_x$ minor with nonzero determinant. The latter is an open condition and hence stable under small perturbation. But now we have the sum of two lower semi-continuous functions that give something locally constant and hence, each summand is locally constant. Finally the above shows that $\ker(P) = (1-P)(E)$ we have the decomposition. ∎

Lemma 3.2.19 (Metric Induced Complement)**.** *Suppose that F is a sub-bundle of E. Then a metric provides a canonical complementary sub-bundle.*

Proof. Let h denote the metric. For each x we have the orthonormal projection $P_x : E_x \to F_x$ inducing a map $P : E \to F$. Let $(f_1, \dots f_n)$ be an orthonormal frame of F. Then fibrewise with $\xi \in E_x$ the orthogonal projection is given by

$$P_x(\xi) = \sum_i h_x(\xi, f_i(x)) f_i(x)\,.$$

And since h is continuous the whole map P is. Furthermore $P^2 = P$ and hence we get a canonical sum decomposition. ∎

Definition 3.2.20. We call a sequence of bundle homomorphisms

$$\cdots \longrightarrow E \longrightarrow F \longrightarrow \cdots$$

exact, if it is fibrewise exact.

Lemma 3.2.21 (Splitting Principle)**.** *Every short exact sequence of bundles over the same base space splits.*

Proof. Let X be the base space and assume that the sequence is given as

$$0 \longrightarrow E' \xrightarrow{\varphi'} E \xrightarrow{\varphi''} E'' \longrightarrow 0$$

Now it is well known that any short exact sequence of vector spaces splits. We just need to ensure that this splitting is continuous. For this we give E a metric. Now since φ' is injective, it is a monomorphism and $\varphi'(E')$ is a subbundle by Lemma 2.1.31. By Lemma 3.2.19 we can split $E \cong \varphi'(E') \oplus L$. However, $\varphi''|_L$ is fibrewise (and hence as a bundle map) an isomorphism. This can be seen as φ'' being surjective and trivial on $\varphi'(E')$ hence the restriction is still surjective. The injectivity is also clear as the kernel by the homomorphism theorem of linear maps. But then we clearly have $E = E' \oplus E''$ and the sequence is isomorphic as a short exact sequence to the splitting one:

$$0 \longrightarrow E' \xrightarrow{i_1} E' \oplus E'' \xrightarrow{\pi_2} E'' \longrightarrow 0$$

□

Definition 3.2.22. A subspace $V \subseteq \Gamma(E)$, where $\Gamma(E)$ is viewed as a $\mathbb{C}$-vector space is called **ample**, if

$$\varphi : X \times V \to E, \quad \varphi(x,s) := s(x)$$

is a surjection.

Corollary 3.2.23. If the subspace V is ample and $\dim(V) = \operatorname{rank}(E)$, then E is trivialisable.

Lemma 3.2.24. *For any vector bundle E over a compact space X, there exists a finite dimensional ample subspace of* $\Gamma(E)$.

Proof. Let $\{U_\alpha\}$ be a finite open cover of X such that $E|_{U_\alpha}$ is trivial. Let $\{s_{\alpha,i}\}$ be the corresponding local frames, and define V_α to be the finite dimensional ample subspace $V_\alpha \subseteq \Gamma(E|_{U_\alpha})$. Let ρ_α be a partition of unity subordinate to the cover $\{U_\alpha\}$. Define

$$\theta_\alpha : V_\alpha \to \Gamma(E)\,, \quad \theta_\alpha(s_\alpha)(x) = \begin{cases} s_\alpha(x) \cdot \rho_\alpha(x) & \text{if } x \in U_\alpha\,, \\ 0 & \text{else.} \end{cases}$$

Then the θ_α give rise to a homomorphism

$$\theta : \Pi_\alpha V_\alpha \to \Gamma(E)\,,$$

and the image of θ is finite dimensional and ample. The surjectivity is clear, as for any x there exists a $\rho_\alpha(x) > 0$ making the map

$$\theta_\alpha(V_\alpha) \to E_x$$

a surjection. □

Corollary 3.2.25. If E is any bundle over X, there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m.

Proof. Just take a basis $(s_1, \dots, s_m)$ of an ample subspace V and define $\varphi(x, e_i) = s_i(x)$, where e_i is the i-th standard basis vector of $\mathbb{C}^m$. $\square$

Corollary 3.2.26. If E is a bundle over X, there exists a bundle F such that $E \oplus F$ is trivial.

Proof. By Corollary 3.2.25 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$. The kernel of φ is a subbundle of $X \times \mathbb{C}^m$ and hence by Lemma 3.2.21 we have a splitting of the short exact sequence

$$0 \longrightarrow \ker(\varphi) \longrightarrow X \times \mathbb{C}^m \xrightarrow{\varphi} E \longrightarrow 0$$

Hence, the middle term becomes isomorphic to the direct sum of the outer terms and we get $X \times \mathbb{C}^m \cong E \oplus \ker(\varphi)$. $\square$

Begründung ergänzen, z.B.: Metriken auf $X \times \mathbb{C}^m$ und E wählen; dann ist $P := \varphi^*(\varphi\varphi^*)^{-1}\varphi$ ein stetiger Projektionsoperator mit $(1 - P)(X \times \mathbb{C}^m) = \ker(\varphi)$, und Lemma 3.2.18 liefert die Zerlegung. ($\varphi\varphi^*$ ist invertierbar, da φ Epimorphismus.)

Definition 3.2.27 (Grassmann Manifold)**.** Let V be a vector space and n be any integer. The **Grassmann manifold** $\mathrm{Gr}_n(V)$ is the set of all subspaces of V with **codimension** n. If V is given a Hermitian metric, we can topologise $\mathrm{Gr}_n(V)$ as a subspace of the set of all projection operators on V and this defines a map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)\,,$$

where $\mathrm{End}(V)$ is the space of all endomorphisms of V with the operator norm topology. We give $\mathrm{Gr}_n(V)$ the subspace topology of $\mathrm{End}(V)$.

Definition 3.2.28. Suppose that E is a bundle over X, V is a vector space and $\varphi : X \times V \to E$ is an epimorphism. Then we can define a map $f : X \to \mathrm{Gr}_{\dim(V)}(V)$ by sending x to the kernel of $\varphi_x : V \to E_x$. We call f **induced by** φ. This map is continuous for any metric on V. To see this notice that the topology on V is metric invariant, as V is finite dimensional. Now the map f is continuous if for the map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)$$

the composition $P \circ f$ is continuous. But $P \circ f$ is given by sending x to the projection onto $\ker(\varphi_x)$, which is continuous as φ is.

Definition 3.2.29 (Classifying Bundle)**.** Let V be a vector space and $F \subseteq \mathrm{Gr}_n(V) \times V$ be the subbundle consisting of all points (g, v) such that $v \in g$. We call F the **tautological bundle** over $\mathrm{Gr}_n(V)$. Furthermore, if $E = (\mathrm{Gr}_n(V) \times V)/F$ is the quotient bundle, we call E the **classifying bundle** over $\mathrm{Gr}_n(V)$.

Corollary 3.2.30. If E' is a bundle over X, and $\varphi : X \times V \to E'$ is an epimorphism, then φ induces a map $f : X \to \mathrm{Gr}_m(V)$ such that $E' \cong f^*E$, where E is the classifying bundle over $\mathrm{Gr}_m(V)$. The isomorphism follows from a fibrewise construction because $E_x \cong V \big/ \ker(\varphi_x)$. The latter however is exactly the fibre over $\ker(x)$ in the classifying bundle and hence the pullback via the φ induced map gives an isomorphism. That motivates the name "classifying bundle" for E and "classifying map" for f.

Lemma 3.2.31. *The topology on* $\mathrm{Gr}_n(V)$ *is independent of the choice of the metric on* V.

Proof. Let h and h' be two metrics on V. Then we denote the topological space $\mathrm{Gr}_n(V)$ with the topology induced by h by $\mathrm{Gr}(V_h)$ and the one induced by h' by $\mathrm{Gr}(V_{h'})$. We need to show that the identity map $\mathrm{id} : \mathrm{Gr}(V_h) \to \mathrm{Gr}(V_{h'})$ is a homeomorphism. For this we need to show that id is continuous. The projection $\mathrm{Gr}_n(V_h) \times V_{h'} \to E$ (where E is the classifying bundle) is an epimorphism and hence induces a classifying map $f : \mathrm{Gr}_n(V_h) \to \mathrm{Gr}_n(V_{h'})$ such that $f^*E \cong E$. However that map is given by the identity on $\mathrm{Gr}_n(V)$ and hence the identity is continuous by Definition 3.2.28. □

Remark 3.2.32. To continue further we need direct limits of directed systems to construct general Grassmannians. However we will also need said construction for topological spaces and groups. Hence, we will give the definitions for arbitrary categories.

Definition 3.2.33 (Directed System)**.** Let $\mathcal{C}$ be a category and $(I, \leq)$ be a partially ordered set where every finite subset is bounded from above. Let $(A_i)_{i \in I}$ be a family of objects from $\mathcal{C}$ and $f_{ij} : A_i \to A_j$ be a morphism for all $i \leq j$ that satisfies:

1. $f_{ii} = \mathrm{id}_{A_i}$ for all $i \in I$.
2. $f_{kj} \circ f_{ik} = f_{ij}$ for all $i \leq k \leq j$.

The pair $\big((A_i)_{i \in I}, (f_{ij})_{i \leq j}\big)$ is called a **direct system**.

Definition 3.2.34 (Direct Limit)**.** Let $\big((A_i)_{i \in I}, (f_{ij})_{i \leq j}\big)$ be a direct system in a category $\mathcal{C}$. A pair $\big(X, (\varphi_i)_{i \in I}\big)$ where X is an object of $\mathcal{C}$ and $\varphi_i : A_i \to X$ are morphisms such that $\varphi_j \circ f_{ij} = \varphi_i$ for all $i \leq j$ is called a direct limit, if the following universal property holds: For any family of morphisms $g_i : A_i \to Y$ that satisfies $g_i \circ f_{ji} = g_j$ for all $i \leq j$ there exists a unique morphism $g : X \to Y$ such that $g_i = g \circ \varphi_i$. We denote the space X by $\mathrm{colim}_i\, A_i$ (or just $\mathrm{colim}\, A_i$), where the maps f_{ij} are omitted from the notation.

Corollary 3.2.35. As always, the direct limit is (if it exists) unique up to unique isomorphism.

Remark 3.2.36 (Construction of Direct Limits)**.** We will now see three constructions of direct limits. Let $\big((A_i)_{i \in I}, (f_{ij})_{i \leq j}\big)$ denote a directed system in the noted category.

In the category of sets we define the direct limit by

$$\mathrm{colim}_i\, A_i := \bigsqcup A_i \big/ \sim,$$

where $x \sim y$ if there exists $i, j, k \in I$ such that $f_{ik}(x) = f_{jk}(y)$. This gives an equivalence relation. The maps $\varphi_i : A_i \to \mathrm{colim}_i\, A_i$ are the composition of the inclusion into the disjoint union with the projection onto the quotient space. The universal property is easy to verify. For a family of suitable maps $g_i : A_i \to B$ we define the map $g : \mathrm{colim}_i\, A_i \to B$

as $g(x) = g_m(x)$ for $x \in A_m$. This is well defined since the g_i are compatible with the f_{ij} and unique by definition.

In the category of topological spaces we define the direct limit to be the set-theoretic direct limit equipped with the final topology of the inclusions, i.e. a subset $U \subseteq \operatorname{colim}_i A_i$ is open if and only if $\varphi_i^{-1}(U)$ is open for all m. Again the universal property is easy to prove by using the construction of the universal map from the set-theoretic limit. To show the continuity we show that $g^{-1}(U)$ is open for any open U which is checked by verifying openness of $\varphi_i^{-1}\big(g^{-1}(U)\big)$ for all i, which in turn equals $g_i^{-1}(U)$ and is open as the g_i are continuous. Uniqueness follows from the uniqueness of g established already in set.

In an algebraic category like semigroups, groups, rings we define the direct limit again as the set-theoretic direct limit, where the algebraic operations are defined such that the φ_i are homomorphisms meaning for example (if + is an operation in the A_i) $a \in A_i$ and $b \in A_j$ we define $a + b$ in $\operatorname{colim} i A_i$ by including the sum $f_{ik}(a) + f_{jk}(b)$ which is defined in A_k. Again the universal morphism g is the one from the set-theoretic limit which now is a homomorphism by construction.

Remark 3.2.37. If we build the direct limit of sets and all the maps of the direct system are injective, then the inclusions into the direct limit are injective. If all the maps of the direct system are closed embeddings, then the inclusions into the direct limit are also closed embeddings.

Proof. Recall that the inclusions $\varphi_i : A_i \to \operatorname{colim} A_i$ are defined to be the composition of the inclusion into the disjoint union with the projection onto the quotient. Now assume that all maps f_{ij} in the direct system are injective and $\varphi_i(x) = \varphi_i(y)$. Then $x \sim y$ in the quotient and hence, there exists $i, j, k \in I$ such that $f_{ik}(x) = f_{jk}(y)$ which implies that $x = y$.

Now assume that all f_{ij} are closed embeddings. Then, the inclusions are injective by the set-theoretic argument. Furthermore, the images of $\varphi_i : A_i \to \operatorname{colim} A_i$ are closed. That follows from subsets in the final topology being closed, if and only if all their preimages are closed. Hence we check if $\varphi_j^{-1}(\varphi(A_i))$ is closed for all $j \in I$:

$$\begin{array}{llll} \text{If } j \le i & \text{then: } \varphi_j^{-1}(\varphi_i(A_i)) = (\varphi_i \circ f_{ji})^{-1}(\varphi_i(A_i)) = f_{ji}^{-1}(A_i) & \text{which is closed.} \\ \text{If } i < j & \text{then: } \varphi_j^{-1}(\varphi_i(A_i)) = \underbrace{\varphi_j^{-1}(\varphi_j}_{\varphi \text{ is injective}} \; f_{ij}(A_i)) = f_{ij}(A_i) & \text{which is closed.} \end{array} \tag{2}$$

To show that $\varphi_i^{-1} : \operatorname{im}(\varphi_i) \to A_i$ is continuous we assume that $B \subset A_i$ is open. Then $\varphi_i(B)$ is open if and only if for any j $\varphi_j^{-1}(\varphi_i(B))$ is open. This follows from the same calculations as in equation (2) using that the f_{ij} are homeomorphisms onto their image. □

Corollary 3.2.38. The so-called inverse system given by the projections

$$\pi : \mathbb{C}^m \to \mathbb{C}^{m-1}, \quad (x_1, \dots, x_m) \mapsto (x_1, \dots, x_{m-1}),$$

induces a direct system of Grassmann manifolds with the induced continuous maps

$$i_{m-1} : \mathrm{Gr}_n(\mathbb{C}^{m-1}) \to \mathrm{Gr}_n(\mathbb{C}^m)\,, \quad U \mapsto \pi^{-1}(U)\,.$$

If we denote the classifying bundle over $\mathrm{Gr}_n(\mathbb{C}^m)$ by E_m, then the pullback of E_m along i_{m-1} is isomorphic to E_{m-1}.

Theorem 3.2.39. *The map*

$$\mathrm{colim}_m[X, \mathrm{Gr}_n(\mathbb{C}^m)] \to \mathrm{Vect}_n(X)\,,$$

induced by $f \mapsto f^*E_m$ is a bijection for all compact spaces X.

Proof. We shall construct an inverse map. If E is a bundle over X, then by Corollary 3.2.25 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m. Let $f : X \to \mathrm{Gr}_n(\mathbb{C}^m)$ be the map induced by φ. If we manage to show that the homotopy class of f as a map to $\mathrm{Gr}_n(\mathbb{C}^{m'})$ for some large enough m' is independent of the choice of φ and m, then we get a well defined map from $\mathrm{Vect}_n(X)$ to $\mathrm{colim}_m[X, \mathrm{Gr}_n(\mathbb{C}^m)]$ that is inverse to the classifying map.

Suppose that $\varphi_i : X \times \mathbb{C}^{m_i} \to E$ for $i = 1, 2$ are two epimorphisms. Let $g_i : X \to \mathrm{Gr}_n(C^{m_i})$ be the maps induced by φ_i. We want to show that g_1 and g_2 are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. For this we define the map

$$\psi_t : X \times \mathbb{C}^{m_1} \times C^{m_2} \to E\,, \quad \psi_t(x, v_1, v_2) = t\varphi_1(x, v_1) + (1-t)\varphi_2(x, v_2)\,.$$

Since the φ_i are epimorphisms, this map is an epimorphism for any t. Let

$$f_t : X \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$$

be the maps induced by ψ_t. (where we identified $\mathbb{C}^{m_1} \times \mathbb{C}^{m_2}$ with $\mathbb{C}^{m_1+m_2}$ by mapping (v_1, v_2) to (v_1, v_2)). Then

$$f_0 = j_0 \circ g_1 \quad \text{and} \quad f_1 = T \circ j_1 \circ g_2\,,$$

where $j_i : \mathrm{Gr}_n(\mathbb{C}^{m_i}) \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$ is the map induced by the inclusion $\mathbb{C}^{m_i} \to \mathbb{C}^{m_1+m_2}$ and T is the map induced by the isomorphism $\mathbb{C}^{m_1+m_2} \to \mathbb{C}^{m_1+m_2}$ that sends (v_1, v_2) to (v_2, v_1). Since T is homotopic to the identity, we can conclude that $j_0 \circ g_1$ and $j_1 \circ g_2$ are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. Hence, the homotopy class of the map induced by an epimorphism is independent of the choice of the epimorphism and the integer m. This concludes the proof. □

Lemma 3.2.40. *For any compact X and direct system $\big((Y_m)_{m\in\mathbb{N}}, (i_m)_{m\in\mathbb{N}}\big)$ where all Y_m are Hausdorff and all i_m are closed embeddings we have a canonical bijection between the set theoretic direct limit of homotopy classes of maps and the* *homotopy classes of maps to the topological direct limit:*

$$\operatorname{colim}_m[X, X_m] \cong [X, \operatorname{colim}_m X_m]\,.$$

Proof. We denote $\operatorname{colim}_m Y_m = Y$ and the corresponding inclusion $\varphi_m : Y_m \to Y$. Since X is compact, any map $f : X \to Y$ has image that lies in some $\operatorname{im}(\varphi_m)$ for m big enough. Assume that this was not the case, then there would exist an increasing sequence $(j_i)_{i\in\mathbb{N}}$ such that for all i there is an x_i with $f(x_i) \in \operatorname{im}\varphi_{j_i} \smallsetminus \operatorname{im}\varphi_{j_i-1}$. With this we define the set

$$A := \{f(x_i) \,|\, i \in \mathbb{N}\} \subseteq \operatorname{im} f \subseteq Y\,.$$

Then A is closed, since for all m the intersection $\operatorname{im}\varphi_m \cap A$ is finite and since $\operatorname{im}(\varphi_m)$ is closed, said intersection is closed. By the final topology we can conclude that the whole A is closed and since it is contained in the image of f it is compact (Here we use that the Y_i are Hausdorff!). Furthermore, the same argument shows that every subset $B \subset A$ is closed, because again $B \cap \operatorname{im}\varphi_m$ is finite. This tells us that A as a subset of Y carries the discrete topology: Let $C \subseteq A$ be an arbitrary subset. Then $A \smallsetminus C \subseteq A$ is closed in Y and hence its complement in Y is open and intersected with A yields C. Hence, C is open in the subspace topology. This however is a contradiction, as A cannot be infinite, compact and carry the discrete topology. Hence any map $f : X \to Y$ has image that lies in some $\operatorname{im}(\varphi_m)$ for m big enough. But then there exists a f_m such that $f = \varphi_m \circ f_m$. This assignment is well defined on homotopy classes because if $f, s : X \to Y$ are homotopic, then there exists a homotopy $H : X \times I \to Y$ between them: By the same argument as before, the image of H is compact and hence contained in some $\operatorname{im}\varphi_m$. Hence, f_m and s_m are homotopic via H_m as maps to $\operatorname{im}\varphi_m$. Hence, we get a well defined canonical map

$$\Phi : [X, \operatorname{colim}_m Y_m] \to \operatorname{colim}_m[X, Y_m]\,.$$

This map is injective, as if f_m and s_m are homotopic as maps to $\operatorname{im}\varphi_m$, then they are also homotopic as maps to the direct limit. Finally Φ is surjective, as any element of the direct limit is represented by some $f_m : X \to Y_m$ and hence $\Phi([\varphi_m \circ f_m]) = [f_m]$. □

Definition 3.2.41. We denote $\operatorname{colim}_m \operatorname{Gr}_n(\mathbb{C}^m)$ by $\mathrm{BU}(n)$ and call it the **classifying space** for complex vector bundles of rank n. The name originates from the functor B that sends a topological group G to the classifying space of vector bundles with structure group G.

Corollary 3.2.42. For any compact space X, there is a canonical bijection between the homotopy classes of maps from X to $\mathrm{BU}(n)$ and the isomorphism classes of vector bundles of rank n over X:

$$[X, \mathrm{BU}(n)] \cong \operatorname{Vect}_n(X)\,.$$

Proof. This is a direct application of Lemma 3.2.40. □

3.3 K-Theory as a Cohomology Theory

Definition 3.3.1. We need the following categories of topological spaces:

- $\mathbf{CWH}$ is the category of compact spaces.
- $\mathbf{CWH}^*_{\mathbf{cof}}$ is the category of well-pointed compact spaces.
- $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ is the category of pairs of compact spaces, where the inclusions are cofibrations.

Here, we have the natural functors:

- $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}} \to \mathbf{CWH}^*_{\mathbf{cof}}$ by sending a pair $(X,Y) \mapsto (X/Y,[Y])$ with obvious morphisms. If $Y = \varnothing$ this is the disjoint union of X with a point.
- $\mathbf{CWH} \to \mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ by sending $X \mapsto (X,\varnothing)$ again with obvious morphisms. The inclusion of $\varnothing$ into X is a cofibration, as the empty set itself is an open neighbourhood and hence the pair is a NDR pair.
- $\mathbf{CWH} \to \mathbf{CWH}^*_{\mathbf{cof}}$ as the composition. Again, the inclusion of $[\varnothing]$ into the disjoint union $X \sqcup [\varnothing]$ is a cofibration, as $\{[\varnothing]\}$ is an open neighbourhood of itself.

If we descend to the category where the morphisms are homotopy classes of maps, we get the categories $\mathbf{hCWH}$, $\mathbf{hCWH}^*_{\mathbf{cof}}$ and $\mathbf{hCWH}^{\mathbf{2}}_{\mathbf{cof}}$. Furthermore, the construction gives a functor from the category of abelian semigroups to the category of abelian groups.

Definition 3.3.2 (Grothendieck Group and Ring)**.** Let A be an abelian semigroup. Then the **Grothendieck group** of A is the abelian group $K(A)$ together with a homomorphism $\varphi : A \to K(A)$ such that for any other group G together with a homomorphism $\psi : A \to G$, there exists a unique homomorphism $\psi : K(A) \to G$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[dr, "\psi"'] & K(A) \arrow[d, dashed, "\tilde{\psi}"] \\ & G \end{tikzcd}$$

If G is a semiring, then $K(G)$ inherits a ring structure and we call $K(G)$ the **Grothendieck ring** of G.

Corollary 3.3.3. The Grothendieck group exists for any abelian semigroup and is unique up to canonical isomorphism. If G is a semiring, then the Grothendieck ring exists and is unique up to canonical isomorphism.

Proof. The construction follows the following idea: We define tuples (a,b) that are to be interpreted as differences "$a-b$". For this construction to yield something useful we want to enforce the rule that "$a-b = a+c-(b+c)$" or said differently $(a+c,b+c) = (a,b)$. Finally, to account for the universal property, we want to use the smallest relation that obeys said rule. This gives us the following construction: Let A be an abelian semigroup and $\Delta : A \to A \times A$ be the diagonal homomorphism that sends a to (a,a). Let $K(A) := A \times A \big/ \Delta(A)$. By this notation we mean: $\Delta(A)$ acts on $A \times A$ via

$$\Delta(A) \times (A \times A) \to A \times A \quad ((a,a),(b,c)) \mapsto (a+b, a+c)\,.$$

With this we have the set of orbits which gives rise to a relation:

$$\mathcal{R} := \{((a,b),(a+c,b+c)) | c \in A\}$$

Now we take the smallest equivalence relation on $A \times A$ that contains $\mathcal{R}$ (This relation is made explicit in Remark 3.3.4). We define $K(A)$ to be $A \times A$ modulo said relation. Then $K(A)$ is a priori an abelian semigroup, but for $a, b \in A$ we have

$$(a,b) + (b,a) = (a+b, b+a) \in \Delta(A)\,,$$

and hence $K(A)$ is an abelian group. Also we can see that

$$\overline{(a,b)} = \overline{(a,0)} - \overline{(b,0)} = \alpha_A(a) - \alpha_A(b)$$

Together with the homomorphism $\alpha_A : A \to K(A)$ that sends a to the class of $(a,0)$, we get the universal property of the Grothendieck group:

Let G be another abelian group and $\psi : A \to G$ a semigroup homomorphism. From this define

$$\tilde{\psi} : K(A) \to G\,, \quad \overline{(a,b)} \to \psi(a) - \psi(b)$$

The well definition of this map can be verified by a calculation. By calculating it also follows that $\tilde{\psi}$ is a group homomorphism and that $\tilde{\psi} \circ \alpha_A = \psi$. Hence, for the universal property we only need to verify uniqueness. So assume that $\theta : K(A) \to G$ is another group homomorphism. Then:

$$\theta\left(\overline{(a,b)}\right) = \underbrace{\theta(\alpha_A(a))}_{\psi(a)} - \underbrace{\theta(\alpha_A(b))}_{\psi(b)} = \tilde{\psi}\left(\overline{(a,b)}\right)$$

This concludes the construction of the Grothendieck group. To see the functoriality we first need to define what $K(\beta)$ means, when $\beta : A \to B$ defines a semigroup homomorphism between two abelian semigroups. For this we use the universal property and define $K(\beta)$ to be the unique group homomorphism coming from $\alpha_B \circ \beta : A \to K(B)$. Now the uniqueness of said definition shows the equalities $K(\mathrm{id}) = \mathrm{id}$ and $K(f \circ g) = K(f) \circ K(g)$. □

Remark 3.3.4. In $K(A)$ we get the following equivalence for any two pairs $\overline{(a,b)}$ and $\overline{(a',b')}$:

$$\overline{(a,b)} = \overline{(a',b')} \Leftrightarrow \text{There exists a } c \in A \text{ such that } (a + b' + c) = (a' + b + c)\,.$$

Proof. We start by proving "$\Leftarrow$". This direction follows from the properties of an equivalence relation, as then, every element of an equivalence class is also a representative:

$$\overline{(a,b)} = \overline{(a+(a'+b+c), b+(a+b'+c))} = \overline{(a'+(a+b+c), b'+(a+b+c))} = \overline{(a',b')}\,. \quad (3)$$

The other direction follows from the minimality of the equivalence relation: To see this notice how the argument above works in every equivalence relation containing the orbits of $\Delta(A)$ and hence works in the smallest one. Furthermore,

$$(a,b) \sim (a',b') \Leftrightarrow \text{ There exists a } c \in A \text{ such that } (a+b'+c) = (a'+b+c)$$

defines an equivalence relation that contains the orbits and hence this is the smallest one. □

Remark 3.3.5 (A simpler Construction of the Grothendieck Group?)**.** One might be wondering, why we defined the relation on the pair $A \times A$ as the smallest equivalence relation starting by identifying the orbits of the operation of the diagonal, when Remark 3.3.4 might look and feel different. The reason for this detour is that I wanted to understand what Atiyah meant when he defined in Chapter II.1 [Ati89] word for word:

A slightly different construction of $K(A)$ *which is sometimes convenient is the following: Let* $\Delta : A \to A$ *be the diagonal homomorphism of semi-groups, and let* $K(A)$ *denote the set of cosets of* $\Delta(A)$ *in* $A \times A$*. It is a quotient semi-group, but the interchange of factors in* $A \times A$ *introduces an inverse in* $K(A)$ *so that* $K(A)$ *is a group.*

Whilst this definition feels easy to the reader, it carries a lot of baggage. First of all, if one is not familiar with commutative monoid or abelian semigroup theory, one might want to generalise the notion of coset from group theory that states that for a subgroup $A \subseteq G$ of an abelian group G the cosets are of the form $g + A$. Whilst this notion can be generalised to semigroups, it would yield something useless. To see this just take $\mathbb{N}$. In the semigroup $\mathbb{N} \times \mathbb{N}/\Delta(\mathbb{N})$ the cosets $(2,0) + \mathbb{N}$ and $(3,1) + \mathbb{N}$ are different, as $(2,0)$ is not contained in the latter. To give Atiyah the benefit of the doubt we could generalise the definition of cosets as orbits of the right translation, which is what we did in our construction. However, the literature on semigroups from that time defines the notion of quotients in the setting of semigroups with the help of *congruence* instead of cosets. The latter refers to an equivalence relation that behaves well with the operation of said semigroup. Compare Chapter 1.5 [CP61]. It was even shown that the notion of congruence and coset are equivalent for groups (compare for example: Chapter 6.1 [Gre65]). Regardless of what Atiyah meant by his notion of coset, the construction became known and cited. ([Bro96], [Lan05], [Cou04]) None of those references define coset in semigroups and hence this little detour into the history should function as a reminder, that we need to be careful when generalising things from the known categories. We are so used to cosets giving equivalence relations, that it's easy to overlook the needed care.

Remark 3.3.6. We will omit the inclusion maps α from now on and interpret (if not said different) the equivalence class $[E]$ of a vector bundle as an element in $K(X)$.

Corollary 3.3.7. In the proof above we have shown that the assignment K is a functor from the category of abelian semigroups to the category of abelian groups. Alternatively, we can view K as a functor from the category of semirings to the category of rings.

Definition 3.3.8. Let X be a compact Hausdorff space. We define the **K-group** of X to be the Grothendieck ring of the semiring $\mathrm{Vect}(X)$ of isomorphism classes of vector bundles over X with the direct sum as addition and the tensor product as multiplication. We denote the K-group of X by $K(X)$. As a whole we get a contravariant functor from the category of compact Hausdorff spaces to the category of rings

$$\mathbf{CWH} \to \mathbf{Ring}\,. \tag{4}$$

Remark 3.3.9. For a continuous map $f : X \to Y$ we denote the induced map $K(f^*) : K(Y) \to K(X)$ by just f^*.

Theorem 3.3.10. *The functor $K : \mathbf{CWH} \to \mathbf{Ring}$ factors over the homotopy category* **hCWH** *of compact Hausdorff spaces together with homotopy classes of maps.*

Proof. This directly follows from Theorem 3.2.4. □

Definition 3.3.11 (Reduced K-Theory)**.** Let X be a compact Hausdorff space and $x_0 \in X$ be a base point. We define the **reduced K-group** $\tilde{K}(X)$ of X to be the kernel of the map $K(X) \to K(\{x_0\})$ induced by the inclusion $\{x_0\} \to X$.

Corollary 3.3.12. If $c : X \to x_0$ is the collapsing map sending each point to the base-point, then we get a natural splitting

$$K(X) \cong \tilde{K}(X) \oplus K(x_0)\,.$$

Naturality here means, that for any homomorphism of pointed spaces $f : (X, x_0) \to (Y, y_0)$ we have a commuting diagram:

$$\begin{tikzcd}
K(X) \arrow[r] \arrow[d, "f^*"] & \tilde{K}(X) \oplus K(x_0) \arrow[d, "\tilde{K}(f)\oplus(\{x_0\}\to\{y_0\})^*"] \\
K(Y) \arrow[r] & \tilde{K}(Y) \oplus K(y_0)
\end{tikzcd}$$

Proof. We have the exact sequence of rings:

$$0 \longrightarrow \tilde{K}(X) \xrightarrow{inclusion} K(X) \xrightarrow{i_X^*} K(x_0) \longrightarrow 0$$

where $c_X^* : K(x_0) \to K(X)$ is a right inverse of i_X^*, and hence the sequence splits and $K(X) \cong \tilde{K}(X) \oplus K(x_0)$. The isomorphism is explicitly given by

$$K(X) \to \tilde{K}(X) \oplus K(x_0)\,, \quad \xi \mapsto (\xi - c_X^* \circ i_X^*(\xi), i_X^*(\xi))$$

and its inverse maps $(a, b) \mapsto a + c_X^*(b)$. The splitting is natural with respect to maps in $\mathbf{CWH}^*_{\mathbf{cof}}$, because for a map of pointed spaces $f : X \to Y$ we get commutativity:

$$\begin{tikzcd}
0 \arrow[r] & \tilde{K}(X) \arrow[r, hook] & K(X) \arrow[r, "i_X^*"'] & K(x_0) \arrow[r] \arrow[l, bend right, "c_X^*"'] & 0 \\
0 \arrow[r] & \tilde{K}(Y) \arrow[r, hook] \arrow[u, "f^*|_{\tilde{K}(Y)}"] & K(Y) \arrow[r, "i_Y^*"] \arrow[u, "f^*"] & K(y_0) \arrow[r] \arrow[u, "(f|_{x_0})^*"'] \arrow[l, bend left, "c_Y^*"] & 0
\end{tikzcd}$$

The well definition of $f^*|_{\tilde{K}(Y)}$ follows immediately from the commutativity. The commutativity itself follows from K being a contravariant functor and all topological maps commute. Finally, define $\tilde{K}(f) := f^*|_{\tilde{K}(Y)}$ and we are done, since the naturality of the splittings, i.e. commutativity of the diagram in the corollary, then follows from commutativity of each square separately: the left square controls the $\tilde{K}$-summand and the right square (together with the sections c_X^*, c_Y^*) controls the $K(x_0)$-summand. ☐

Remark 3.3.13. For a map $f : (X, x_0) \to (Y, y_0)$ we denote $\tilde{K}(f)$ and $K(f)$ by f^*. The context will make clear if we work in the reduced theory or not.

Corollary 3.3.14. $\tilde{K}$ is a functor from $\mathbf{CWH}^*_{\mathbf{cof}}$ to **Ring**.

Proof. The proof follows immediately from the naturality of the splitting in Corollary 3.3.12. ☐

Corollary 3.3.15. Let X be a compact Hausdorff space, and let

$$F : \mathbf{CWH} \to \mathbf{CWH}^*_{\mathbf{cof}}$$

be the functor that sends X to $(X, [\varnothing])$ with obvious homomorphisms. Then there is a natural isomorphism between the functors $\tilde{K} \circ F$ and K.

Proof. This follows from the fact that disjoint unions give rise to direct sums of semirings: $\mathrm{Vect}(X \sqcup \{\varnothing\}) \cong \mathrm{Vect}(X) \oplus \mathrm{Vect}(\{\varnothing\})$ and the isomorphism is given by gluing the bundles and hence natural. Furthermore, the Grothendieck ring of a direct sum gives the direct sum of rings (again in a natural manner). (This can be verified using the universal property). Hence we get the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] \arrow[d, equal] & \tilde{K}(F(X)) \arrow[r, "inclusion"] & K(F(X)) \arrow[r, "i^*"] \arrow[d, "\Phi"] & K([\varnothing]) \arrow[r] \arrow[d, equal] & 0 \arrow[d, equal] \\
0 \arrow[r] & K(X) \arrow[r, "i_1"] & K(X) \oplus K([\varnothing]) \arrow[r, "\pi_2"] & K([\varnothing]) \arrow[r] & 0
\end{tikzcd}$$

But then we can define the map $\Phi|_{\tilde{K}(F(X))}$ and diagram yoga shows that this is an isomorphism. Furthermore, since Φ is natural, the new map is also natural and defines a natural transformation between the two functors. ☐

Definition 3.3.16. Let $F : \mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}} \to \mathbf{CWH}^*_{\mathbf{cof}}$ be the functor that collapses the second space. Then the composition $\tilde{K} \circ F$ defines a contravariant functor.

Definition 3.3.17. Let (X, x_0) and (Y, y_0) be two pointed spaces. Then we define the **pointed sum** as:

$$X \vee Y := (X \sqcup Y) \big/ (x_0 \sim y_0) \,.$$

With this, we can define the **smash product** as:

$$X \wedge Y := (X \times Y) \big/ (X \vee Y)$$

Definition 3.3.18. For the sake of intuition, I want to give a bit of insight into tensor products without being too technical and opening Pandora's box of monoidal categories. We define a tensor product in a category simply as the space over which bilinear maps factor uniquely. To capture bilinearity in such a category we define the following. Let $\pi_i : X_0 \times X_1 \to X_i$ be the canonical projection for $i = 0, 1$. Any reasonable product should come with such surjective maps. Since those are surjective, the π_i admit a family of right inverses $r_i : X_i \to X_0 \times X_1$. We call such a right inverse "product preserving" if the compositions $\pi_2 \circ r_1$ and $\pi_1 \circ r_2$ are constant. Hence the right inverse maps into just one fibre. Now bilinearity in this general setting states, that for any product preserving right inverse of the projections r_i, the composition $f \circ r_i$ is structure preserving.

Remark 3.3.19 (What is the Smash Product)**.** We can think of the smash product as the tensor product in the pointed topological spaces as follows. Let $f : (X, x_0) \times (Y, y_0) \to (Z, z_0)$ be bilinear. For any $y \in Y$ we have the product preserving right inverse of π_1 given by

$$r_1^y : (X, x_0) \to (X, x_0) \times (Y, y_0), \quad x \mapsto (x, y) \,.$$

Analogously we define r_2^x for any $x \in X$ as

$$r_2^x : (Y, y_0) \to (X, x_0) \times (Y, y_0), \quad y \mapsto (x, y) \,.$$

Now for any $x \in X$ we have $f(x, y_0) = f \circ i_Y^x(y_0) = z_0$ as the composition is said to be basepoint preserving. Similarly we see that $f(x_0, y) = z_0$ for any $y \in Y$. Hence f factors over the quotient space

$$X \times Y \big/ (X \times \{y_0\} \cup (Y \times \{x_0\})) \,,$$

where the numerator is the canonically embedded $X \vee Y$ into the product. Furthermore by the universal property of quotient spaces, this factorisation is unique.

Example 3.3.20. Let S^n be the n-dimensional sphere given by all points of norm one in $\mathbb{R}^{n+1}$. Then a general fact is that $S^n \cong I^n \big/ \partial I^n$ and hence for all $n, m > 0$ we have $S^n \wedge S^m \cong S^{n+m}$. Later we will make this homeomorphism canonical. ref

Definition 3.3.21. Let (X, x_0) be a pointed topological space. We define the **reduced suspension** of X to be

$$\Sigma X := S^1 \wedge X \,.$$

Furthermore, we define inductively $\Sigma^n X := \Sigma(\Sigma^{n-1} X) \cong S^n \wedge X$ and using the principle of permanence we extend this definition to $\Sigma^0 X := S^0 \wedge X \cong X$. The last homeomorphism is canonical.

Definition 3.3.22. Let (X, x_0) be a pointed topological space. We define the **suspension** of X to be the space

$$S(X) := X \times I \big/ \sim \tag{5}$$

where $(x,t) \sim (x',t')$ if $t = 1$ or $t = 0$. Here we identify the top to a point and the bottom to a different point. $((x,1) \nsim (0,0))$. We view $(S(X),(x,1))$ as a pointed space.

In general the reduced and unreduced suspension differ. However, under the following condition there will be a natural homotopy equivalence:

Lemma 3.3.23. *If (X, x_0) is well-pointed, then the inclusion $\{x_0\} \times I \hookrightarrow S(X)$ is a cofibration.*

Proof. This proof is omitted but is Proposition 5.1.16 in [tom08]. □

Corollary 3.3.24. If (X, x_0) is well pointed, then $S(X)$ is homotopically equivalent to $\Sigma(X)$.

Proof. By lemma 3.3.23 we conclude that $(S(X), x_0 \times I)$ satisfies the homotopy extension property. Hence, by Lemma 1.2.4 we know that $S(X) \simeq S(X)\big/x_0 \times I$ and the homeomorphism

$$S(X)\big/x_0 \times I \cong \Sigma(X)$$

concludes the proof. □

Definition 3.3.25 (Lower K-groups)**.** For $n \geq 0$ we define:

$$\begin{array}{llll} \tilde{K}^{-n}(X, x_0) & := & \tilde{K}(\Sigma^n X) & \text{for} \quad (X, x_0) \in Ob(\mathbf{CWH}^*_{\mathbf{cof}}), \\ K^{-n}(X, Y) & := & \tilde{K}^{-n}(X/Y, [Y]) & \text{for} \quad (X, Y) \in Ob(\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}) \\ K^{-n}(X) & := & K^{-n}(X, \varnothing) & \text{for} \quad X \in Ob(\mathbf{CWH}) \ . \end{array}$$

All those definitions give contravariant and homotopically invariant functors into the category of rings.

Corollary 3.3.26. Let X be a topological space. Then we have a natural isomorphism between the functors $K^0(X)$ and $K(X)$.

Proof. When disentangling the definitions, we see that

$$K^0(X) = \tilde{K}(\Sigma^0(X, [\varnothing])) \cong \tilde{K}(X, [\varnothing]) \cong K(X)$$

where the first isomorphism comes from Σ^0 being canonically isomorphic to the identity and the second isomorphism is given in Corollary 3.3.15. □

Remark 3.3.27. If we have pointed topological spaces (X, x_0) or pairs (X, A) and want to denote $K^0(X, pt)$ or $K^0(X, A)$ we will often omit the exponent and just write $K(X, x_0)$ and $K(X, A)$.

Theorem 3.3.28 (Excision)**.** *For any pair (X, A) and $U \subseteq A$ open such that $\bar{U} \subseteq \mathring{A}$ the inclusion*

$$i : (X \setminus U, A \setminus U) \to (X, A)$$

induces an isomorphism

$$i^* : K^{-n}(X, A) \to K^{-n}(X \setminus U, A \setminus U)$$

Proof. Since U is open and A is closed, $(X \setminus U, A \setminus U)$ is a compact pair. The inclusion i induces a continuous bijection

$$\varphi : (X \setminus U)\Big/(A \setminus U) \to X\big/A$$

which is a homeomorphism as a continuous bijection from a compact space to a Hausdorff space. Hence i^* is an isomorphism. ☐

3.4 Homotopy-Theoretic definition of K

Definition 3.4.1. For any natural number $n \in \mathbb{N}$ we define $\underline{n}$ to be the n-dimensional trivial bundle over a base space X. With this we define the inclusions

$$i_n : \mathrm{Vect}_n(X) \to \mathrm{Vect}_{n+1}(X)\,, \quad \xi \mapsto \xi \oplus [\underline{1}],$$

and this gives rise to the direct limit $\mathrm{colim}_n \, \mathrm{Vect}_n(X)$.

Definition 3.4.2 (Stable Equivalence)**.** Two vector bundles E, F over the same base space are called **stably equivalent** if there exists an n such that E and F become isomorphic after adding the trivial bundle:

$$E \oplus \underline{n} \cong F \oplus \underline{n}\,.$$

Lemma 3.4.3. *In $K(X)$ two equivalence classes $[E]$ and $[F]$ are the same, if and only if E and F are stably equivalent.*

Proof. Assume that two vector bundles satisfy

$$E \oplus \underline{n} \cong F \oplus \underline{n}\,,$$

for some $n \in \mathbb{N}$. By abuse of notation we let E (and $\underline{n}$) denote a vector bundle and its isomorphism class simultaneously. Then by Remark 3.3.4

$$[E] = \overline{(E, 0)} = \overline{(E + \underline{n}, \underline{n})} = \overline{(F + \underline{n}, \underline{n})} = \overline{(F, 0)} = [F]\,.$$

On the other hand, if $[E] = [F]$, then by the same Remark we can conclude:

$$(E, 0) \sim (F, 0) \Leftrightarrow \text{ there exists } G \text{ such that} E \oplus G = F \oplus G\,.$$

Now by Corollary 3.2.26 we can find a vector bundle G' such that $G \oplus G' = \underline{n}$ for some n. Then

$$E \oplus G \oplus G' = F \oplus G \oplus G' \Rightarrow E \oplus \underline{n} = F \oplus \underline{n}$$

□

Definition 3.4.4. We call two vector bundles E, F over the same base space **eventually equivalent**, if there exist $n, m \in \mathbb{N}$ such that $E \oplus \underline{n} \cong F \oplus \underline{m}$.

Lemma 3.4.5. *The set of equivalence classes using the notion of eventual equivalence is the limit*

$$\operatorname{colim}_n \operatorname{Vect}_n(X)$$

Proof. Let $\operatorname{EVect}(X)$ be the set of equivalence classes using the notion of eventual equivalence. This satisfies the universal property of the direct limit. To see this let $f_m : \operatorname{Vect}_m(X) \to Y$ be a family of maps, that commute with the inclusions i_n from Definition 3.4.1. From those we define the map

$$f : \operatorname{EVect}(X) \to Y\,, \quad [\xi] \mapsto f_{\operatorname{rank}(\xi)}(\xi)\,.$$

This map is well defined, because f commuting with the inclusions i_n is equivalent to f being invariant under eventual equivalence. Uniqueness follows from every element of $\operatorname{EVect}(X)$ being represented by a vector bundle. □

Lemma 3.4.6. *The set of equivalence classes using the notion of eventual equivalence together with the direct sum is an abelian group*

Proof. Everything except inverses is clear. The neutral element is the class of $\underline{0}$. For inverses we take a bundle E and by Corollary 3.2.26 there exists a G such that $E \oplus G \cong \underline{n}$. Now $[E] + [G] = [\underline{n}] = [\underline{0}]$. □

Lemma 3.4.7. *For a compact connected space X there is a natural group isomorphism.*

$$K(X) \cong \mathbb{Z} \times \operatorname{EVect}(X) \cong \mathbb{Z} \times \operatorname{colim}_n \operatorname{Vect}_n(X)\,.$$

Furthermore if (X, p) is pointed, there is a natural group isomorphism

$$\tilde{K}(X) \cong \operatorname{EVect}(X) \cong \operatorname{colim}_n \operatorname{Vect}_n(X)\,.$$

Proof. Let K be an element of $K(X)$. Then by Corollary 3.3.3 we can write K as the difference of two classes of vector bundles $K = [E] - [F]$. We abuse the notation of equivalence classes in this proof meaning we won't differentiate between isomorphism classes of vector bundles in Vect_n, eventually or stable isomorphism classes or equivalence classes of naturally included vector bundles in $K(X)$. The context should always be clear. With this we define the function

$$\mathrm{rank}(K) : K(X) \to \mathbb{Z}\,, \quad K = [E] - [F] \mapsto \mathrm{rank}(E) - \mathrm{rank}(F)\,.$$

This is well defined as if $K = [E] - [F] = [E'] - [F']$, then by the relation in $K(X)$ we know that there is an n such that

$$E \oplus F' \oplus \underline{n} \cong E' \oplus F \oplus \underline{n}\,.$$

and hence:

$$\mathrm{rank}(E) - \mathrm{rank}(F) = \mathrm{rank}(E') - \mathrm{rank}(F')$$

Furthermore, rank has a natural right inverse given by

$$r : \mathbb{Z} \to K(X)\,, \quad m \mapsto \begin{cases} [m] - [0] & \text{if } m \geq 0 \\ [0] - [-m] & \text{if } m < 0\,. \end{cases}$$

Now we claim that the kernel $\ker(\mathrm{rank}) \cong \mathrm{EVect}(X)$. For this we define the map

$$\varphi : \mathrm{EVect}(X) \to K(X)\,, \quad [E] \mapsto [E] - [\underline{\mathrm{rank}(E)}]$$

To see the well-definedness we assume that E and E' are eventually isomorphic and hence there are $n, n' \in \mathbb{N}$ such that $E \oplus \underline{n} \cong E' \oplus \underline{n'}$. But then in $K(X)$ we have:

$$[E] - [\underline{\mathrm{rank}(E)}] = [E \oplus \underline{n}] - [\underbrace{\underline{\mathrm{rank}(E)} \oplus \underline{n}}_{=\underline{\mathrm{rank}(F)} \oplus \underline{n'}}] = [E' \oplus \underline{n'}] - [\underline{\mathrm{rank}(F)} \oplus \underline{n'}] = [F] - [\underline{\mathrm{rank}(F)}]\,.$$

Obviously $\mathrm{rank} \circ \varphi \equiv 0$. To get the other inclusion we first notice the following: Let $K \in K(X) = [E] - [F]$ be arbitrary. By Corollary 3.2.26 there exists a G such that $F \oplus G \cong \underline{n}$ for some $n \in \mathbb{N}$. Hence if K is in the kernel of rank we have:

$$K = [E] - [F] = [E \oplus G] - [\underline{n}] = [E \oplus G] - [\underline{\mathrm{rank}(E \oplus G)}] = \varphi([E \oplus G])$$

This leads to the naturally split short exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{EVect}(X) \arrow[r, "\varphi"] & K(X) \arrow[r, "\mathrm{rank}"'] & \mathbb{Z} \arrow[l, bend right, "r"'] \arrow[r] & 0 \end{tikzcd}$$

This already proves the isomorphisms

$$K(X) \cong \mathbb{Z} \times \mathrm{EVect}(X) \cong \mathbb{Z} \times \mathrm{colim}_n \,\mathrm{Vect}_n(X)\,.$$

Now we define the group isomorphism

$$l : \mathbb{Z} \to K(p)\,, \quad m \mapsto \begin{cases} [m] - [0] & \text{if } m \geq 0 \\ [0] - [-m] & \text{if } m < 0\,. \end{cases}$$

This gives the commutative diagram:

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{SVect}(X) \arrow[r, "\varphi"] & K(X) \arrow[r, "\mathrm{rank}"'] \arrow[d, "\mathrm{id}"] & \mathbb{Z} \arrow[r] \arrow[l, bend right, "r"'] \arrow[d, "l"] & 0 \\ 0 \arrow[r] & \tilde{K}(X) \arrow[r, "inclusion"] & K(X) \arrow[r, "i^*"] & K(p) \arrow[r] \arrow[l, bend left, "c_p^*"] & 0 \end{tikzcd}$$

Where i is the inclusion $\{p\} \to X$ and c_p is the collapse $X \to \{p\}$. Then by some diagram chasing the map

$$\tilde{\varphi} : \mathrm{SVect}(X) \to \tilde{K}(X)$$

is well defined and by the five lemma (add zeros to the short sequences on the left) it is an isomorphism. □

Remark 3.4.8. In the proof above we have seen that every element $k \in K(X)$ is given as the difference of a vector bundle and a trivial bundle $k = [\xi] - [\underline{n}]$ for some vector bundle ξ and $n \in \mathbb{N}$. We also saw that $\tilde{K}(X)$ is in the kernel of rank and hence any $k \in \tilde{K}(X)$ is given by $k = [\xi] - [\underline{\mathrm{rank}(\xi)}]$ where ξ is a vector bundle.

Definition 3.4.9. We define the **unitary group** U to be the direct limit of the unitary group of rank n under the standard inclusions

$$U(n) \to U(n+1)\,, \quad A \mapsto A \oplus \mathbb{1}_1 = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition 3.4.10. We now want to make $BU(n)$ into a direct system. Notationally, this proof will have many maps with two indices. The convention is that lower indices show for which colimit the maps form a directed system. As an example, the maps i_m^n are used for the colimit over m. We start with the inclusion

$$j_m : \mathbb{C}^m \to \mathbb{C}^{m+1}\,, \quad (x) \mapsto (x,0)\,,$$

which gives rise to the maps

$$f_n^m : \mathrm{Gr}_n(\mathbb{C}^m) \to \mathrm{Gr}_{n+1}(\mathbb{C}^{m+1})\,, \quad V \mapsto j_m(V)\,.$$

Those inclusions commute with the maps

$$i_m^n : \mathrm{Gr}_n(\mathbb{C}^m) \to \mathrm{Gr}_n(\mathbb{C}^{m+1})\,, \quad V \mapsto V \oplus \langle e_{m+1} \rangle$$

from the direct system from Corollary 3.2.38. If we denote the inclusion by φ_m^{n+1} : $\mathrm{Gr}_{n+1}(\mathbb{C}^m) \to \mathrm{colim}_m(\mathrm{Gr}_{n+1}(\mathbb{C}^m))$ we get the following commuting diagram, for all $n \le m$:

$$\begin{tikzcd} \mathrm{Gr}_n(\mathbb{C}^m) \arrow[r, "f_n^m"] \arrow[d, "i_m"] & \mathrm{Gr}_{n+1}(\mathbb{C}^{m+1}) \arrow[r, "\varphi_{m+1}^{n+1}"] \arrow[d, "i_m^n"] & \mathrm{colim}_m(\mathrm{Gr}_{n+1}(\mathbb{C}^m)) = BU(n+1) \\ \mathrm{Gr}_n(\mathbb{C}^{m+1}) \arrow[r, "f_n^{m+1}"] & \mathrm{Gr}_{n+1}(\mathbb{C}^{m+2}) \arrow[ur, "\varphi_{m+2}^{n+1}"] & \end{tikzcd}$$

Now for all n we use the universal property of the direct limit $\operatorname{colim}_m(\mathrm{Gr}_n(\mathbb{C}^m))$ on the family

$$\left(\varphi^{n+1}_{m+1} \circ f^m_n : \mathrm{Gr}_n(\mathbb{C}^m) \to BU(n+1)\right)_m$$

to induce maps $f_n : BU(n) \to BU(n+1)$, which in turn give us a directed system and let us define:

$$BU := \operatorname{colim}_n BU(n) = \operatorname{colim}_n \left(\operatorname{colim}_m \mathrm{Gr}_n(\mathbb{C}^m)\right).$$

Lemma 3.4.11. *The maps $f_n : BU(n) \to BU(n+1)$ are closed embeddings.*

Proof. Since $\mathrm{Gr}_n(\mathbb{C}^m)$ is compact and Hausdorff for any $n \leq m$ the maps f^m_n are closed embeddings. Furthermore, the natural maps $\varphi^n_m : \mathrm{Gr}_n(\mathbb{C}^m) \to BU(n)$ are also closed embeddings. Now assume that $f_n(x) = f_n(x')$. Then for r big enough (omitting the inclusions) $f^r_n(x) = f^r_n(x')$ which contradicts the injectivity of the f^r_n. Now we want to show that f_n has a closed image. For this we let $C := \mathrm{im} f_n \subset BU(n+1)$ and call the natural inclusions $\varphi^{n+1}_m : \mathrm{Gr}_{n+1}(\mathbb{C}^m) \to BU(n+1)$. Then C is closed, if and only if $C \cap \mathrm{im}\varphi^{n+1}_m$ is closed for all $m > n$. To show this we first show that $C \cap \mathrm{im}\varphi^{n+1}_m = \mathrm{im}\, \varphi^{n+1}_m \circ f^{m-1}_n$. The first inclusion $\subseteq$ is clear. The other inclusion follows directly from the construction of f_n that satisfies $f_n \circ \varphi^n_{m-1} = \varphi^{n+1}_m \circ f^{m-1}_n$. By Remark 3.2.37 we know that $\varphi^{n+1}_m \circ f^{m-1}_n$ is a closed embedding hence $C \cap \mathrm{im}\varphi^{n+1}_m$ is closed for all $m > n$ and hence C is closed. Now finally we need to show that f_n are homeomorphisms onto their image. For this we show that they are closed maps, meaning they map closed sets to closed sets. This property together with injectivity implies the closed embedding property. Let $U \subset BU(n)$ be closed. By construction of the direct limit we can write $U = \bigcup_m(\mathrm{im}(\varphi^n_m) \cap U) = \bigcup_m(\varphi^n_m(U_m))$, where $U_m := (\varphi^n_m)^{-1}(U)$. Now by the same argument as before we have the equality $f_n(U) \cap \mathrm{im}\varphi^{n+1}_{m+1} = \varphi^{n+1}_{m+1} \circ f^m_n(U_m)$. This is closed for all m because the maps φ^{n+1}_{m+1} and f^m_n are closed embeddings and hence closed. But then $f_n(U)$ is closed in the final topology and hence f_n is a closed embedding. ☐

Theorem 3.4.12. *For a connected pointed space X we have a natural homomorphism*

$$\tilde{K}(X) \cong [X, BU]$$

Proof. We can now interchange the homotopy classes of maps and the limit by Lemma 3.2.40:

$$\begin{aligned} [X, BU] &= [X, \operatorname{colim}_n \left(BU(n)\right)] & \\ &\cong \operatorname{colim}_n \mathrm{Vect}_n(X) & \text{by Corollary 3.2.42,} \\ &\cong \tilde{K}(X) & \text{by Lemma 3.4.7.} \end{aligned}$$

☐

Lemma 3.4.13. *The general linear group* $\mathrm{GL}_n(\mathbb{C})$ *retracts onto the unitary group* $U(n)$ *and the retraction commutes with the inclusions* $A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$.

Proof. Let $A \in \mathrm{GL}_n(\mathbb{C})$. Then A^*A is a positive-definite Hermitian matrix, since for $v \in \mathbb{C}^n$:

$$v^*A^*Av = (Av)^*(Av) = \|Av\|^2 \geq 0\,.$$

Hence, it is diagonalisable with real non-negative eigenvalues $\lambda_1, \ldots \lambda_n > 0$. We define $P_A := \sqrt{A^*A}$ to be the matrix, that scales all eigenspaces by $\sqrt{\lambda_i}$. Then $P_A^2 = A^*A$, and $\sqrt{A^*A}$ is also positive definite Hermitian, as all eigenvalues are positive and hence, it is invertible. Notice, that

$$\sqrt{\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}^* \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}} = \begin{pmatrix} \sqrt{A^*A} & 0 \\ 0 & 1 \end{pmatrix}.$$

We then define $U_A := A(P_A)^{-1}$. By definition $A = U_A P_A$, and since P_A is Hermitian, P_A^{-1} is also Hermitian letting us conclude that U_A is unitary.

$$U_A^* U_A = ((P_A)^{-1})^* \underbrace{A^*A}_{P_A^2}(P_A)^{-1} = \mathbb{1}\,.$$

Notice again, that for $\tilde{A} = A \oplus \mathbb{1}_1 \in GL_{n+1}(\mathbb{C})$, we have $U_{\tilde{A}} = U_A \oplus \mathbb{1}_1 \in U(n+1)$. Now the space of positive-definite Hermitian matrices is convex. This is just a calculation of all the axioms for the matrices on the connecting line of two positive-definite Hermitian matrices. But as a convex space, it is contractible. □

Lemma 3.4.14. *For a connected pointed space* X *with well sitting basepoint, we have a natural homomorphism*

$$\tilde{K}(\Sigma X) \cong [X, U]$$

The group structure in $[X, U]$ *comes from viewing* $[X, U]$ *as the direct limit of abelian groups* $\mathrm{colim}[X, \mathrm{GL}_n(\mathbb{C})]$, *where the group structure is derived from the one in* $\mathrm{GL}_n(\mathbb{C})$.

Proof. If the base point is well sitting, we can apply Corollary 3.3.24, which gives us the homotopy equivalence

$$\Sigma X \simeq S(X)\,.$$

Now $S(X)$ itself can be cut into two contractible cones

$$\begin{aligned} C_+(X) &:= \{(x,t) \in S(X) \,|\, t \geq \tfrac{1}{2}\} \\ C_-(X) &:= \{(x,t) \in S(X) \,|\, t \leq \tfrac{1}{2}\}. \end{aligned}$$

Those cones intersect in $C_+ \cap C_- = X \times \{\frac{1}{2}\} \cong X$. Now denote the trivial n bundle over $C_+(X)$ by $\underline{n}_+$ and over $C_-(X)$ by $\underline{n}_-$. With this notation we know by the proof of Lemma 3.2.14 that the map

$$\begin{aligned} \varphi_n : [X, \mathrm{GL}_n(\mathbb{C})] &\to \mathrm{Vect}_n(\Sigma X) \\ f &\mapsto \underline{n}_+ \textstyle\bigcup_f \underline{n}_- \end{aligned} \tag{6}$$

is a bijection for all n. Furthermore, the inclusion of $\mathrm{GL}_n(\mathbb{C})$ into $\mathrm{GL}_{n+1}(\mathbb{C})$ gives an inclusion $[X, \mathrm{GL}_n(\mathbb{C})]$ into $[X, \mathrm{GL}_{n+1}(\mathbb{C})]$ which commutes with the inclusion $\mathrm{Vect}_n(\Sigma X)$ into $\mathrm{Vect}_{n+1}(\Sigma X)$ under the bijection (6) meaning that for any $n \leq m$ the diagram

$$\begin{tikzcd} {[X, \mathrm{GL}_n(\mathbb{C})]} \arrow[r, "\varphi_n"] \arrow[d, "i_{nm}"] & \mathrm{Vect}_n(\Sigma X) \arrow[d, "i_{nm}"] \\ {[X, \mathrm{GL}_m(\mathbb{C})]} \arrow[r, "\varphi_m"] & \mathrm{Vect}_m(\Sigma X) \end{tikzcd}$$

commutes. Lemma 3.4.7 gives us

$$\tilde{K}(X) \cong \mathrm{colim}_n[X, \mathrm{GL}_n(\mathbb{C})] \cong [X, \mathrm{colim}_n \mathrm{GL}_n(\mathbb{C})].$$

The latter bijection follows from Lemma 3.2.40. Finally, Lemma 3.4.13 gives us a retraction $\mathrm{GL}_n(\mathbb{C}) \to U(n)$ that commutes with the inclusions and hence we get the same direct limit.

$$[X, \mathrm{colim}_n \mathrm{GL}_n(\mathbb{C})] \cong [X, \mathrm{colim}_n U(n)] = [X, U]$$

concluding that we have a natural bijection. To get the group structure we need to look into said mapping. For this we inspect the mapping (6) closer. Let

$$\Phi_n : [X, \mathrm{GL}_n(\mathbb{C})] \to \mathrm{colim}\, \mathrm{Vect}_n(\Sigma X)$$

be the composition of φ_n with the inclusion $\mathrm{Vect}_n(\Sigma X) \to \mathrm{colim}\, \mathrm{Vect}_n(\Sigma X)$. By the commutative diagram above, the maps φ_n induce a map from $\mathrm{colim}[X, \mathrm{GL}_n(\mathbb{C})]$ which is a group homomorphism, if all Φ_n are one. Hence take two functions $f, g : X \to \mathrm{GL}_n(\mathbb{C})$. Then by definition

$$f \oplus g : X \to GL_n(\mathbb{C}), \quad x \mapsto f(x) \times g(x)$$

Now $f \oplus g$ can be included into $[X, \mathrm{GL}_{2n}(\mathbb{C})]$:

$$f \oplus g : X \to GL_{2n}(\mathbb{C}), \quad x \mapsto \begin{pmatrix} f(x) \times g(x) & 0 \\ 0 & \mathbb{1}_n \end{pmatrix}.$$

Now we have that $f \oplus g$ is homotopic to

$$\widetilde{f \oplus g} : X \to GL_{2n}(\mathbb{C}), \quad x \mapsto \begin{pmatrix} f(x) & 0 \\ 0 & g(x) \end{pmatrix}$$

via the homotopy $\rho_t(x)$ for $t \in (0, \frac{\pi}{2})$:

$$\rho_t(x) = \begin{pmatrix} f(x) & 0 \\ 0 & \mathbb{1}_n \end{pmatrix} \begin{pmatrix} \cos(t) & \sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} \mathbb{1}_n & 0 \\ 0 & g(x) \end{pmatrix} \begin{pmatrix} \cos(t) & \sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \tag{7}$$

Finally we can look at the image of $[f \oplus g] = [\widetilde{f \oplus g}]$ under Φ_{2n} which is given by inclusion of

$$\varphi_n([\widetilde{f \oplus g}]) = \underline{n}_+ \oplus \underline{n}_+ \bigcup_{\widetilde{f \oplus g}} \underline{n}_- \oplus \underline{n}_- = [\varphi_n(f) \oplus \varphi_n(g)]$$

into the colimit, where the equation

$$[\varphi_n(f) \oplus \varphi_n(g)] = [\varphi_n(f)] \oplus [\varphi_n(g)]$$

holds. The second last equality follows from the properties of the gluing construction given in Corollary 3.2.11. □

Remark 3.4.15. The proof of Lemma 3.4.14 gives a slightly more general statement: For a connected well pointed topological space (X, x_0) with a splitting into two contractible parts X^+ and X^- with $x_0 \in A := X^+ \cap X^-$ and $X^+ \cup X^- = X$ and the inclusion of A into the two contractible parts are cofibrations, there is a natural group isomorphism

$$[A, U] \cong \tilde{K}(X)$$

Lemma 3.4.16. *Let Y be a pointed space. Define*

$$\begin{aligned} T : S^1 &\to S^1 \\ t &\mapsto 1 - t \end{aligned}$$

and let $\Sigma T : SY \to SY$ be the map induced by T on S^1 and the identity on Y. Then:

$$\begin{aligned} (\Sigma T)^* : \tilde{K}(SY) &\to \tilde{K}(SY) \\ y &\mapsto -y \end{aligned}$$

Proof. We have the following commutative diagram:

$$\begin{tikzcd} \tilde{K}(SX) \arrow[d, "(\Sigma T)^*"'] & \operatorname{colim}[Y, \mathrm{GL}(n, \mathbb{C})] \arrow[l] \arrow[d, "f \mapsto f^{-1} (pointwise)"] \\ \tilde{K}(SX) & \operatorname{colim}[Y, \mathrm{GL}(n, \mathbb{C})] \arrow[l] \end{tikzcd}$$

To see why this commutes, notice how we defined the map by gluing via f, where ΣT changes the place of the upper and lower vector bundle. Hence the gluing function needed is now the inverse. The horizontal maps are natural group homomorphisms and on the right side the inverse is taken. Thereby the commutativity tells us that the left side is also the same as taking the inverse. □

Example 3.4.17. Let l and l' be line bundles over S^2. Then the clutching function for the vector bundle $l \otimes l'$ is given by the tensor product of clutching functions. So let f and f' be two clutching functions

$$f, f' : S^1 \to \mathrm{GL}_1(\mathbb{C})$$

Then the product bundle $l \otimes l'$ is given by the clutching function

$$f \otimes f' : S^1 \to \mathrm{GL}(\mathbb{C} \otimes \mathbb{C}) \cong \mathrm{GL}_1(\mathbb{C}), \quad x \mapsto f(x) f'(x)$$

Hence, the product in $\tilde{K}(S^2)$ is given by $f \otimes f' = f \cdot f'$. Now, the homotopy (7) of the proof above shows that $f \cdot f' \oplus \mathrm{id}_1 \simeq f \oplus f'$ and hence $l \otimes l' \oplus \underline{1} \cong l \oplus l'$.

Proposition 3.4.18. For compactly generated pointed Hausdorff spaces $(X, x_0), (Y, y_0)$ there is a natural bijection

$$[SX, Y] \to [X, \Omega Y],$$

where Ω denotes the loop space

$$\Omega Y := \{ f : S^1 \to Y \mid f \text{ continuous and } f(1) = y_0 \}.$$

together with the compact open topology.

Proof. For any three spaces $(X, x_0), (Y, y_0), (Z, z_0)$ we have that a basepoint preserving map $f : X \wedge Y \to Z$ can be lifted to a map $\tilde{f} : X \times Y \to Z; f(z) = \tilde{f}(\overline{z})$ that maps the whole $X \vee Y$ to z_0. This map is continuous by the universal property of the quotient topology. Hence we get a bijection

$$C(X \wedge Y, Z) \to \{ f \in C(X \times Y, Z) \mid \ f|_{X \vee Y} \equiv z_0 \}$$

its inverse is given by the quotient induced maps. We can compose this map with the natural currying map $C(X \times Y, Z) \to C(X, C(Y, Z))$. For $Y = S^1$ we gain a map

$$C(SX, Z) \to C(X, \Omega Z)$$

that is again a bijection, since any $f : X \to \Omega Z$ gives a map $\tilde{f} : X \times S \to Z$ $\tilde{f}(x, s) \mapsto f(x)(s)$. If $(x, s) \in X \vee S \subseteq X \times S$, then either $x = x_0$ or $s = 1$. In the first case, $f(x_0)$ is the constant loop and hence $f(x_0)(s) = z_0$ and in the second case we evaluate a based loop at its base and hence $f(x)(1) = z_0$. Hence, the $\tilde{f}$ factors over $X \vee S$ giving a well defined element in $C(SX, Z)$. This is its inverse. The continuity part of the argument follows from Proposition 1.3.3. □

Remark 3.4.19. By Theorem 3.4.12, and the Proposition above we know that for a connected pointed compact space we have

$$\tilde{K}(SX) \cong [SX, BU] \cong [X, \Omega BU]$$

On the other hand we have Lemma 3.4.14 which tells us that

$$\tilde{K}(SX) \cong [X, U].$$

This hints at a connection that states that ΩBU is homotopically equivalent to U:

3.5 K-Theory as a Cohomology Theory

This chapter aims to give the definitions and theorems that make K theory an axiomatic cohomology theory as defined by Eilenberg and Steenrod. For this we start trying to find a long exact sequence and on the way we will prove the additivity:

Lemma 3.5.1. *Let* $(X, Y) \in ob(\mathbf{CWH^2_{cof}})$ *and define* $i : (Y, \varnothing) \to (X, \varnothing)$ *and* $j : (X, \varnothing) \to (X, Y)$ *to be the inclusions. Then we have an exact sequence:*

$$K(X, Y) \xrightarrow{j^*} K(X) \xrightarrow{i^*} K(Y)$$

Here, we identify $K(X) = K(X, \varnothing)$ *and the same with* $K(Y) = K(Y, \varnothing)$ *such that we have well defined induced functions from the same functor in the sequence. This is just a formal note, as* $K(X, \varnothing) \cong K(X)$ *by Corollary 3.3.26.*

Proof. First, we show that $\mathrm{im}(i^*) \subseteq \ker(j^*)$. To see this, notice that $i^* \circ j^* = (j \circ i)^*$, and $j \circ i$ factors over (Y, Y):

$$\begin{tikzcd}
(Y, \varnothing) \arrow[r, "i"] \arrow[d] & (X, \varnothing) \arrow[d, "j"] \\
(Y, Y) \arrow[r] & (X, Y)
\end{tikzcd}$$

Hence, $j^* \circ i^*$ factors over the trivial ring $K(Y, Y)$ and thereby is trivial. Now suppose $k \in \ker(i^*)$. The intuition now is that if we have a vector bundle that gets mapped to 0 this means that it is trivial over Y which lets us define a vector bundle over the quotient X/Y by a trivialisation: We write $k = [\xi] - [\underline{n}]$ for ξ being a vector bundle over X. (Compare Remark 3.4.8) Since $i^*\xi = 0$ it follows that $i^*([\xi] - [\underline{n}]) = [\xi|_Y] - [\underline{n}|_Y] = 0$ and hence ξ and $\underline{n}$ are stably equivalent over Y giving us an m such that

$$(\xi \oplus \underline{m})|_Y \cong \xi|_Y \oplus \underline{m} \cong \underline{n} \oplus \underline{m}$$

Hence, there is a trivialisation $\alpha : (\xi \oplus \underline{m})|_Y \to Y \times \mathbb{C}^{\mathrm{rank}(\xi)+m}$ that defines a bundle $(\xi \oplus \underline{m})|_Y \big/ \alpha$ as shown in the proof of Lemma 3.2.7 over $X\big/Y$. Now we define

$$\eta = [\,(\xi \oplus \underline{m})|_Y \big/ \alpha] - [\underline{n + m}] \in \tilde{K}(X\big/Y) = K(X, Y)$$

and since the trivialisation acts as a right inverse of the pullback of the restriction (compare Corollary 3.2.8)

$$j^*(\eta) = [\xi \oplus \underline{m}] - [\underline{n+m}] = k\,.$$

□

Lemma 3.5.2. *Let X, Y be pointed topological spaces and define the maps*

$$i : X \to X \vee Y \text{ the inclusion, and } j : X \vee Y \to Y \text{ the collapse of } X.$$

Then the sequence

$$0 \longrightarrow \tilde{K}(X) \xrightarrow{j^*} \tilde{K}(X \vee Y) \xrightarrow{i^*} \tilde{K}(Y) \longrightarrow 0$$

is exact and naturally splits via the map q^*, that is a right inverse of j^*, where $q : X \vee Y \to Y$ is the collapse.

Proof. Notice how there is a natural homeomorphism $X \cong (X \vee Y)/Y$. Hence, Lemma 3.5.1 provides the exactness at $K(X \vee Y)$. Now the map q is a left inverse of j as continuous maps, and hence q^* is a right inverse of j^* in the K group, proving the injectivity of j^*. Now only the surjectivity of i^* is missing. However the inclusion $l : Y \to X \vee Y$ is a topological right inverse of i and thereby l^* is a left inverse of i^* giving us the injectivity and hence proving the exactness and splitting of the short exact sequence. □

Theorem 3.5.3 (Pointed Additivity)**.** *Let X, Y be pointed topological spaces and denote the inclusions of X (resp. Y) into $X \vee Y$ by i_X (resp. i_Y). Then the map*

$$i_X^* \oplus i_Y^* : \tilde{K}(X \vee Y) \to \tilde{K}(X) \oplus \tilde{K}(Y)$$

is a natural isomorphism with inverse

$$\psi : \tilde{K}(X) \oplus \tilde{K}(Y) \to \tilde{K}(X \vee Y)\,, \quad (x, y) \mapsto c_X^*(x) + c_Y^*(y)\,,$$

where c_X (resp. c_Y) denotes the collapse of X (hence onto Y) respectively the collapse of X and hence onto Y.

Proof. This is a direct consequence of Lemma 3.5.2 and the algebraic machinery of splitting short exact sequences. □

Corollary 3.5.4 (Unpointed Additivity)**.** Let X, Y be topological spaces, then there is a natural isomorphism

$$K(X \sqcup Y) \to K(X) \oplus K(Y)$$

Proof. This follows from $K(A) \cong \tilde{K}(A, [\varnothing])$ for all topological spaces. Furthermore, $(X \sqcup Y, [\varnothing]) \cong ((X, [\varnothing]) \vee (Y, [\varnothing]))$. Hence, using those isomorphisms, the statement is a direct consequence of the pointed additivity. $\square$

Corollary 3.5.5. If $(X, Y) \in ob(\mathbf{CWH^2_{cof}})$ and $Y \in ob(\mathbf{CWH^*_{cof}})$ such that X is pointed by using the base point y_0 from Y. Then the sequence

$$K(X, Y) \xrightarrow{j^*} \tilde{K}(X) \xrightarrow{i^*} \tilde{K}(Y)$$

is exact.

Proof. We have the commutative diagram with exact top row:

$$\begin{tikzcd}
K(X,Y) \arrow[r] \arrow[ddr] & K(X) \arrow[r] \arrow[d, "\cong"] & K(Y) \arrow[d, "\cong"] \\
& \tilde{K}(X) \oplus K(y_0) \arrow[r] \arrow[d, two heads] & \tilde{K}(Y) \oplus K(y_0) \arrow[d] \\
& \tilde{K}(X) \arrow[r] & \tilde{K}(Y)
\end{tikzcd}$$

The middle horizontal map is a map that respects the grading, since we have the same base point. With this we can do a bit of diagram yoga concluding the statement. $\square$

Definition 3.5.6. Let X be a topological space and $I = [0, 1]$ be the unit interval. We define the cone over X to be the space

$$CX := (X \times I) \big/ X \times \{0\}.$$

Let (X, Y) be a pair of compact topological spaces. We define the cone over the subspace Y to be

$$\left(X \sqcup CY\right) \big/ Y \sim Y \times \{1\}$$

and denote that with $(X \cup CY)$.

Proposition 3.5.7. If we define for $f : X \to Y$ the induced maps on the cones Cf by

$$Cf : CX \to CY \quad \overline{(x,t)} \mapsto \overline{(f(x),t)},$$

the cone defines a functor $\mathbf{CGWH} \to \mathbf{TopP}$. The cone over $Y \subseteq X$ defines in similar fashion a functor from $\mathbf{CGWH^2_{cof}}$ to $\mathbf{TopP}$. There is a natural isomorphism:

$$T : (X \cup CY) \big/ X \to CY \big/ Y$$

Here, the left and right hand side both define functors from $\mathbf{CGWH^2_{cof}}$ to $\mathbf{TopP}$ with the obvious morphisms. Furthermore, we have that CY/Y is homotopically equivalent to ΣY which is homotopically equivalent to SY as long as (Y, y_0) is well pointed. All the equivalences are natural up to homotopy.

Corollary 3.5.8. From the considerations above we get a natural isomorphism

$$\theta : K(X \cup CY, Y) \to \tilde{K}(SY) = \tilde{K}^{-1}(Y)\,.$$

Definition 3.5.9 (Connecting Homomorphism)**.** Let (X, Y) be a topological pair of compact spaces and Y be pointed. Define the two maps

$$m : (X \cup CY, \varnothing) \to (X \cup CY, X) \tag{8}$$

$$\pi : (X \cup CY, \varnothing) \to X\big/Y \tag{9}$$

where m is the inclusion in the second space, and π is the collapse of CY. Since CY is contractible, the induced map $\pi^* : \tilde{K}(X\big/Y) \to K(X \cup CY, \varnothing)$ is an isomorphism by Lemma 3.2.9. Together with the isomorphism

$$\theta : K(X \cup CY, X) \to \tilde{K}(SY) = \tilde{K}^{-1}(Y)\,.$$

We define the connecting homomorphism

$$\delta := (\pi^*)^{-1} \circ m^* \circ \theta^{-1} : \tilde{K}^{-1}(Y) \to K(X, Y)$$

This map defines also the higher degrees connecting homomorphisms by replacing the spaces with their suspensions.

Remark 3.5.10. The map $\delta : K^{-n}(A) \to K^{-n+1}(X, A)$ is a natural transformation between the functors

- $K^{-n} \circ R$ from $\mathbf{CWH^2_{cof}}$ to $\mathbf{Ab}$, where R is an endofunctor in $\mathbf{CWH^2_{cof}}$ sending $(X, A) \mapsto (A, \varnothing)$.
- K^{-n+1} from $\mathbf{CWH^2_{cof}}$ to $\mathbf{Ab}$.

Proof. Let $f : (X, A) \to (Y, B)$ be a map of pairs. Then we need to check if the diagram

$$\begin{tikzcd} (X, A) \arrow[d, "f"] & K^{-n}(A, \varnothing) \arrow[r, "\delta"] & K^{-n+1}(X, A) \\ (Y, B) & K^{-n}(B, \varnothing) \arrow[r, "\delta"] \arrow[u, "(f|_A)^*"] & K^{-n+1}(Y, B) \arrow[u, "f^*"'] \end{tikzcd}$$

commutes. This however is the case since δ is defined via the pullback of the inclusion (m^*) together with the natural isomorphisms θ and π^*. $\square$

Theorem 3.5.11 (Long Exact Sequence)**.** *For* $(X, Y) \in ob(\mathbf{CWH^2_{cof}})$ *we have a natural exact sequence that is infinite to the left:*

$$\cdots \longrightarrow K^{-2}(Y) \xrightarrow{\delta} K^{-1}(X, Y) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(Y)$$

$$\xrightarrow{\delta} K^{0}(X, Y) \xrightarrow{j^*} K^{0}(X) \xrightarrow{i^*} K^{0}(Y)$$

The maps i^* are induced by the inclusion $i : Y \to X$, j^* are induced by the inclusion $j : (X, \varnothing) \to (X, Y)$ and δ is the connecting homomorphism. For the lower degrees, δ is the connecting homomorphism where X and Y are replaced by the reduced suspensions of X and Y.

Proof. We will prove the pointed version of the theorem, that states that for a compact pair (X, Y) such that Y is itself compact pointed (and X is pointed with the same basepoint) we have the exact sequence (infinite to the left):

$$\cdots \longrightarrow \tilde{K}^{-2}(Y) \xrightarrow{\delta} K^{-1}(X,Y) \xrightarrow{j^*} \tilde{K}^{-1}(X) \xrightarrow{i^*} \tilde{K}^{-1}(Y)$$

$$\xrightarrow{\delta} K^{0}(X,Y) \xrightarrow{j^*} \tilde{K}^{0}(X) \xrightarrow{i^*} \tilde{K}^{0}(Y)$$

The pointed sequence induced the statement for unpointed spaces because for any topological space A we have a natural isomorphism $\tilde{K}(A, [\varnothing]) \cong K(A)$.

We will furthermore just prove the exactness of the segment:

$$\tilde{K}^{-1}(X) \xrightarrow{i^*} \tilde{K}^{-1}(Y) \xrightarrow{\delta} K^{0}(X,Y) \xrightarrow{j^*} \tilde{K}^{0}(X) \xrightarrow{i^*} \tilde{K}^{0}(Y) \tag{10}$$

From this we regain the long sequence by replacing (X) and Y with $S^n(X)$ and S^nY.

In $\tilde{K}^0(X)$, the exactness was shown in Corollary 3.5.5. To get the exactness at the other spots we apply the same corollary to the pairs:

- $(X \cup CY, X)$
- $((X \cup CY) \cup CX, X \cup CY)$

For the middle part we have the commutative diagram where the naming of the maps is the same as in the definition of the connecting homomorphisms (Definition 3.5.9).

$$\begin{tikzcd}
K(X \cup CY, X) \arrow[r, "m^*"] \arrow[d, "\theta"] & \tilde{K}(X \cup CY) \arrow[r, "k^*"] & \tilde{K}(X) \arrow[d, "\mathrm{id}"] \\
K^{-1}(Y) \arrow[r, "\delta"] & K(X,Y) \arrow[r, "j^*"] \arrow[u, "\pi^*"] & \tilde{K}(X)
\end{tikzcd}$$

The commutativity in the left square is the definition of δ and in the right square follows from $\pi \circ k = j : X \to X/Y$. Since all vertical maps are isomorphisms, Corollary 3.5.5 implies exactness in the bottom row $K(X,Y)$. Now only the exactness in the first three terms is missing. For this we look at the pair $((X \cup C_1Y) \cup C_2X, X \cup C_1Y)$, where the C_i are cones and denoted with the index to distinguish the cones. First we get again an exact sequence:

$$K\big((X \cup C_1Y) \cup C_2X, X \cup C_1Y\big) \xrightarrow{l^*} \tilde{K}\big((X \cup C_1Y) \cup C_2X\big) \xrightarrow{s^*} \tilde{K}(X \cup C_1Y)$$

Our goal is now to show that we commute with the start of the exact sequence (10), hence to find vertical maps between the top rows (denoted with ψ in the diagram below) such that the diagram commutes:

$$\begin{tikzcd} K\big((X\cup C_1Y)\cup C_2X, X\cup C_1Y\big) \arrow[r, "l^*"] \arrow[d, "\psi_1"] & \tilde{K}\big((X\cup C_1Y)\cup C_2X\big) \arrow[r, "s^*"] \arrow[d, "\psi_2"] & \tilde{K}(X\cup C_1Y) \arrow[d, "\psi_3"] \\ \tilde{K}^{-1}(X) \arrow[r, "i^*"] & \tilde{K}^{-1}(Y) \arrow[r, "\delta"] & K(X,Y) \end{tikzcd}$$

Diagram 11

The bottom row is added for clarification on what inclusion induces i^*.. besser erklähren

We start with the right square and notice that by definition we get a commutative diagram:

$$\begin{tikzcd} K(X\cup C_1Y\cup C_2X) \arrow[r, "s^*"] \arrow[d, "(\tilde{\pi}^*)^{-1}"] & \tilde{K}(X\cup C_1Y) \arrow[d, "\mathrm{id}"] \\ K(X\cup C_1Y, X) \arrow[r, "m^*"] \arrow[d, "\theta"] & \tilde{K}(X\cup C_1Y) \arrow[d, "(\pi^*)^{-1}"] \\ K^{-1}(Y) \arrow[r, "\delta"] & K(X,Y) \end{tikzcd}$$

The bottom square commutes by definition. In the top square, the map $\tilde{\pi}$ denotes the collapsing map $X\cup C_1Y\cup C_2X \to (X\cup C_1Y)/X$ which is invertible, as the quotiented out space C_2X is contractible. To see the commutativity in the top square notice how $\tilde{\pi}^*\circ s^* = (s\circ\tilde{\pi})^*$ and the maps m and $s\circ\tilde{\pi}$ coincide and hence the top square commutes. With $\psi_2 := \theta\circ(\tilde{\pi}^*)^{-1}$ and $\psi_3 := (\pi^*)^{-1}\circ\mathrm{id}$. For the left square, a natural choice is the homotopical equivalence given by quotienting the second space of the pair $\big((X\cup C_1Y)\cup C_2X, X\cup C_1Y\big)$ to get:

$$\Big((X\cup C_1Y)\cup C_2X\Big/X\cup C_1Y\Big) \cong (C_2X/X) \simeq S(X)\,.$$

We now have all the unknown maps, but we run into a problem, since the map l^* induces a map λ by the "wrong" inclusions (compared to i^* which is induced by the inclusion $C_2Y\to C_2X$, whereas we have C_1Y). This is depicted in the non-commutative diagram:

$$\begin{tikzcd} K\big((X\cup C_1Y)\cup C_2X, X\cup C_1Y\big) \arrow[r, "l^*"] \arrow[d, "\psi_1"] & \tilde{K}\big((X\cup C_1Y)\cup C_2X\big) \arrow[d, "\psi_2"] \\ \tilde{K}(C_2X/X)\cong\tilde{K}^{-1}(X) \arrow[r, "\lambda"] & \tilde{K}(C_1Y/Y)\cong\tilde{K}^{-1}(Y) \\ K(C_2X,X) \arrow[r, "i^*"] \arrow[u] & K(C_2Y,Y) \arrow[u] \end{tikzcd}$$

To resolve that problem we introduce the double cone on Y namely $C_1Y \cup C_2Y$, that fits into the commutative diagram where all unnamed maps are the natural ones:

$$\begin{tikzcd}
X \cup C_1Y \cup C_2X \arrow[rr, Rightarrow] \arrow[dd] & & C_1Y/Y \arrow[r, Rightarrow] & \Sigma Y \arrow[dd, dotted, "\Sigma T"] \\
& C_1Y \cup C_2Y \arrow[ul, Rightarrow] \arrow[ur, Rightarrow] \arrow[dl] \arrow[dr, Rightarrow] & & \\
C_2X/X & & C_2Y/Y \arrow[ll, "i"] \arrow[r, Rightarrow] & \Sigma Y
\end{tikzcd}$$

The inclusion induced map i is the one, that induces the bottom left map in Diagram 11. The double arrows always induce isomorphisms in the K-rings. To see that all those maps induce isomorphism we note that we always, with one exception, collapse something contractible. The exception is the map $C_1Y \cup C_2Y \to X \cup C_1Y \cup C_2X$. Here, we get that the induced map is invertible, by noticing the commutative triangle on the top, where two maps induce isomorphisms and thereby also the third. The map ΣT is one we want to construct below. For said construction we formulate the following Lemma:

Lemma 3.5.12. *Let Y be a well-pointed space and denote the points of CY and ΣY by (x,t) such that CY has the apex $[\{0\} \times Y]$ and base $\{1\} \times Y \cong Y$. Let*

$$D := C_1Y \cup C_2Y = \left(C_1Y \sqcup C_2Y\right)\Big/\left((x,1)_1 \sim (x,1)_2\right)$$

denote the double cone glued along a common base. Now define the maps

- $c_i : D \to C_iY\big/Y$ *the collapse of the complementary cone C_jY $j \neq i$.*
- $\overline{\pi_1} : C_iY\big/Y \to \Sigma Y \quad [(x,t)] \mapsto (x,t)$.
- $p_i := \overline{\pi_1} \circ c_i : D \to \Sigma Y$.
- $R : D \to D \quad (x,t)_i \mapsto (x,t)_j$ *where $j \neq i$, the cone swap.*
- $\Sigma T : \Sigma Y \to \Sigma Y\,, \quad (x,t) \mapsto (x, 1-t)$ *the suspension flip.*
- $\Phi : D \to \Sigma Y\,, \quad (x,t)_1 \mapsto (x, 1-\frac{t}{2}), \quad (x,t)_2 \mapsto (x, \frac{t}{2})$.

Then we have that

$$p_2 = p_1 \circ R \quad \text{and} \quad \Phi \circ R = \Sigma T \circ \Phi\,.$$

Proof of Lemma 3.5.12. The first statement $p_2 = p_1 \circ R$ is clear, as:

$$\begin{aligned}
p_2(x,t)_2 &= [(x,t)] = p_1(x,t)_1 = p_1 \circ R(x,t)_2\,, \qquad \text{and} \\
p_2(x,t)_1 &= [\{1\} \times Y] = p_1(x,t)_2 = p_1 \circ R(x,t)_1\,.
\end{aligned}$$

Similarly, the second statement is also just a calculation:

$$\Sigma T \circ \Phi(x,t)_1 = \Sigma T(x, 1 - \frac{t}{2}) = (x, \frac{t}{2}) = \Phi(x,t)_2 = \Phi \circ R(x,t)_1$$
$$\Sigma T \circ \Phi(x,t)_2 = \Sigma T(x, \frac{t}{2}) = (x, 1 - \frac{t}{2}) = \Phi(x,t)_1 = \Phi \circ R(x,t)_2 \,.$$

□

By the lemma above, we know that ΣT makes the diagram commute and by Lemma 3.4.16 we know that

$$(\Sigma T)^* : \tilde{K}(SY) \to \tilde{K}(SY)$$

is multiplication by minus one. Hence, the diagram where all maps are either topologically induced or topologically induced and inverted:

$$\begin{tikzcd}
\tilde{K}(X \cup C_1 Y \cup C_2 X) \arrow[rr, "\psi_2"] \arrow[dr] & & \tilde{K}(C_1 Y / Y) \arrow[r] & \tilde{K}^{-1}(Y) \\
& \tilde{K}(C_1 Y \cup C_2 Y) \arrow[dr] & & \\
\tilde{K}^{-1}(X) \arrow[uu, "\rho"] \arrow[rr, "i^*"] & & \tilde{K}(C_2 Y / Y) \arrow[r] & \tilde{K}^{-1}(Y) \arrow[uu, Rightarrow, "(-1)^*"']
\end{tikzcd}$$

is commutative. By construction (just composition of collapsing maps) we have that

$$l^* = \rho \circ \psi_1 \,, \quad \text{and by the diagram} \quad \psi_2 \circ \rho = -i^* \,.$$

Which gives us in total:

$$i^* \circ \psi_1 = -\psi_2 \circ \rho \circ \psi_1 = -\psi_2 \circ l^*$$

and thereby the diagram commutes up to sign:

$$\begin{tikzcd}
K\big((X \cup C_1 Y) \cup C_2 X, X \cup C_1 Y\big) \arrow[r, "l^*"] \arrow[d, "\psi_1"] & \tilde{K}\big((X \cup C_1 Y) \cup C_2 X\big) \arrow[r, "s^*"] \arrow[d, "\psi_2"] & \tilde{K}(X \cup C_1 Y) \arrow[d, "\psi_3"] \\
\tilde{K}^{-1}(X) \arrow[r, "i^*"] & \tilde{K}^{-1}(Y) \arrow[r, "\delta"] & K(X, Y)
\end{tikzcd}$$

The exactness in the top row now proves exactness in the bottom. □

Corollary 3.5.13. If (X, Y) is a compact pair such that $Y \subseteq X$ is a retract, i.e. the inclusion has a continuous up to homotopy left inverse. Then for all $n \geq 0$, the sequence

$$K^{-n}(X, Y) \xrightarrow{j^*} K^{-n}(X) \xrightarrow{i^*} K^{-n}(Y)$$

is a splitting short exact sequence giving us an isomorphism

$$K^{-n}(X) \cong K^{-n}(X, Y) \oplus K^{-n}(Y) \,.$$

This splitting is naturally dependent on the retraction map.

Corollary 3.5.14. Let X, Y be objects in $\mathbf{CWH}^*_{\mathbf{cof}}$. Then we have the projection maps

$$\pi_X : X \times Y \to X\,, \quad \pi_Y : X \times Y \to Y\,.$$

Those maps induce isomorphisms for all $n \geq 0$:

$$\tilde{K}^{-n}(X \times Y) \cong \tilde{K}^{-n}(X \wedge Y) \oplus \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y)$$

Proof. X is a retract of $X \times Y$ as the projection yields a continuous left inverse of the natural inclusion, and Y is a retract of $(X \times Y)/X$ (for the quotient to be natural, we include X at the base point of Y). With this we apply corollary 3.5.13 twice:

$$\begin{aligned}\tilde{K}^{-n}(X \times Y) \;&\cong \tilde{K}^{-n}(X \times Y/X) && \oplus \tilde{K}^{-n}(X)\\ &\cong \tilde{K}^{-n}(Y) \oplus \underbrace{\tilde{K}^{-n}((X \times Y/X)/Y)}_{\tilde{K}^{-n}(X \wedge Y)} && \oplus \tilde{K}^{-n}(X)\end{aligned}$$

□

Corollary 3.5.15. Let's inspect the isomorphism induced by the splitting: Since we apply the splitting twice, we know that the final isomorphism

$$\Phi : \tilde{K}^{-n}(X \times Y) \to \underbrace{\tilde{K}^{-n}((X \times Y/X)/Y)}_{\tilde{K}^{-n}(X \wedge Y)} \oplus \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y)$$

is of the form

$$\Phi = (\, g \circ l\,,\, \tilde{i_Y}^{\,*} \circ l\,,\, i_X{}^*\,)\,,$$

where:

- $i_Y : Y \to X \times Y$
- $\tilde{i}_Y : Y \to (X \times Y/X)$.
- $j : (X \times Y) \to (X \times Y/X)$.
- $l : \tilde{K}^{-n}(X \times Y) \to \tilde{K}^{-n}(X \times Y/X)$ is the unique left inverse of j^* such that $j^* \circ l = \mathrm{id} - \pi_X^* \circ i_X^*$.
- $g : \tilde{K}^{-n}(X \times Y\big/X) \to K(X \times y\big/X, Y)$ is a left inverse of the map $K(X \times Y\big/X, Y) \to \tilde{K}^{-n}(X \times Y\big/X)$.

We now want to show that $\tilde{i_Y}^{\,*} \circ l$ is induced by the inclusion of Y into $X \times Y$.

We have that $\tilde{i}_Y = j \circ i_Y$ and with this we can deduce:

$$\tilde{i_Y}^{\,*} \circ l = i_Y{}^* \circ j^* \circ l = i_Y{}^*\big(\mathrm{id} - \pi_X^* \circ i_X^*\big) = i_Y^* - i_Y{}^* \circ \pi_X^* \circ i_X^* = i_Y^* - \underbrace{(\pi_X \circ i_Y)}_{\text{constant}}{}^* \circ i_X{}^* = i_Y^*$$

Hence, in the proof above the isomorphism has the form:

$$\begin{aligned}\Phi : \tilde{K}^{-n}(X \times Y) \to \tilde{K}^{-n}(X \wedge Y) \oplus \tilde{K}^{-n}(Y) \oplus \tilde{K}^{-n}(X)\\ \Phi = g \oplus i_Y^* \oplus i_X^*\end{aligned}$$

or said differently, the short exact sequence that splits looks like:

$$0 \longrightarrow \tilde{K}^{-n}(X \wedge Y) \xrightarrow{q^*} \tilde{K}^{-n}(X \times Y) \xrightarrow{(i_X)^* \oplus (i_Y)^*} \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y) \longrightarrow 0$$

Before we start constructing a cohomology theory, we extend the K groups to be defined not only over the negatives, but over the whole $\mathbb{Z}$ which is done via an extension. The extension relies on periodic behaviour. So-called Bott periodicity.

Definition 3.5.16 (External Product)**.** The **external product** $\mu : K(X) \otimes K(Y) \to K(X \times Y)$ is defined via the two projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ by:

$$\begin{aligned} \mu : K(X) \otimes K(Y) &\to K(X \times Y) \\ (a \otimes b) &\mapsto (\pi_X)^*(a) \cdot (\pi_Y)^*(b) , \end{aligned}$$

where $\cdot$ denotes the multiplication in the K-group, which was induced by the tensor product. This μ is a ring homomorphism, since:

$$\begin{aligned} \mu((a \otimes b)(c \otimes d)) &= \mu(ac \otimes bd) \\ &= \pi_X^*(ac)\pi_Y^*(bd) \\ &= \pi_X^*(a)\pi_X^*(c)\pi_Y^*(b)\pi_Y^*(d) \\ &= \pi_X^*(a)\pi_Y^*(b)\pi_X^*(c)\pi_Y^*(d) \\ &= \mu(a \otimes b)\mu(c \otimes d) \end{aligned}$$

Proposition 3.5.17. Let (X, x_0) and (Y, y_0) be well-pointed compact spaces and let $q: X \times Y \to X \wedge Y$ denote the quotient map that appears in the exact sequence of Corollary 3.5.15. Then:

1. For $a \in \tilde{K}(X)$ and $b \in \tilde{K}(Y)$ we have $\mu(a \otimes b) \in q^* \tilde{K}(X \wedge Y)$. Since q^* is injective, this defines the **reduced external product**

$$\tilde{\mu} := (q^*)^{-1} \circ \mu|_{\tilde{K}(X) \otimes \tilde{K}(Y)} : \tilde{K}(X) \otimes \tilde{K}(Y) \to \tilde{K}(X \wedge Y) .$$

2. With respect to the decompositions

$$\begin{aligned} K(X) \otimes K(Y) &\cong \big(\tilde{K}(X) \otimes \tilde{K}(Y)\big) \oplus \tilde{K}(X) \oplus \tilde{K}(Y) \oplus \mathbb{Z} , \\ K(X \times Y) &\cong q^* \tilde{K}(X \wedge Y) \oplus \pi_X^* \tilde{K}(X) \oplus \pi_Y^* \tilde{K}(Y) \oplus \mathbb{Z} \cdot 1_{X \times Y} \end{aligned}$$

the external product is block diagonal:

$$\mu = (q^* \circ \tilde{\mu}) \oplus \pi_X^*|_{\tilde{K}(X)} \oplus \pi_Y^*|_{\tilde{K}(Y)} \oplus \mathrm{id}_{\mathbb{Z}} .$$

In particular μ is an isomorphism if and only if $\tilde{\mu}$ is.

Proof. (i): Let $i_X : X \to X \times Y$, $x \mapsto (x, y_0)$, and i_Y analogously. Then

$$i_X^* \mu(a \otimes b) = i_X^*\big(\pi_X^*(a) \cdot \pi_Y^*(b)\big) = (\underbrace{\pi_X \circ i_X}_{= \mathrm{id}_X})^*(a) \cdot (\underbrace{\pi_Y \circ i_X}_{= \text{constant}})^*(b) = 0 ,$$

since $b \in \ker\left(i_{y_0}^*: K(Y) \to K(\{y_0\})\right)$ and the constant map $\pi_Y \circ i_X$ factors through $i_{y_0}: \{y_0\} \hookrightarrow Y$. Analogously $i_Y^* \mu(a \otimes b) = 0$, so by exactness of the sequence from Corollary 3.5.15

$$\mu(a \otimes b) \in \ker\left((i_X^*, i_Y^*)\right) = \operatorname{im}(q^*) = q^* \tilde{K}(X \wedge Y)\,.$$

(ii): The first decomposition arises by tensoring the splittings $K(X) \cong \tilde{K}(X) \oplus \mathbb{Z} \cdot 1_X$, the second by combining the analogous splitting of $K(X \times Y)$ with the splitting $\pi_X^* \oplus \pi_Y^*$ of the exact sequence from Corollary 3.5.15. On the first summand μ equals $q^* \circ \tilde{\mu}$ by (i) and the definition of $\tilde{\mu}$. Further $\mu(a \otimes 1_Y) = \pi_X^*(a) \cdot 1_{X \times Y} = \pi_X^*(a)$, and $\pi_X^*|_{\tilde{K}(X)}$ is an isomorphism onto $\pi_X^* \tilde{K}(X)$. The third summand is symmetric, and $\mu(1_X \otimes 1_Y) = 1_{X \times Y}$. Finally, a direct sum of homomorphisms is an isomorphism if and only if each block is one, and $q^* \circ \tilde{\mu}$ is an isomorphism onto $q^* \tilde{K}(X \wedge Y)$ if and only if $\tilde{\mu}$ is one. □

By example 3.4.17 we know that for any line bundle l, $[l \otimes l \oplus \underline{1}] = [l]^2 + [\underline{1}] = [l] + [l]$ which is equivalent to $([l] - 1)^2 = [l^2] - [l] + [l] + [\underline{1}] = 0$. Now $\{[\underline{1}], [l]\}$ is an additive basis of $\mathbb{Z}/([l]^2 - [\underline{1}])$. The mapping that is defined on a basis of the tensor product as follows

$$\begin{aligned} \left(\mathbb{Z}([l])\big/([l]^2 - [\underline{1}])\right) \otimes K(X) &\to K(S^2 \times X) \\ [\underline{1}] \otimes [\xi] &\mapsto \mu([\underline{1}], [\xi]) \\ [l] \otimes [\xi] &\mapsto \mu([l], [\xi]) \end{aligned}$$

is well defined.

Theorem 3.5.18 (The Fundamental Product Theorem)**.** *For any compact X, and the tautological line bundle H on $\mathbb{CP}^1 \cong S^2$, the mapping*

$$\left(\mathbb{Z}([H])\big/([H]^2 - [\underline{1}])\right) \otimes K(X) \to K(S^2 \times X)$$

is an isomorphism of rings.

Proof. The proof is omitted here. There are elementary proofs given in Chapter 2.1 [Hat02] or Chapter 2.3 [Ati89]. □

Corollary 3.5.19. If we take X to be a point, then Theorem 3.5.18 implies that

$$\Phi : \left(\mathbb{Z}([H])\big/([H]^2 - [\underline{1}])\right) \to K(S^2)$$

is an isomorphism. Hence, $K(S^2)$ is generated by $\{[\underline{1}], [H]\}$. Furthermore together with the map

$$\operatorname{rank} : \left(\mathbb{Z}([H])\big/([H]^2 - [\underline{1}])\right) \to \mathbb{Z}$$

that maps the generators to their rank, and

$$\nu : \mathbb{Z} \to K(pt); \,, \quad n \mapsto [\underline{n}]$$

we get the commutative diagram:

$$\begin{tikzcd} \mathbb{Z}([H])\big/([H]^2-[\underline{1}]) \arrow[r, "\mathrm{rank}"] \arrow[d, "\Phi"] & \mathbb{Z} \arrow[d, "\nu"] \\ K(S^2) \arrow[r, "i_{pt}^*"] & K(pt) \end{tikzcd}$$

By the commutativity, we get an isomorphism induced by Φ:

$$\tilde{K}(S^2) = \ker(i_{pt}^*) \cong \ker(\mathrm{rank}) = \mathrm{span}_{\mathbb{Z}}([H]-[\underline{1}])$$

Theorem 3.5.20 (Bott periodicity)**.** *Let X be a well-pointed compact space. The isomorphism from the fundamental product theorem applied in its reduced form gives a natural isomorphism:*

$$\beta_X : \tilde{K}^{-n}(X) \to \tilde{K}^{-n}(S^2 \wedge X) \cong \tilde{K}^{-n-2}(X)$$

Proof. We start with $n = 0$. By Proposition 3.5.17 we know that the reduced external product applied to the spaces $\tilde{K}(S^2) \cong \mathrm{span}_{\mathbb{Z}}([H]-[\underline{1}])$ and $\tilde{K}(X)$ is an isomorphism due to the non-reduced external product being one, by the fundamental product theorem. Hence we have that

$$\mathbb{Z}([H]-[\underline{1}]) \otimes \tilde{K}(X) \to \tilde{K}^{-2}(X)$$

is an isomorphism. And since $\tilde{K}(X) \cong \mathbb{Z}([H]-[\underline{1}]) \otimes \tilde{K}(X)$ via $[\xi] \mapsto ([H]-[\underline{1}]) \otimes [\xi]$ we have the isomorphism

$$\beta_X : \tilde{K}^{-2}(X) \otimes \tilde{K}(X), \quad [\xi] \mapsto \tilde{\mu}([H]-[\underline{1}], [\xi]).$$

Naturality follows from μ being natural, q^* being natural (and thereby its inverse) and $\tilde{\mu}$ being the composition of both of them. Furthermore, for Replacing X with $S^n X$ proves the lower degree. ☐

natürlich im zweiten faktor mit festem ersten

Definition 3.5.21. The Bott periodicity mapping β_X is natural and hence commutes with all maps in the long exact sequence. With this map we can expand the notion inductively via $K^{-n}(X) = K^{-n-2}(X)$. Said differently, we define $K^n(X)$ to be $K(X)$ for even n and $K^{-1}(X)$ for odd n. With this we have the exact sequence of six terms (in the top row we identified $K^0 \equiv K^{-2}$ and the vertical map on the left is $\beta \circ \delta$):

$$\begin{tikzcd} K^0(X,Y) \arrow[r] & K^0(X) \arrow[r] & K^0(Y) \arrow[d] \\ K^1(Y) \arrow[u] & K^1(X) \arrow[l] & K^1(X,Y) \arrow[l] \end{tikzcd}$$

We now define the whole long exact sequence to be the one from Theorem 3.5.11 for negative indices and the sequence of six terms above, for positive indices. With this we have gained a cohomology theory over all whole numbers.

4 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse theory, we will always deal with a smooth function $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 4.0.1 (Critical Points)**.** A point $p \in M$ at which the smooth map $\mathrm{d}f_p = 0$ is called **critical**. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 4.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear form

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. We then define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = \Xi\left(Z(f)\right)(p)$$

See 2.1.41 for the notation of applying a function to a vector field. In corollary 4.0.5 we will give a local description telling us that this definition is well defined, meaning independent of the extensions Ξ and Z.

Definition 4.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point, the Hessian is a symmetric bilinear and non-degenerate form.

For such a form there exists the following theorem:

Theorem 4.0.4 (Sylvester's Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that:*

$$T^\top \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^\top \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^\top \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^\top \circ A \circ T') = k' + l'\,. \tag{12}$$

Hence it suffices to show that $k = k'$. Which we will do by proving the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^\top A x > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing "$\leq$" : Denote the first k columns of T with $x_1, ..., x_k$. They form a basis of the subspace $\mathbb{R}^k \subseteq \mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda_i x_i$ we have that by bilinearity

$$x^\top Ax = \sum_{i=1}^k \lambda_i x_i^\top Ax = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^\top Ax_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{13}$$

This concludes the first inequality.

Now let U be any k-dimensional subspace such that for all non zero $x \in U$, $x^\top Ax > 0$. By a calculation analogous to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots, x_n)$ the number $x^\top Ax$ is less than or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k\,. \tag{14}$$

This completes the proof. $\square$

Corollary 4.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

Proof. This follows by direct calculation of $\operatorname{Hess}_p(f)\,(\partial_i(p), \partial_j(p))$. Denote $x := \varphi(p)$, then:

$$\begin{aligned}
\Big(\partial_i(p)\,(\partial_j(f))_p\Big) &= \left(\partial_i(p) \left(p \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_{\varphi(p)} \right)_p \right) \\
&= \left.\frac{\partial}{\partial x_i} \left(y \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_y \right)\right|_x \\
&= \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\,.
\end{aligned}$$

$\square$

Remark 4.0.6. If we have two different charts, then the matrix $M_p(f)$ of the Hessian transforms from one basis to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = (\mathrm{d}\varphi_{12})^\top \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 4.0.7. By Sylvester's Law of inertia, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is an invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each critical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted by λ_p. If omitting the function would cause ambiguity, we write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

Theorem 4.0.8 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse function $f : M \to \mathbb{R}$. There exists a centred chart $\varphi : U \to \mathbb{R}^m$ around p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2 \,.$$

Proof. We refer to Lemma 3.11 in [BH04] for a proof. □

Definition 4.0.9. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x))\,, \quad \text{and}$$
$$\varphi_0(x) = x\,.$$

We call that group the **negative gradient flow** of f with respect to the metric g. The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of the negative gradient. Notice how the Morse lemma is a statement about smooth manifolds, no metric needed nor mentioned. In fact there are many Morse charts and we can find one where the Riemannian metric in p is diagonal with non-zero entries. To prove such a statement we first need a somewhat obvious lemma:

Lemma 4.0.10. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the tangent bundle in a point p that come from a basis in said point induced from a chart χ by q_χ. Then the diagram*

$$\begin{tikzcd} T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\ & \mathbb{R}^m \end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1}\circ\psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_pM$:

$$\begin{aligned}
\left[(q_{\psi\circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f\circ\psi^{-1}\circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f\circ\psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p)\cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p)\cdot\lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\big(\sum_{i,j} v_i\lambda_{ji}e^j\big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can prove that a Morse chart exists where the gradient is of a particularly easy form:

> **Theorem 4.0.11** (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ around p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a diagonal matrix* $\operatorname{diag}(\mu_1,\dots\mu_m)$ *with $\mu_i > 0$ for all i.*

Proof. The quadratic form of $f\circ\psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T Hv$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T Gv$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precise this means for $v, w \in \mathbb{R}^m$:

$$g\circ(q_\psi\big|_p \oplus q_\psi\big|_p)(v,w) = v^T Gw \text{ and } f\circ\psi^{-1}(v) = f(p) + v^T Hv\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalised eigenvalue problem $Hv = \lambda Gv$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1,\dots,\lambda_m$ and corresponding eigenvectors $v_1,\dots,v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram–Schmidt to make the bases of the eigenspaces orthogonal and, by rescaling, orthonormal.

Let $L = [v_1|\dots|v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$q_G(y) = y^T G y = (Lz)^T G (Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^{m} z_i^2$$
$$q_H(y) = y^T H y = (Lz)^T H (Lz) = z^T L^T H L z$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive). This is true by Sylvester's law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0\ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p\right)$. By the lemma 4.0.10

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 4.0.12. We call a Morse chart where the matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulas around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 4.0.13 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\mathrm{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \ldots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\mathrm{grad}(f))_\psi(x) := J\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial (q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij} .$$

Lemma 4.0.14. *Let ψ be a coordinate system around a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \ldots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM. Then the Jacobian of the gradient reads*

$$J(\mathrm{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} , \tag{15}$$

where g^{ij} is defined as in definition 2.1.45.

Proof. Let q_ψ denote the coordinate function $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \ldots, \frac{\partial}{\partial x_m}\big|_x\right)$. Then the gradient in ψ reads:

$$q_\psi \circ \mathrm{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \ldots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j \, \partial x_k}\right]\right)_{ij} . \end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \, \partial x_k}\right]\right)_{ij} .$$

□

Now that we know how the gradient locally looks, it is reasonable to assume that the flow locally looks like its exponential:

Lemma 4.0.15. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ is a critical point. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{dt}}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{dt}}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceding Lemma 4.0.14 now finishes the proof.

□

Hence in a refined Morse chart we have:

Remark 4.0.16. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}
\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\mathrm{grad}(f))(p)\cdot t) \\
&= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}}\cdot t \right) \\
&= \mathrm{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p} t},-e^{2\mu_{\lambda_p+1} t},\dots,-e^{2\mu_m t}\right)
\end{aligned}$$

In a refined Morse chart we have that the (un-)stable manifold embedding:

Definition 4.0.17 (Stable and Unstable Manifold)**.** Let $p \in M$ be a critical point of $f : M \to \mathbb{R}$. Then we define:

1. The **stable set** of p as
$$W(\to p) := \{x \in M | \lim_{t\to\infty} \varphi_t(x) = p\}.$$
That is the points that flow towards p.

2. The **unstable set** of p as
$$W(p \to) := \{x \in M | \lim_{t\to-\infty} \varphi_t(x) = p\}.$$
That is the points that came from p.

Those sets will turn out to be manifolds, which explains the names **(un-)stable manifolds**.

Theorem 4.0.18 (Stable and unstable manifold theorem)**.** *Let $f : M \to \mathbb{R}$ be a smooth function on a smooth compact Riemannian manifold (M, g). Let p be a critical point of f. Then the tangent space splits as:*

$$T_pM = T_p^s(M) \bigoplus T_p^u M$$

such that the Hessian is positive definite on $T_p^s M$ and negative definite on $T_p^u M$. Moreover, the stable and unstable manifolds are submanifolds and images of smooth embeddings:

$$\begin{aligned}
E_p^s : T_p^s M &\to W(\to p) \subset M, \\
E_p^u : T_p^u M &\to W(p \to) \subset M.
\end{aligned}$$

Proof. I will summarise the proof given in Chapter 4.3 [BH04]. We work in the discrete setting meaning that we take the negative gradient flow φ_t for a fixed positive t and the limit-behaviour of the flow is captured by $\lim_{n\to\infty}\varphi_t^n$ instead of the t-limit. First, one shows the splitting of spaces locally in a small neighbourhood U around a critical point p and then extends the splitting globally. Around U we already know by remark 4.0.16 that the tangent space T_pM splits into $T_p^sM \oplus T_p^uM$ such that the differential of the flow is a contraction around T_p^sM and an expansion in T_p^uM. We can now map a small neighbourhood U around p into T^pM using the map $\Psi : U \to T_pM$ given by a refined Morse chart and the identification of $\mathbb{R}^n$ with T_pM via $e_i \mapsto \partial_i(p)$, where the latter uses the same refined Morse chart. Now inspect the image of the stable manifold $W(\to p)$ under said map. This looks to be a small perturbation of the subspace T_p^sM and the local stable manifold theorem supports that intuition:

Lemma 4.0.19 (Local (Un-)stable Manifold Theorem)**.** *We define:*

$$T := \mathrm{d}\varphi_t\big|_p, \quad T_s := T\big|_{T_p^sM}, \quad T_u := T\big|_{T_p^uM}.$$

Let $\lambda < 1$ such that $\|T_s\| < \lambda$ and $\left\|T_u^{-1}\right\| < \lambda$, where $\|T\| = \sup\{T(x)|\,\|x\| \leq 1\}$ denotes the operator norm. Now for this setting there exists an ε_λ depending only on λ, and there exists a $\delta > 0$ for every $r > 0$ which satisfy: For all $\theta : T_p^sM \times T_p^uM \to T_pM$ that satisfies $\theta - T$ is Lipschitz with Lipschitz constant

$$\mathrm{Lip}(\theta - T) < \varepsilon, \text{ and } \|\theta(0)\| < \delta.$$

In this setting,

$$W_r^s(\theta) := \{x \in T_pM \mid \forall n \geq 0, \theta^n(X) \text{ is defined and } \|\theta^n(x)\| \leq r\}$$

is the graph of a Lipschitz function $g : T_p^sM \to T_p^uM$ with Lip$(g) \leq 1$ and:

1. *If θ is C^k (k-times continuously differentiable), then so is g.*
2. *If θ is smoothly differentiable, $\theta(0) = 0$ and $\mathrm{d}\theta\big|_0 = T$, then $g(0) = 0$ and $\mathrm{d}g|_0 = 0$. Hence, $T_pW_r^s(\theta) = T_p^sM$.*

Proof. For a proof we refer to Chapter 4.2 [BH04]. □

We can now define $\theta := \Psi \circ \varphi \circ \Psi^{-1}$. The local (un-)stable manifold theorem now tells us that $W_r^s(\theta)$ is the graph of a smooth function $g : T_p^sM \to T_p^uM$. Furthermore, it is easy to see that in a small neighbourhood of the origin we have the the equiality of $W_r^s(\theta)$ and $Psi(W(\to p) \cap U)$. Hence the projection π of said graph to its domain gives us a chart of the stable manifold around p. A whole atlas of $W(\to p)$ is obtained by $U^n := \varphi^{-n}(U)$ which covers the whole $W(\to p)$ and with this the maps $\chi^n : \pi \circ \Psi \circ \varphi^n$ give us an atlas. The smooth embeddings are obtained as follows: We define $h := \chi \circ \varphi \circ \chi^{-1}$ and $B := W(\to p) \cap U$. Now $\mathrm{d}\big(\,\varphi_p|_{T_p^sM}\,\big)$ having an operator norm less than one translates to $\mathrm{d}h_p$ as they are similar as matrices. Let $0 < \alpha < 1$ and $B_0 \subseteq B$ be a ball centred at p such that $\|\mathrm{d}h_x\| \leq \alpha$ for all $x \in B_0$. Then $h|_{B_0}$ is a contraction, which can be extended to a global contraction $\tilde{h} : T_p^s M \to T_p^s M$. Now we define the embedding:

$$\begin{aligned} E^s : T_p^s M &\to W(\to p) \\ x \mapsto \varphi^{-n} \circ \chi^{-1} \circ \tilde{h}^n(x) &\quad n\text{such that } \tilde{h}^n(x) \in B_0 \,. \end{aligned}$$

All required properties and the well-definedness of E^s are now standard calculations. The stable manifold theorem follows from the unstable one by replacing f with $-f$. □

Remark 4.0.20. The proof also gives us the equality $T_p W(\to p) = T_p^s M$: near p, the stable manifold agrees with $\Psi^{-1}(W_r^s(\theta))$, and by part 2 of Lemma 4.0.19 (applicable since $\theta(0) = 0$ and $\mathrm{d}\theta|_0 = T$) the set $W_r^s(\theta)$ is the graph of a function g with $g(0) = 0$ and $\mathrm{d}g|_0 = 0$, hence has tangent space $T_p^s M$ at the origin.

We want to refine the unstable embedding $E_q^u : T_q^u M \to W(q \to)$. For this we will make use of the Hartman–Grobman conjugation theorem:

Theorem 4.0.21 (Hartman-Grobman Theorem). *Let E be an open subset of R^n containing the origin, let $f \in \mathcal{C}$, and let φ_t be a flow given by f. (i.e. $\frac{\partial}{\partial t}(\varphi_t(x)) = f(x)$). Assume that $f(0) = 0$, i.e. that 0 is a fixed point of φ_t and furthermore assume that fixed point to be hyperbolic, i.e. Df has no eigenvalue with zero real part. Then there exists a homeomorphism H of an open set U containing the origin onto an open set V containing the origin such that for each $x_0 \in I$, there is an open interval $I_0 \subseteq \mathbb{R}$ containing zero such that for all $x_0 \in U$ and $t \in I_0$ the diagram*

$$\begin{array}{ccc} U & \xrightarrow{H} & V \\ \downarrow{\scriptstyle \varphi_t} & & \downarrow{\scriptstyle \mathrm{e}^{At}} \\ U & \xrightarrow{H} & V \end{array}$$

commutes, meaning that locally the flow is topologically conjugate to its linearisation.

Proof. A proof is given in chapter 2.8 in [Per01]. □

Theorem 4.0.22 (Refined Stable Embedding). *In the setting of the stable manifold theorem, we can refine the embedding*

$$E_q^u : T_q^u M \to W(q \to) \,,$$

such that it resembles the flow, meaning that for $x = E_p^s(v)$ the flow is given by

$$\varphi_t(x) = E_p^s(A_t v)\,,$$

where A_t is a linear map defined as: $A_t = B_{\mathfrak{b}} \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \ldots, e^{-2\mu_{\lambda_p} t}\right) \circ B_{\mathfrak{b}}^{-1}$, where $\mathfrak{b}$ is the basis of $T_q^u M$ that comes from the basis of $T_q M$ which is induced by a refined Morse chart. The diagonal matrix is the stable part of the Jacobian of the flow in said refined Morse chart ψ.

Proof. In the proof of the unstable manifold theorem from [BH04] which was summarised above, we started with a local map $\rho' : U \to V$, where $U \subseteq W(q \to)$ is a neighbourhood of q and $V \supseteq T_q^u$. (This was a refined Morse chart together with a projection once realised that the stable part gets mapped to the graph of a smooth function). Now we define

$$\begin{aligned} \theta_t : V &\to V \\ v &\mapsto \rho' \circ \varphi_t \circ \rho'^{-1}(v) \end{aligned}$$

for all t such that the map is well defined. Now with a basis $\mathfrak{b}_p^s$ of $T_p^s M$ we can push these considerations into $\mathbb{R}^n$ and by abuse of notation assume that ρ maps to $\mathbb{R}^n$ and hence θ is a flow around the origin of $\mathbb{R}^n$. Now by the Hartman–Grobman Theorem we find a homeomorphism H around the origin, such that $H \circ \theta_t = e^{At} \circ H$, where $A = \mathrm{d}\theta_t =$ To calculate said differential we look into the map ρ' again. For this we give names to the functions:

- $\psi : U \to V$ is the refined Morse chart around q.
- $g : T_p^s M \to T_p^u M$ is the map that defines the image of the stable part under ψ to be a graph. I.e. $\psi\left(W(\to p)\right) = \Gamma_g$. We know that $g(0) = 0$ and $\mathrm{d}g\big|_0 = 0$.
- $\pi : \Gamma_g \to T_p^s M$ is the projection. Its inverse is obviously $\mathrm{id} \times g$.

We know from the local stable manifold theorem 4.0.19, that $\mathrm{d}g\big|_0 = 0$. With this we can calculate:

$$\begin{aligned} \mathrm{d}\theta_t\big|_0 &= \mathrm{d}\left(\pi \circ \psi \circ \varphi_t \circ \psi^{-1} \circ \pi^{-1}\right)\big|_0 \\ &= \mathrm{d}\pi\big|_{(0,0)} \circ \mathrm{d}\psi \circ \varphi_t \circ \psi^{-1}\big|_{0,0} \circ \mathrm{d}\,\mathrm{id} \times g\big|_0 \\ &= \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right) \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \ldots, e^{-2\mu_{\lambda_p} t}, e^{2\mu_{\lambda_q+1} t}, \ldots, e^{2\mu_m t}\right) \circ \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right)^T \\ &= \operatorname{diag}\left(e^{-2\mu_1 t}, \ldots, e^{-2\mu_{\lambda_p} t}\right) \end{aligned}$$

Now define for a small neighbourhood $V \subseteq \mathbb{R}^{m-\lambda_p}$ the map

$$\rho \coloneqq \rho' \circ H : V \to U\,.$$

U is chosen such that the map is onto. Then we have that $\varphi_t \circ \rho^{-1} = \rho^{-1} \circ e^{At}$, in that small neighbourhood of p. Now finally we can extend this property to be true for the whole stable set: For this we define the embedding

$$\begin{aligned} E_p^s : T_p^s M &\to W(\to p) \\ x &\mapsto \varphi_{-n} \circ \rho^{-1} \circ A_n(x) \quad \text{ where } n \in \mathbb{N} \text{ is big enough such that } A_n(x) \in V\,. \end{aligned}$$

This map is in fact well defined since

$$\begin{aligned} \varphi_n \circ \rho^{-1} \circ A_n(x) &= \varphi_{-n} \circ \varphi_{-1} \circ \varphi_1 \circ \rho^{-1} \circ A_n(x) \\ &= \varphi_{-n} \circ \varphi_{-1} \circ \rho^{-1} \circ A_{n+1}(x) \\ &= \varphi_{-(n+1)} \circ \rho^{-1} \circ A_{n+1}(x) \end{aligned}$$

And obviously the flow compatibility that is given locally by ρ is now true globally: Let $x \in W(\to p)$ and assume $x = E_p^s(y)$. Then:

$$\begin{aligned} \varphi_t(x) = \varphi_t(E_p^s(y)) &= \varphi_t \circ \varphi_{-n} \circ \rho^{-1} \circ A_n(y) \\ &= \varphi_{-n} \circ \varphi_t \circ \rho^{-1} \circ A_n(y) \\ &= \varphi_{-n} \circ \circ A_n(A_t(y)) \end{aligned}$$

The details of the proof are exactly the same as in the standard (un-)stable manifold theorem. □

Definition 4.0.23. We say that a flow line γ with respect to the negative gradient flow of a Morse function f starts in q, if $\lim_{t\to-\infty} \gamma(t) = q$, and ends in p, if $\lim_{t\to\infty} \gamma(t) = p$. We denote the space of flow lines starting in q by $\mathcal{M}(q \to)$, the ones ending in p by $\mathcal{M}(\to p)$ and the flow lines from q to p $\mathcal{M}(q \to p)$.

4.1 Orientations

The idea of Morse homology is now to find an invariant under manipulations of the Morse–Smale function. Obviously neither the number of critical points nor the number of flow lines stays invariant. However certain linear combinations stay invariant, when one establishes a sophisticated way to count flow lines. For this count we want to introduce signs to the flow lines between critical points with index one apart.

Definition 4.1.1 (Morse-Smale)**.** A Morse function $f : M \to \mathbb{R}$ such that all stable and unstable manifolds intersect transversally is called a **Morse-Smale** function.

Definition 4.1.2 (Orientation)**.** Let V be a finite dimensional $\mathbb{R}$ vector space and $\det(V) := \Lambda^{\dim(V)}$ the maximal exterior algebra (or Grassmann algebra). By the principle of permanence we define $\Lambda^0(V) = \mathbb{R}$. An orientation of V is a choice of one of the two connected components of $\det(V) \setminus \{0\}$.

Remark 4.1.3. Classically, an orientation of a vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1, \ldots b_m)$ of V and hence the product $b_1 \wedge \cdots \wedge b_m$ is an element of the maximal exterior product $\det(V) = \Lambda^m V$. Two ordered bases define different elements in $\det(V)$ and their difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence a choice of equivalence class of bases modulo base change matrix of positive determinant is the same as a choice of one connected component of $\det(V) \smallsetminus \{0\}$.

Definition 4.1.4 (Orientation and Exact Sequences)**.** A short exact sequence of vector spaces FIX

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

induces a canonical isomorphism

$$\begin{gathered} \det(A) \otimes \det(C) \to \det(B) \\ (a_1 \wedge \cdots \wedge a_l) \otimes (c_1 \wedge \cdots \wedge c_k) \mapsto (f(a_1) \wedge \ldots f(a_l) \wedge \tilde{c_1} \wedge \cdots \wedge \tilde{c_k}), \end{gathered}$$

where $\tilde{c_j} \in g^{-1}(c_j)$. The well-definedness of this map (invariance under chosen lift of c) can be calculated, since two different choices differ in an element in the kernel of g which equals the image of f and hence the different choices vanish in the wedge product. By this isomorphism we get an orientation induced in the middle from the outside. Notice the convention, that the basis from A is placed in front of the basis from C. Notice how the definition is consistent with the edge cases of trivial vector spaces A or C. If $A = 0$ we get an isomorphism $B \cong C$ and a basis from C induced to B, that is either the induced one from g, or exactly the opposite one, if the trivial vector space A is negatively oriented.

Remark 4.1.5. Let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be a short exact sequence of vector spaces and $\mathfrak{a}$ be a basis of A, $\mathfrak{c}$ be a basis of C and $\tilde{\mathfrak{c}}$ be a lift of $\mathfrak{c}$ via g. Replacing the basis $\mathfrak{a}$ with $P\mathfrak{a}$ and $\mathfrak{c}$ with $Q\mathfrak{c}$ where P, Q are invertible matrices, gives us a new induced basis on B, which differs from the original one by a matrix $\begin{pmatrix} P & * \\ 0 & Q \end{pmatrix}$, where $*$ depends on the chosen lift. Hence the corresponding elements in $\det(B)$ differ by the product $\det(P)\det(Q)$.

Lemma 4.1.6. *For a rank r vector bundle $E \to X$ we set $\det(E) := \Lambda^r(E)$. An orientation is an equivalence class of a non-vanishing section into the determinant bundle, where two sections are equivalent if they differ by multiplication of a positive scalar valued function. E is orientable if such a section exists, i.e. if* det(E) *is trivial. An orientation of a bundle restricts to an orientation in each fibre.*

Lemma 4.1.7. *Let $E \to X$ be an orientable vector bundle and X be connected. For each $x \in X$ we get the bijection induced from the restriction to the fibre:*

$$\{\textit{Orientation on } E\} \to \{\textit{Orientation on } E_x\}.$$

Hence, an orientation in one fibre can introduce an orientation in every other fibre.

Proof. Two orientations over E are given as nowhere-vanishing sections s, s'. The quotient $x \mapsto \frac{s}{s'}$ is continuous and its image is contained in $\mathbb{R} \setminus \{0\}$. Hence the sign of the quotient is a continuous function into $\{-1, 1\}$. Since X is connected we conclude that the sign of said quotient is a constant function. Thereby, if the sections induce the same orientation, they differ by multiplication of the positive number $\frac{s(x)}{s'(x)}$ and then the sign of $\frac{s}{s'}$ is globally one. The surjectivity is clear by mapping the sections s and $-s$. □

Proposition 4.1.8. Let

$$0 \longrightarrow E' \xrightarrow{f} E \xrightarrow{g} E'' \longrightarrow 0$$

be a short exact sequence of orientable vector bundles. Then the fibrewise isomorphisms $\det(E'_p) \otimes \det(E''_p) \to \det(E_p)$ from the vector space case assemble into an isomorphism of line bundles

$$\det(E') \otimes \det(E'') \xrightarrow{\cong} \det(E).$$

In particular, orientations of E' and E'' induce a canonical orientation of E.

Proof. Fibrewise the map is the canonical isomorphism from the vector space case: a lift $\tilde{c} \in g_p^{-1}(c)$ exists since g_p is surjective, and the resulting element $f(a_1) \wedge \cdots \wedge f(a_l) \wedge \tilde{c}_1 \wedge \cdots \wedge \tilde{c}_k$ is independent of the choice of lift.

It remains to check that this bundle map is continuous (resp. smooth) to ensure that it maps sections smoothly into sections, which is a local statement: locally choose a frame $(a_1, \dots, a_l)$ of E' and a frame $(c_1, \dots, c_k)$ of E''; since g is an epimorphism it admits local sections, so we may choose local lifts $\tilde{c}_j$ of the c_j which are (smooth) local sections of E. Then the map is given in the induced local frames of the determinant bundles by wedging these sections, hence continuous (resp. smooth). By this, two non-vanishing sections get mapped to a non-vanishing section and hence, orientations on E' and E'' which are represented by sections give a orientation over E. □

Definition 4.1.9. We call a manifold orientable, if its tangent bundle is orientable.

4.2 Signed Flow Lines

Definition 4.2.1. Let $f : M \to \mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M, g). Furthermore let $\mathfrak{o}$ be a choice of orientations for all unstable tangent spaces ($T_q^u M$ for all $q \in \mathrm{Crit}(f)$). The tuple $\mathfrak{A} := (g, f, \mathfrak{o})$ is a **Morse datum**.

Proposition 4.2.2 (Oriented Unstable Manifolds)**.** All unstable manifolds of a manifold with Morse function are orientable which can be deduced from the (un-)stable embedding theorem. Hence, the orientations from a Morse datum $\mathfrak{A} := (g, f, \mathfrak{o})$ induce orientations in all unstable manifolds by enforcing Lemma 4.1.7.

Lemma 4.2.3. *Let M, N be smooth manifolds and $f : N \to M$ be a smooth embedding. Then the tangent bundle $T(f(N))$ of the smooth submanifold $f(N)$ is a smooth subbundle of $TM|_{f(N)}$.*

Proof. Write $S := f(N)$, and let $n = \dim N$ and $m = \dim M$. Since f is an embedding, S satisfies the local slice condition: around every $p \in S$ there is a chart $\varphi: U \to \mathbb{R}^m$ of M with $\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^n \times \{0\})$, and the restriction $\varphi|_{S \cap U} : S \cap U \to \mathbb{R}^n \times \{0\} \cong \mathbb{R}^n$ is a chart for S. With respect to these two charts, the inclusion $S \cap U \hookrightarrow U$ has the coordinate representation $v \mapsto (v, 0)$, hence for every $q \in S \cap U$ we have

$$T_q S \ = \ \operatorname{span}\big(\partial_1(q), \dots, \partial_n(q)\big) \ \subset \ T_q M,$$

where $\partial_1, \dots, \partial_m$ denotes the coordinate frame of φ. Hence the trivialisation of $TM|_U$ induced by φ maps $TS|_{S \cap U}$ onto $(S \cap U) \times (\mathbb{R}^n \times \{0\})$, i.e. TS is locally the standard subbundle. Moreover, any transition map between two slice charts sends slices to slices, so its Jacobian along S is block upper triangular; the transition functions of TS are the smooth upper-left $n \times n$ blocks. Thus TS is a smooth subbundle of $TM|_S$. □

Proposition 4.2.4. Let $\mathfrak{A} := (g, f, \mathfrak{o})$ be a Morse Datum. Then for all critical points p the quotient bundles $TM|_{W(\to p)} / TW(\to p)$ inherit canonical orientations from .

Proof. By lemma 4.2.3 we know that the bundel $TW(\to p)$ is a smooth subbundle of $TM|_{W(\to p)}$ and by Lemma 2.1.35 we know that the quotient bundle $TM|_{W(\to p)} / TW(\to p)$ is smooth. Finally, we have a natural vectorspace isomorphism

$$\Big(TM|_{W(\to p)} / TW(\to p) \Big)_p = T_p M / T_p W(\to p) \cong T_p^u M \,,$$

beacuse $T_p M \cong T_p^u M \oplus T_p^s M$ and by Remark 4.0.20 $T_p^s M = T_p W(\to p)$. This lets us induce an orientation in the quotient bundle over p and by Lemma 4.1.7 this orientation orients the whole bundle. □

Definition 4.2.5. Let $\mathfrak{A} := (g, f, \mathfrak{o})$ be a Morse Datum. Let q, p be two critical points with $\mathrm{ind}(q) - 1 = \lambda(p)$. Then we have for all $\gamma \in \mathcal{M}(q \to p$ the exact sequence of bundles

$$0 \longrightarrow T\gamma \xrightarrow{i} TW(q\to)|_\gamma \xrightarrow{p} TM\big/TW(\to p)\big|_\gamma \longrightarrow 0$$

where the surjectivity of p follows from the smale condition.

Remark 4.2.6 (Notiz für Mich)**.** Given a Morse datum $\mathfrak{A} := (g, f, \mathfrak{o})$, there are multiple ways to inflict a sign to a flow line $\gamma \in \mathcal{M}(q \to p)$ between two critical points with index one apart. All of them introduce two different orientations (meaning basis of $T_xW(q\to)$) in a point $x \in \gamma$ the first notion depends on the orientation T_q^u and is introduced either via the embedding of the unstable manifold of q E_p^u, or by trivialising the bundle $TW(q\to)$ which is possible by contractibility of $W(q\to)$. Both ways yield the same, as they introduce the same orientation in the fibre $T_qW(q\to)$. Now the difficulty appears when introducing a basis of $T_xW(q\to)$ that depends on T_p^uM. For this we can build the quotient bundle $TM\big/\big(TW(\to p)\big)$. The fibre above p should be naturally isomorphic to T_p^uM by the direct sum decomposition and contractibility of $W(\to p)$ then gives us a trivialisation and hence lets us induce an orientation over $T_xM\big/T_xW(\to p)$. By transversality we get the short exact sequence of vector spaces that introduces a basis in the middle.

$$0 \longrightarrow T_x\gamma \longrightarrow T_xW(q\to) \longrightarrow T_xM\big/T_xW(\to p) \longrightarrow 0$$

Another way to get the orientation in $T_xW(q\to)$ from T_p^uM is by transporting the orthogonal complement of $-\gamma_x(f)$ in $T_xW(q\to)$ towards p along the curve γ. Here we have the problem, that we will never reach T_pM as $p \notin \gamma$. To introduce an orientation that way we need to show that the tangent spaces converge into one another which is a corollary of the inclination or λ lemma.

Definition 4.2.7. Let $f : M \to \mathbb{R}$ be a Morse function and q, p two critical points and assume that $\operatorname{ind}(q) - 1 = \operatorname{ind}(p)$. Now take a point $x \in \gamma \in \mathcal{M}(q \to p)$ that lives on a flow line between those points. Then we define the quotient bundle $\nu W(\to p)$ over the stable set of p as $TM|_{W(\to p)}\big/T(W \to p)$.

Umbedingt Orientierungen Sortieren! Hier geht einiges schief aktuell

Definition 4.2.8 (Morse Homology)**.** Let $\mathfrak{A} := (g, f, \mathfrak{o})$ be a Morse datum. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M, \mathfrak{A}, \Lambda) := \bigoplus_{\operatorname{Crit}(f)} \Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q \to p)$. For each $\gamma \in \mathcal{M}(q \to p)$ we assign a sign $n_\gamma \in \{\pm 1\}$. Let $x \in \gamma$. Then the tangent space in x splits $T_xW(q\to) = N_x(\gamma) \oplus \langle -\operatorname{grad}(f)(x)\rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to T_p^uM. (To be precise, we transport alongside the compactification of γ such that it contains p. This compactified version of γ has T_p^uM as the fibre above p which can be seen by inspecting it locally and applying the refined unstable embedding.) Hence, the orientation in $T^u_q M$ can be compared to the one in $T^u_p M$ and the sign n_γ can be chosen accordingly. The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q) := \sum_{p \in \mathrm{Crit}_k(f)} \sum_{\gamma \in \mathcal{M}(q \to p)} n_\gamma$$

Theorem 4.2.9. *The Morse chain groups form a chain complex, meaning that* $\partial^2 = 0$.

Proof. This will be a corollary of theorem . □

reference zu commuting boundary homomorphisms einfügen

Definition 4.2.10 (Orientation and Exact Sequences)**.** An orientation of a vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1, \dots b_m)$ of V and hence the product $b_1 \wedge \dots \wedge b_m$ is an element of the maximal exterior product $\det(V) = \Lambda^m V$. Two ordered bases define different elements in $\det(V)$ where their difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence an orientation is a choice of one connected component of $\det(V) \setminus \{0\}$. Now a short exact sequence of vector spaces

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

induces a canonical isomorphism

$$\det(A) \otimes \det(C) \to \det(B)$$
$$(a_1 \wedge \dots \wedge a_l) \otimes (c_1 \wedge \dots \wedge c_k) \mapsto (f(a_1) \wedge \dots f(a_l) \wedge \tilde{c_1} \wedge \dots \wedge \tilde{c_k}),$$

where $\tilde{c_j} \in g^{-1}(c_j)$. The well-definedness of this map (invariance under chosen lift of c) can be calculated, since two different choices differ in an element in the kernel of g which equals the image of f and hence the different choices vanish in the wedge product. By this isomorphism we get an orientation induced in the middle from the outside.

Remark 4.2.11. Let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be an exact sequence of vector spaces and $\mathfrak{a}$ be a basis of

Remark 4.2.12 (The Counting of Flow Lines)**.** There are several ways to formulate the counting of flow lines that happens in the boundary operator. Let $\gamma \in \mathcal{M}(q \to p)$ and $x \in \gamma$. Instead of using the parallel transport, we can introduce two different ordered bases in $T_x W(q \to)$. We use the following notation: Let $\gamma \in \mathcal{M}(q \to p)$. We extend γ to also contain the q and p by compactifying the flow line. Now assume that $x \in \gamma \setminus \{q, p\}$. For this we define $N_x^{W(q\to)}\gamma$ be the normal bundle of γ as a submanifold of $W(q \to)$. Since γ is contractible we can trivialise said normal bundle to get a bundle isomorphism:

$$\nu : N^{W(q\to)}\gamma \to \gamma \times T^u_p M .$$

Let $\nu\big|_x$ denote the isomorphism $N_x^{W(q\to)}\gamma \to T_p^u M$. An orientation of $T_p^u M$ induces then one on $N_x^{W(q\to)}\gamma$ and if we define the basis $(-\mathrm{grad}_x(f))$ of its own span to be positive, we get the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \langle -\mathrm{grad}_x(f)\rangle \arrow[r] & T_x W(q\to) \arrow[r] & N_x^{W(q\to)}\gamma \arrow[r] & 0 \\ & & & T_p^u M \arrow[u, "(\nu_x)^{-1}"'] & \end{tikzcd}$$

That introduces an orientation in the middle that can be represented by the basis $(-\mathrm{grad}_x(f), i\circ \nu|_x)(\mathfrak{b}^p)$, where i is the inclusion $N_x^{W(q\to)}\gamma \to T_x W(q\to)$ and $(\mathfrak{b}^p)$ is an ordered basis of $T_p^u M$. Another way to get an orientation there is to trivialise the bundle $TW(q\to)$ via the isomorphism:

$$\mu : TW(q\to) \to W(q\to)\times T_q^u M\,;\xi_y \mapsto \left(y, Q^{-1}_{E_q^{u-1}(y)} \circ \mathrm{d}E_q^{u-1}\big|_y(\xi_y)\right).$$

The map Q_p is the map from $\mathbb{R}^n \to T_p\mathbb{R}^n$ from definition 2.1.40 and the map E_q^u is an embedding from the refined unstable manifold theorem. This trivialisation together with an orientation on $T_q^u M$ orients the middle in the above exact sequence. Now the deciding factor that gives γ its sign is whether the two orientations agree. This can be checked by calculating the determinant of a certain map. To formulate the map somewhat readably, we assume that $\mathfrak{b}^q$ is a basis of $T_q^u M$ such that for $y = E_q^{u-1}(x)$ we have $\mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q) = -\mathrm{grad}_x(f)$ for $i = 1$, and for all $i > 1$ the image lives in the orthogonal complement of $-\mathrm{grad}_x(f)$ and hence in the domain of ν_x. Then we can define the map

$$\begin{aligned} B^{-1}_{\mathfrak{b}^p}\circ \nu_x \circ \mu_x^{-1} \circ B_q : \mathbb{R}^{\lambda_q} \supseteq \mathbb{R}^{\lambda_q - 1} = \langle e_2, \dots e_{\lambda_q}\rangle &\to \mathbb{R}^{\lambda_q-1} \\ e_i &\mapsto B^{-1}_{\mathfrak{b}^p}\circ \nu_x \circ \mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q), \end{aligned}$$

and the determinant of said map gives us n_γ.

Now we want to understand the flow in an unstable set and analyse two maps:

Remark 4.2.13. Let q be a critical point and $c < f(q)$ such that no other critical point lies in $f^{-1}[c, f(q)]$ then, the map

Bild malen!

$$\begin{aligned} \tau : W(q\to)\setminus\{q\} &\to \mathbb{R} \\ x &\mapsto \tau(x) \text{ such that } f\circ\varphi_{\tau(x)}(x) = c \end{aligned}$$

is smooth.

Proof. We know that such a map exists and is well defined since every flow line passes through $f^{-1}(c)$. Now let $x \in W(q\to)\setminus\{q\}$, V be a neighbourhood of x and U be a neighbourhood of $\varphi_{\tau(x)}(x)$. Define the map

$$\begin{aligned} G : V\times I &\to \mathbb{R} \\ (x,t) &\mapsto f\circ(\varphi_t(x)) . \end{aligned}$$

Now the partial derivative reads

$$\frac{\partial}{\partial t} G\Big|_{x,t} = \mathrm{d}f\Big|_{\varphi_t(x)} \circ \frac{\partial}{\partial t}\varphi_t(x)\Big|_t = \mathrm{d}f\Big|_{\varphi_t(x)} \circ \left(-\mathrm{grad}_{\varphi_t(x)}(f)\right) .$$

Since the flow line is transversal to the level set or in our case, the gradient is even orthogonal to the level sets, it is not contained in $T_{\varphi_{\tau(x)}(x)} f^{-1}(c) = \ker \mathrm{d}f\big|_{\varphi_{\tau(x)}(x)}$ and hence:

$$\frac{\partial}{\partial t} G\big| x, \tau(x) = \mathrm{d}f\Big|_{\varphi_{\tau(x)}(x)} \left(-\mathrm{grad}_{\varphi_{\tau(x)}(x)}(f)\right) \neq 0 .$$

This lets us use the implicit function theorem to deduce that the level set $G = c$ is locally the graph of a function $\tau : U_x \to \mathbb{R}$ where U_x is a neighbourhood of x. Hence, the function τ is globally smooth since this argument can be done everywhere. □

Lemma 4.2.14. *Let $\gamma_j \in \mathcal{M}(q \to)$ and let $x_j = \gamma_j \cap f^{-1}(c)$ denote the intersection of the image of γ_j with the levelset $f^{-1}(c)$. Then there exists a frame $(l_1, \ldots, l_m)$ of the restricted bundle $TW(q \to)|_{\gamma_j}$ such that for all $x \in \gamma_j$ $l_1(x) = -\mathrm{grad}_x(f)$ and*

$$\mathrm{d}\,\tau|_x \left(-\mathrm{grad}_x(f)\right) = -1 ,$$

and for any $i > 1$ we have that

$$\mathrm{d}\,\tau|_x \left(l_i(x)\right) = 0 .$$

Proof. We denote the connected component of $f^{-1}(c)$ that contains x_j by V_j. Now we choose a chart of $W(q \to)$ around $x_j \in W(q \to) \cap V_j$ such that the induced basis of $T_{x_j} W(q \to) = \langle -\mathrm{grad}_{x_j}(f)\rangle \oplus T_{x_j} V_j$ respects that splitting by setting $\frac{\partial}{\partial x^1}\big|_{x_j} = -\mathrm{grad}_{x_j}(f)$ and $\frac{\partial}{\partial x^i}\big|_{x_j} \in T_{x_j} V_j$ for i bigger than one. With this we can start our calculations:

Notice how $x \mapsto f \circ \varphi_{\tau(x)}(x)$ is a constant function for all $x \in W(q \to)$ and hence its differential is 0. For the next calculations we denote the derivative in x by d_x and in t by d_t:

$$\begin{aligned} 0 = \mathrm{d}_x \left(f \circ \varphi_\tau\right) = \overbrace{\mathrm{d}f}^{\text{1-form}} \left(\mathrm{d}\varphi_\tau\right) &= \mathrm{d}f \left(\mathrm{d}_t \varphi_t \circ \mathrm{d}_x \tau(x) + \mathrm{d}_x \varphi_{\tilde{t} = \tau(x)}\right) \\ &= \mathrm{d}f \left(\mathrm{d}_t \varphi_t \circ \mathrm{d}_x \tau(x)\right) + \mathrm{d}f \left(\mathrm{d}_x \varphi_{\tilde{t} = \tau(x)}\right) \\ &= \mathrm{d}_t \left(f \circ \varphi_t\right) \circ \mathrm{d}_x \tau + \mathrm{d}f \left(\mathrm{d}_x \varphi_{\tilde{t} = \tau(x)}\right) \end{aligned}$$

Hence we know that

$$\mathrm{d}_x \tau = -\frac{\mathrm{d}f \left(\mathrm{d}_x \varphi_{\tilde{t} = \tau(x)}\right)}{\mathrm{d}_t \left(f \circ \varphi_t\right)} = -\frac{\mathrm{d}f \left(\mathrm{d}_x \varphi_{\tilde{t} = \tau(x)}\right)}{-\left\|\mathrm{grad}_x(f)^2\right\|} \tag{16}$$

Now we can calculate in x_j where $\tau(x_j) = 0$

$$\mathrm{d}_x\tau\big|_{x_j}\frac{\partial}{\partial x^i}\Big|_{x_j} = \frac{\mathrm{d}f\big|_{x_j}\frac{\partial}{\partial x^i}\big|_{x_j}}{\|\mathrm{grad}(f)^2\|} = \frac{g\left(\mathrm{grad}_{x_i}(f), \frac{\partial}{\partial x^i}\big|_{x_j}\right)}{\|\mathrm{grad}(f)^2\|} = \begin{cases} -1 & \text{if } i = 1 \\ 0 & \text{else.} \end{cases}$$

This result can be extended to also work outside of x_j as follows: Notice the identity for all $s \in \mathbb{R}$ and $x \in W(q \to) \setminus \{q\}$:

$$\tau(\varphi_s(x)) = \tau(x) - s\,.$$

Differentiating this in the variable s in the point s' yields:

$$\begin{aligned} & \mathrm{d}\tau(\varphi_s(x))|_{s'} = \mathrm{d}(\tau(x) - s)|_{s'} = -1 \\ \text{and} \quad & \mathrm{d}\tau(\varphi_s(x))|_{s'} = \mathrm{d}\tau|_{\varphi_{s'}(x)} \circ \underbrace{\mathrm{d}\varphi_s(x)|_{s=s'}}_{=-\mathrm{grad}_{\varphi_s(x)}(f)} \end{aligned}$$

And hence in $s' = 0$ we have for all $x \in W(q \to) \setminus \{q\}$:

$$\mathrm{d}\tau|_x\,(-\mathrm{grad}(x)(f)) = -1\,.$$

We can furthermore define a basis of $T_xW(q \to)$ with $\mathfrak{b}_1 := -\mathrm{grad}_x(f)$ and $\mathfrak{b}_i = \left(\mathrm{d}\varphi_{\tau(x)}\big|_x\right)^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j}V_j$. Hence w_i is orthogonal on $-\mathrm{grad}_{x_j}(f)$ for all $i > 1$. Hence by the equation (16) we can calculate:

$$\mathrm{d}_x\tau(\mathfrak{b}_i) = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}(\mathfrak{b}_i)\right)}{-\|\mathrm{grad}(f)^2\|} = -\frac{\mathrm{d}f\,(w_i)}{-\|\mathrm{grad}(f)^2\|} = 0 \quad \text{for all } i > 1\,.$$

Finally after choosing a basis (w_i), the map $x \mapsto \mathfrak{b}_i$ is smooth for all i and any $x \in W(q \to) \setminus \{q\}$ and hence altogether we get the desired frame. □

Definition 4.2.15. Let q be a critical point and c such that $c < f(q)$ and $f^{-1}[c, f(q)] = q$. Then we define the map

$$\begin{aligned} \rho : W(q \to) \setminus \{q\} &\to W(q \to) \cap f^{-1}(c) \\ x &\mapsto \varphi_{\tau(x)}(x)\,. \end{aligned}$$

Lemma 4.2.16. *Assume that $\mathfrak{d} = (\mathfrak{d}_i)$ is a basis of T^u_qM. Let $y := E^{u-1}_q(x_j)$ and $y' \in (\rho \circ E^u_q)^{-1}(x_j)$. Then the map*

$$\mathrm{d}E^u_q\big|^{-1}_y \circ \mathrm{d}\rho \circ \mathrm{d}E^u_q\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q} \to T_yT^u_qM$$

hier muss x vs x_j nochmal angeschaut wertden und was woraus kommet. auch sollte Q nochmal genannt werden und E^u_q

can be written as

$$\pi_{-\mathrm{grad}_{x_j}(f)^\top} \circ (B_{\mathfrak{b}^q_{x_j}}) : \mathbb{R}^{\lambda_q} \to T_{x_j} V_j$$
$$e_i \mapsto \begin{cases} 0 & \textit{for } i = 1 \\ (\mathfrak{b}^q_{x_j})_i & \textit{for } i \neq 1 \,. \end{cases}$$

where $(\mathfrak{b}^q_{x_j})_j$ *is basis of* $T_{x_j} W(q \to)$ *that is induced from* $\mathfrak{d}$ *and* $(\mathfrak{b}^q_{x_j})_1 = -\mathrm{grad}_{x_j}(f)$. *In other words, this is a basis compatible with the one used in the counting of flow lines in the Morse boundary operator.*

Proof Strategy:
To prove this, we first remind ourselves how the basis $(\mathfrak{b}^q_{x_j})_{j>1}$ is constructed. It is a basis that when extended with $-\mathrm{grad}_{x_j}(f)$ as the first vector is compatible with $\mathfrak{d}$. Hence, we can construct it from $\mathfrak{d}$, by assuming that $\mathrm{d}E^u_q\big|_y \circ Q_y \circ \mathfrak{d}_1 = -\mathrm{grad}_y(f)$. Then, if we push $\mathfrak{d}$ to $T_{x_j} W(q \to)$ via $\mathrm{d}E^u_q\big|_y \circ Q_y$ and delete the first vector, we get the basis we wanted. So what we need to check in order to prove this lemma is, if the maps

$$\mathrm{d}E^u_q\big|_y \circ Q_y \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q - 1} \to \left(-\mathrm{grad}_{x_j}(f)\right)^\top$$
$$\mathrm{d}E^u_q\big|_y^{-1} \circ \mathrm{d}\rho \circ \mathrm{d}E^u_q\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q - 1} \to \left(-\mathrm{grad}_{x_j}(f)\right)^\top$$

induce the same orientation when restricted to the same domain such that they become vector space isomorphisms. To prove this we will work from the bottom up: We adapt the basis $\mathfrak{d}$ to work with the lower map and show, later, that the adapted basis fits the requirements above:

Proof. We start by defining the space $\left(\tau \circ E^u_q\right)^{-1}(\tilde{t})$, where $\tilde{t} := \tau \circ E^u_q(y')$. As a level set, this is a submanifold of $T^u_q M$ and its tangent space in y' is given by the basis $E^u_q\big|_{y'}^{-1}(\mathfrak{b}_{\mathrm{i}})$ for $i > 1$, where $\mathfrak{b}_i = \mathrm{d}\varphi_{\tilde{t}}\big|_{y'}^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j} V_j$. Now we have the equality of maps:

$$A_{\tilde{t}}\big|_{(\tau \circ E^u_q)^{-1}(\tilde{t})} = \left(E^{u-1}_q \circ \rho \circ E^u_q\right)\big|_{(\tau \circ E^u_q)^{-1}(\tilde{t})}$$

Hence for a basis $\mathfrak{d}_i$ of $T^u_q M$ such that $\mathrm{d}E^u_q\big|_{y'} \circ Q_{y'}(\mathfrak{d}_{\mathrm{i}})$ is $-\mathrm{grad}_{y'}(f)$ for $i = 1$ and $\mathfrak{b}_{\mathrm{i}}$ else, we have:

$$\mathrm{d}\left(E^{u-1}_q \circ \rho \circ E^u_q\right)\big|_{y'} \circ Q_{y'}(\mathfrak{d}_i) = Q_y \circ A_{\tilde{t}}(\mathfrak{d}_i).$$

For $i = 1$, the left side maps to 0, and the right side:

$$\begin{aligned}
Q_y \circ A_{\tilde{t}}(\mathfrak{d}_1) &= \mathrm{d}A_{\tilde{t}}\big|_{y'} \circ Q_{y'}(\mathfrak{d}_1) \\
&= \mathrm{d}A_{\tilde{t}}\big|_{y'} \circ \mathrm{d}E_q^u\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\
&= \mathrm{d}A_{\tilde{t}} \circ \mathrm{d}E_q^u\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\
&= \mathrm{d}(E_q^u)^{-1} \circ \varphi_{\tilde{t}}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\
&= \mathrm{d}E_q^u\big|_{x_j}^{-1} \circ \mathrm{d}\varphi_{\tilde{t}}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\
&= \mathrm{d}E_q^u\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right)
\end{aligned}$$

The last equation is the general fact proven in remark **??**, that for fixed t the identity holds:

$$\mathrm{d}_x\varphi_t\big|_y(-\mathrm{grad}_y(f)) = -\mathrm{grad}_{\varphi_t(y)}(f)$$

Hence if π denotes the projection onto the complement of $\mathrm{d}E_q^u\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right)$ we have an equality of maps

$$\mathrm{d}\left(E_q^{u-1} \circ \rho \circ E_q^u\right)\big|_{y'} \circ Q_{y'} = \pi \circ Q_y \circ A_{\tilde{t}}.$$

Or by the above calculation, we can throw out π, if we exclude $\mathfrak{d}_1$. Furthermore, the equation shows that our requirement $\mathrm{d}E_q^u\big|_{y'} \circ Q_{y'}(\mathfrak{d}_1) = -\mathrm{grad}_{y'}(f)$ implies that $\mathfrak{d}$ is compatible with a basis where $\mathrm{d}E_q^u\big|_y \circ Q_y \circ \mathfrak{d}_1 = -\mathrm{grad}_y(f)$, since the sign of the determinant of $A_{\tilde{t}}$ is always positive. This concludes the proof! $\square$

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